

Saccharomyces cerevisiae Rev7 promotes non-homologous end-joining by blocking Mre11 nuclease and Rad50's ATPase activities and homologous recombination

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Sugith Badugu, Kshitiza M Dhyani, Manoj Thakur, Kalappa Muniyappa

Department of Biochemistry, Indian Institute of Science, Bengaluru, India • Sri Venkateswara College, University of Delhi, New Delhi, India

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eLife Assessment

This manuscript reports **important** data providing evidence that a 42 amino acid region of Rev7 is necessary and sufficient for interaction with the Rad50-Mre11-Xrs2 complex in budding yeast. The authors conclude that Rev7 inhibits the Rad50 ATPase and the Mre11 nuclease with the exception of ssDNA exonuclease activity. The **convincing** data largely support the conclusions, although the effect of Rev7 on homologous recombination is less well documented and the observed effect on resection is moderate. Specifically, the result that the Rev7 C-terminal truncation lacking the 42 amino acid region still suppresses homologous recombination is unexpected and unexplained.

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Abstract

Recent studies have shown that, in human cancer cells, the tetrameric Shieldin complex (comprising REV7, SHLD1, SHLD2, and SHLD3) facilitates non-homologous end-joining (NHEJ) while blocking homologous recombination (HR). Surprisingly, several eukaryotic species lack SHLD1, SHLD2 and SHLD3 orthologs, suggesting that Rev7 may leverage an alternative mechanism to regulate the double-strand break (DSB) repair pathway choice. Exploring this hypothesis, we discovered that *Saccharomyces cerevisiae* Rev7 physically interacts with the Mre11-Rad50-Xrs2 (MRX) subunits, impedes G-quadruplex DNA synergised-HU-induced toxicity and facilitates NHEJ, while antagonizing HR. Notably, we reveal that a 42-amino acid C-terminal fragment of Rev7 binds to the subunits of MRX complex, protects *rev7Δ* cells from G-quadruplex DNA-HU-induced toxicity, and promotes NHEJ by blocking HR. By comparison, the N-terminal HORMA domain, a conserved protein-protein interaction module, was dispensable. We further show that the full-length Rev7 impedes Mre11 nuclease and Rad50's ATPase activities without affecting the latter's ATP-binding ability. Combined, these results provide unanticipated insights into the functional interaction between the MRX subunits and

Rev7 and highlight a mechanism by which Rev7 facilitates DSB repair via NHEJ, and attenuation of HR, by blocking Mre11 nuclease and Rad50's ATPase activities in *S. cerevisiae*.

Impact statement

The mechanisms steering DNA double-strand break repair pathway choice is a topic of intense investigation, but remains incompletely understood. Our findings suggest that yeast Rev7 promotes DSB repair via NHEJ and inhibits homologous recombination by blocking Mre11 nuclease and Rad50's ATPase activities.

Introduction

A hallmark of low fidelity DNA polymerases, also known as DNA translesion synthesis polymerases (TLS polymerases), with no detectable proofreading activity, is their ability to catalyse DNA synthesis across a variety of bulky, helix-distorting DNA lesions (Prakash et al., 2005 [↗](#); Vaisman et al., 2017 [↗](#); Maiorano et al., 2021 [↗](#); Ling et al., 2022 [↗](#)). The TLS polymerases are also involved in a plethora of cellular processes, including but not limited to epigenetics, immune signalling, viral mutagenesis and cancer development (Paniagua and Jacobs, 2023 [↗](#)). Indeed, translesion DNA synthesis is a source of mutagenesis, potentially contributing to the development of cancer and drug resistance (Lange et al., 2011 [↗](#); Baranovskiy et al., 2012 [↗](#); Pilzecker et al., 2019 [↗](#)). The *Saccharomyces cerevisiae* TLS Polζ (henceforth referred to as ScPolζ) is a four-subunit enzyme, comprised of catalytic subunit Rev3, two regulatory subunits of Rev7 and the accessory subunits Pol31 and Pol32 (Johnson et al., 2012 [↗](#); Makarova et al., 2012 [↗](#)). Current evidence suggests that the error rate during ScPolζ-catalysed replication of undamaged DNA templates is much higher than that of DNA polymerases, as it lacks 3'-to-5' proofreading exonuclease activity (Lawrence et al., 1985a [↗](#); 1985b [↗](#); Huang et al., 2002 [↗](#); Northam et al., 2006 [↗](#); Zhong et al., 2006 [↗](#); Kochenova et al., 2017 [↗](#)). Consistent with this, *S. cerevisiae* *rev3*, *rev7* or *pol32* mutant strains show greatly reduced spontaneous mutation frequencies, confirming that ScPolζ is responsible for DNA damage-induced mutagenesis (Quah et al., 1980 [↗](#); Lawrence et al., 1985a [↗](#); 1985b [↗](#); Morrison et al., 1989 [↗](#); Nelson et al., 1996 [↗](#); Makarova et al., 2015 [↗](#)).

The TLS Polζ exists in a wide range of unicellular and multicellular eukaryotes, including fungi, plants, and animals (Maiorano et al., 2021 [↗](#); Ling et al., 2022 [↗](#); Paniagua and Jacobs, 2023 [↗](#)). While the catalytic subunit Rev3 alone is capable of replicating damaged DNA, Rev7 enhances its catalytic efficiency by twenty-to thirtyfold (Quah et al., 1980 [↗](#); Morrison et al., 1989 [↗](#); Nelson et al., 1996 [↗](#); Makarova et al., 2015 [↗](#)) and the accessory subunits Pol31 and Pol32 further raise it by three-to tenfold (Johnson et al., 2012 [↗](#); Makarova et al., 2012 [↗](#)), suggesting that they abet the processivity of translesion DNA synthesis by Polζ (Acharya et al., 2006 [↗](#); Bezalel-Buch et al., 2020 [↗](#)). Indeed, Rev7 (also known as MAD2B and MAD2L2) is an adapter protein, which acts as a bridge between Rev3 and Rev1 (Haracska et al., 2001 [↗](#); Kikuchi et al., 2012 [↗](#); Pustovalova et al., 2012 [↗](#)). Various structural and biochemical investigations have uncovered unique structural features of Polζ, wherein its subunits interact with each other to form an highly proficient, multi-subunit TLS holoenzyme (Gómez-Llorente et al., 2013 [↗](#); Malik et al., 2020 [↗](#); Du Truong et al., 2021 [↗](#)). The high-resolution cryo-EM structures of ScPolζ holoenzyme have revealed that the subunits Rev3, Rev7, Pol31, and Pol32 assemble into a pentameric ring-shaped structure in which they are maintained by a chain of uninterrupted protein-protein interaction networks (Gómez-Llorente et al., 2013 [↗](#); Malik et al., 2020 [↗](#); Du Truong et al., 2021 [↗](#)). Precise details of how Polζ holoenzyme achieves its substrate specificity have not been fully understood. Shedding light on this, A. Aggarwal's lab has recently provided insights into the mechanism by which the active site of ScPolζ responds to the A:C mismatched duplex DNA distortion (Malik et al., 2022 [↗](#)).

A flurry of research has documented that the tetrameric Shieldin complex – comprising REV7, SHLD1, SHLD2 and SHLD3 – binds single-stranded DNA (ssDNA), blocks 5' end resection and homologous recombination (HR), antagonizes the recruitment of BRCA1 to the DSB, while facilitating non-homologous end-joining (NHEJ) (Xu et al., 2015 [\[1\]](#); Boersma et al., 2015 [\[2\]](#); Mirman et al., 2018 [\[3\]](#); Findlay et al., 2018 [\[4\]](#); Gupta et al., 2018 [\[5\]](#); Ghezraoui et al., 2018 [\[6\]](#); Dev et al., 2018 [\[7\]](#); Tomida et al., 2018 [\[8\]](#); Gao et al., 2018 [\[9\]](#); Noordermeer et al., 2018 [\[10\]](#); Liang et al., 2020 [\[11\]](#)). Investigations have also shown that the N-terminus of SHLD3 interacts with REV7 using a stereotypical “safety-belt” interaction mechanism (Gupta et al., 2018 [\[5\]](#); Ghezraoui et al., 2018 [\[6\]](#); Dev et al., 2018 [\[7\]](#); Tomida et al., 2018 [\[8\]](#); Gao et al., 2018 [\[9\]](#); Noordermeer et al., 2018 [\[10\]](#); Liang et al., 2020 [\[11\]](#); Clairmont et al., 2020 [\[12\]](#); Dai et al., 2020 [\[13\]](#)). In the alternative pathway, Shieldin-53BP1-RIF1 counteracts DSB resection and recruits the CST-Polα-primase complex to promote fill-in at the resected DNA ends (Mirman et al., 2018 [\[3\]](#); 2022 [\[14\]](#); 2023 [\[15\]](#)). Cells derived from Fanconi anaemia patients, caused by biallelic REV7 mutations, display hypersensitivity to DNA cross-linking agents, accumulate chromosome breaks during S/G2 phase of the cell cycle and activation of the p53/p21 axis (Bluteau et al., 2016 [\[16\]](#)), revealing its critical role in providing protection against FA disease. It is worth pointing out that Rev7 inhibits the anaphase-promoting complex by sequestering Cdh1, and modulates the transition from metaphase to anaphase, thereby contributing to mitotic fidelity (Listovsky and Sale, 2013 [\[17\]](#); Vaisman et al., 2017 [\[18\]](#); Ling et al., 2022 [\[19\]](#)). However, a mechanistic understanding of how Rev7 regulates cell cycle events and how such roles differ or relate to its role in TLS remains underexplored. Although the emphasis of the findings differ, the fact that Rev7 functions as an anti-resection factor has spurred a new wave of experiments on DNA repair pathways (Setiাপutra and Durocher, 2019 [\[20\]](#); Clairmont and D’Andrea, 2021 [\[21\]](#)).

At DSBs in *S. cerevisiae*, the Mre11–Rad50–Xrs2 (MRX) in conjunction with Sae2 first catalyses endonucleolytic cleavage of 5'-terminated DNA strands and then its 3' → 5' exonucleolytic activity produces a short 3'-ssDNA overhang, which is followed by resection in a 5' → 3' direction by either Exo1 or Dna2–Sgs1 complex to produce long tracks of ssDNA that are critically important for HR (Cejka and Symington, 2021 [\[22\]](#)). As such, the mechanism by which cells restrain over-resection of DSBs remains incompletely understood, although hyper-resection could potentially hinder optimal HR and trigger genomic instability. Surprisingly, however, Shieldin orthologs are absent in different organisms such as yeast, fruit fly, nematode worm, zebrafish and frog (Setiাপutra and Durocher, 2019 [\[20\]](#)). Consequently, we hypothesized that Rev7 in *S. cerevisiae* (hereafter referred to as ScRev7) may recruit an unknown functional equivalent(s) of Shieldin orthologs to regulate the DSB repair pathway choice between HR and NHEJ. Thus, an alternative mechanism might involve the MRX complex on the basis of current knowledge that it plays multiple roles in signalling, processing and repair of DSBs (Cejka and Symington, 2021 [\[22\]](#)). In this study, we provide robust evidence that ScRev7, via its 42-aminoacid C-terminal fragment in the “safety belt” region, physically interacts with the Mre11, Rad50 or Xrs2 subunits, protects *rev711* cells from G-quadruplex DNA/HU-induced toxicity and facilitates DSB repair via NHEJ while antagonizing HR. Mechanistic studies revealed that ScRev7 binds to the MRX subunits with sub-micromolar affinity, attenuates Mre11 nuclease and Rad50's ATPase activities, without affecting the ability of the latter to bind ATP. Collectively, our findings not only provide evidence for functional interaction between Rev7 and the MRX subunits, but also reveal novel insights into how Rev7 regulates the pathway choice between HR and NHEJ in *S. cerevisiae*.

Results

ScRev7 interacts with the MRX subunits

As noted above, studies in cancer cells have shown that the Rev7-Shieldin effector complex facilitates NHEJ by blocking 5' end resection and HR (Xu et al., 2015 [\[1\]](#); Boersma et al., 2015 [\[2\]](#); Mirman et al., 2018 [\[3\]](#); Findlay et al., 2018 [\[4\]](#); Gupta et al., 2018 [\[5\]](#); Ghezraoui et al., 2018 [\[6\]](#); Cejka and Symington, 2021 [\[22\]](#)). Since there are no identifiable Shieldin orthologs in *S. cerevisiae*

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Given that Rev3 has also been implicated in HR-mediated DSB repair (Sonoda et al., 2003 [↗](#)), we asked whether the MRX subunits interact with ScRev7 in the *rev311* mutant strain. To address this question, the *S. cerevisiae rev311* strain was co-transformed with a combination of bait (pGBKT7 or pGBKT7-REV7) and prey vectors (pGADT7, pGADT7-REV7, pGADT7-MRE11, pGADT7-RAD50, pGADT7-XRS2 or pGADT7-SAE2). Interestingly, we observed binary interactions between the subunits of MRX complex and ScRev7 in the *rev311* mutant in the Y2H system (Figure 1B [↗](#)), indicating that Rev3 is dispensable for the binding of ScRev7 to the MRX subunits. Analogously ScRev7 showed binary interactions with the subunits of MRX complex in the *mre11Δ rad50Δ xrs2Δ* triple mutant strain (Figure 1C [↗](#)), thereby confirming that their association is independent of endogenous MRX subunits.

ScRev7 physically interacts with the MRX subunits

Microscale thermophoresis (MST) allows for quantitative analysis of protein-protein interactions in free solution (Wienken et al., 2010 [↗](#)). Since the binding of ScRev7 to the MRX subunits was unanticipated, we sought to validate their interaction by an orthogonal assay and determine their binding affinities using purified proteins. To this end, we purified eGFP-tagged ScRev7 (Figure 1—figure supplement 1A [↗](#)), confirmed its identity (Figure 1—figure supplement 2 [↗](#)), and leveraged MST titration approach to measure its binding affinity to purified Mre11, Rad50, Xrs2 and Rev1 subunits, and also to the Mre11-Rad50 complex (Figure 1—figure supplement 1C, D [↗](#)). The MST signals were plotted as a function of ligand concentration (Figure 1—figure supplement 3A-F [↗](#)). Normalized MST data were fitted to a logistic binding curve, resulting in an apparent dissociation constants (K_d) of 0.16 ± 0.07 , 0.23 ± 0.06 and 0.18 ± 0.03 μ M for Rad50, Mre11 subunits and Mre11-Rad50 complex, respectively, which is two-to threefold greater as compared with Xrs2 (Figure 1D [↗](#)). *S. cerevisiae* Rev1 was used as a positive control to ensure the accuracy of the Y2H assay. The binding kinetics measured demonstrated that purified Rev1 (Figure 1—figure supplement 1D [↗](#)) bound to Rev7 with an affinity (Figure 1D [↗](#) and Figure 1—figure supplement 3D [↗](#)), comparable to previously reported value (Rizzo et al., 2018 [↗](#); Guo et al., 2003 [↗](#)). On the other hand, negative controls such as purified Sae2 (Figure 1—figure supplement 1E [↗](#)) and eGFP showed no significant binding to the MRX subunits and GFP-tagged Rev7, respectively (Figure 1—figure supplement 3F [↗](#)). We next tested the affinity of Rev7-C1 for the Mre11, Rad50 and MR complex. The results showed that Rev7C1 binds to Mre11 with

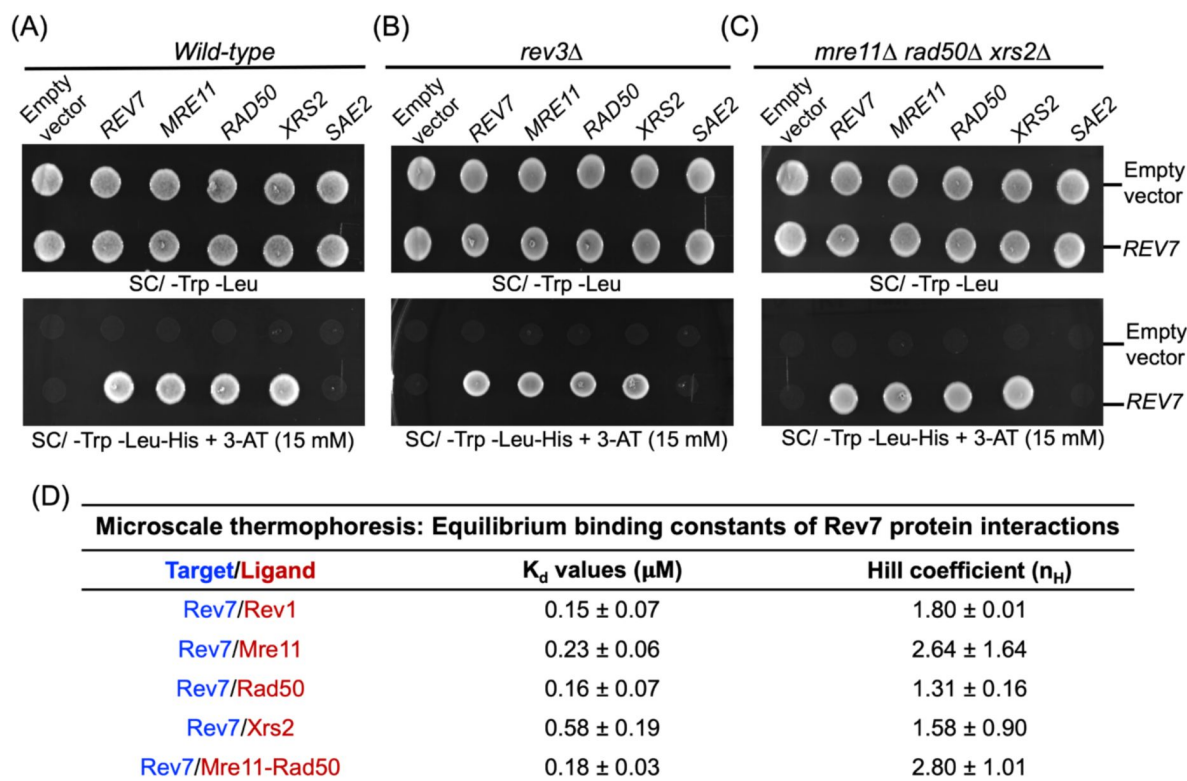


Figure 1.

Y2H screens suggest interaction between ScRev7 and the MRX subunits.

The Y2H assay was performed in: (A) wild-type, (B) *rev311*, and (C) *mre1111 rad5011 xrs211* mutant strains in PJ69-4A background. These strains were cotransformed with pairwise combinations of empty vector, bait (pGBKT7-REV7) and prey (pGADT7/MRE11, RAD50, XRS2, REV7 or SAE2) plasmids. Equal number of mid-log phase cells were spotted onto the SC/-Trp -Leu agar plates (upper panels) or SC/-Trp -Leu-His agar plates containing 3-AT (bottom panels). Cells were imaged after 48 h of growth at 30°C. The images shown in panels (A), (B) and (C) are representative of three independent experiments. (D) Quantitative parameters for interaction between ScRev7 and Rev1, Mre11, Rad50, Xrs2 or Mre11-Rad50 proteins.

approximately 3-fold reduced affinity and to the Rad50 and MR complex with 10-fold reduced affinity (**Figure 1—figure supplement 4**). Collectively, these results confirm the specificity of interaction between Rev7 and the MRX subunits. It is also noteworthy that the Hill coefficients (n_H) indicate larger than one, implying positive cooperativity (**Figure 1D**). The quantitative assessment of binding affinities together with Y2H data suggest that Rev7 robustly interacts with the MRX subunits.

A 42-aminoacid C-terminal segment of Rev7 is critical for its interaction with the MRX subunits

Since the data obtained from the Y2H screening system and MST based protein-protein interaction assay suggested pairwise association between the subunits of MRX complex and Rev7, we sought to identify the functional domain(s) in the ScRev7 required for interaction with the MRX subunits. For this purpose, we generated three N-terminally truncated and an equal number of C-terminally truncated variants. We refer to these variants as Rev7-N1; Rev7-N2; Rev7-N3, and Rev7-C1, Rev7-C2 and Rev7-C3, respectively (**Figure 2A**). We then asked whether these variants bind to the MRX subunits and enable the growth of yeast cells on selection nutrient medium. Our experiments surprisingly revealed that cells expressing the ScRev7 N-terminally truncated variants showed robust cell growth (bottom panel of **Figure 2B**) similar to the WT (**Figure 1A-C**), indicating that they interact with the subunits of MRX complex. Further, these results indicated that the Rev7's N-terminal HORMA domain (residues 1-149), an evolutionarily conserved protein-protein interaction module, is dispensable for binding to the MRX subunits (**Figure 2B**). Notwithstanding, we do not exclude the possibility that it may play a role that is undetectable by the Y2H assay.

Next, we performed Y2H experiments using the C-terminally truncated species (**Figure 2A**) and found that deletion of C-terminal forty-two amino acid residues (i.e., 203-245) resulted in loss of cell proliferation and growth (bottom panel of **Figure 2C**). Similarly, deletion of the C-terminal 150 to 203 amino acid residues of ScRev7 abrogated yeast cell growth (**Figure 2C**). These results indicate that the C-terminal 42-residue segment of ScRev7 is critical for its interaction with the MRX subunits. To further confirm these results, an Y2H experiment was carried out with cells co-expressing 42 aa peptide and the Mre11, Rad50 or Xrs2 subunits. Such an analysis showed that the ScRev7's 42 residue peptide alone was sufficient for interaction with each subunit of the MRX complex (**Figure 2—figure supplement 1**). However, it remained possible that the inability of cells expressing the C-terminally truncated variants of ScRev7 (with appropriate prey proteins) to grow in selection medium (**Figure 2C, bottom panel**) may be due to altered expression or decreased abundance of truncated species. To explore this possibility, whole cell lysates derived from cells expressing the N- and C-terminally truncated variants, tagged with c-Myc epitope at the N-terminus, were resolved by SDS-PAGE and probed with anti-c-Myc antibody. Reassuringly, the results revealed comparable levels of N- and C-terminally truncated species of ScRev7 in the whole cell lysates of strains that were employed for Y2H analyses (**Figure 2—figure supplement 2**).

Models predicted by AlphaFold-Multimer reveal that Mre11 and Rad50 subunits independently associate with Rev7

To further characterize the interaction between the MRX subunits and ScRev7, a structure prediction algorithm, AF2-multimer (Evans et al., 2022), was leveraged to construct structural models of ScRev7-Mre11 and ScRev7-Rad50 heterodimers. The models with the high pLDDT scores were considered for further analysis. Strikingly, the models indicated that Mre11 and Rad50 binding surfaces overlap with ScRev7, suggesting that the latter can bind these subunits (**Figure 2—figure supplement 3**). In the model of Rev7-Mre11 complex, residues Asp 206 and Ile 240 in the Rev7 C-terminal “safety belt region” (green) mediate dimerization with Mre11 through

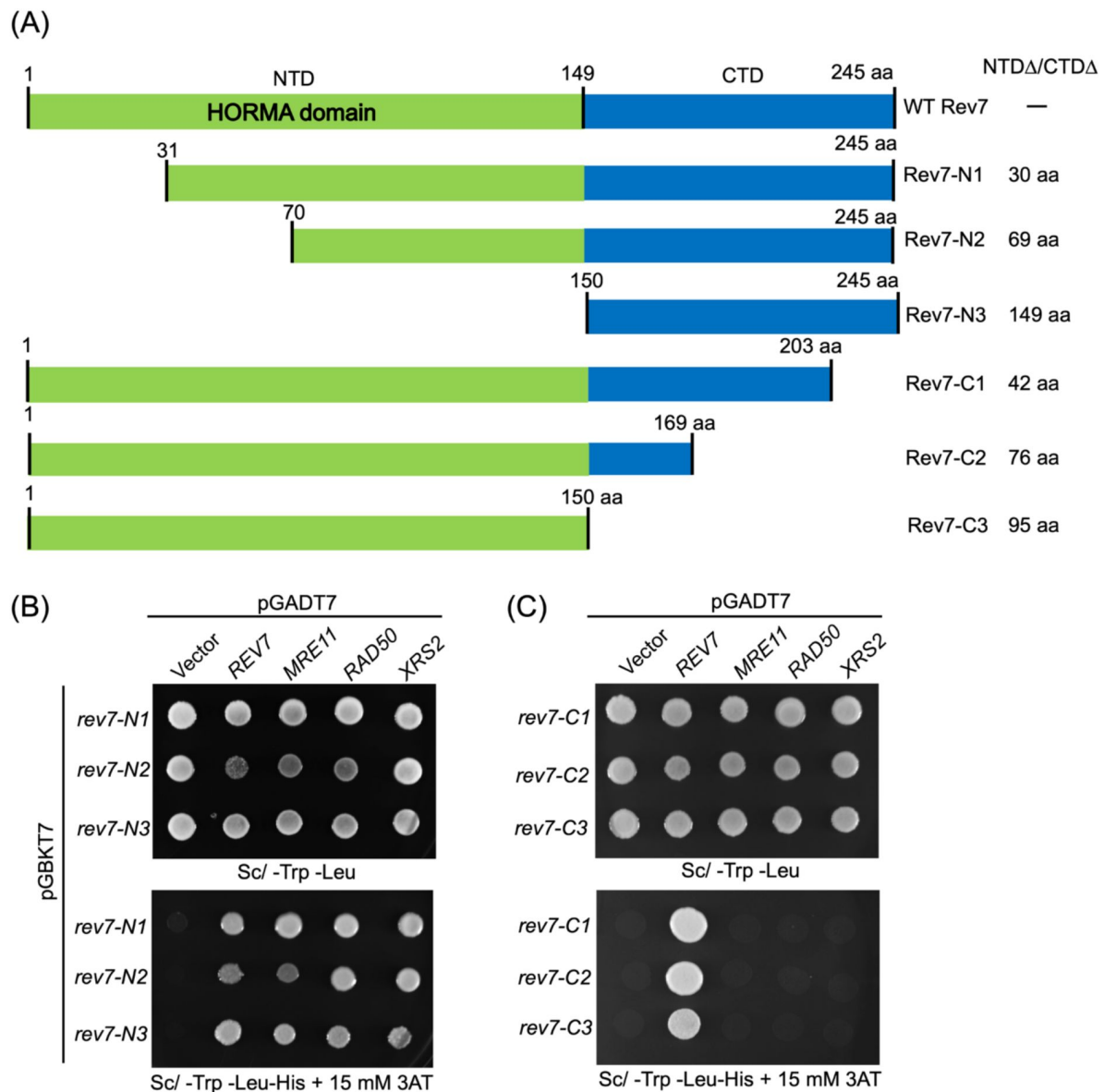


Figure 2.

Deletion analysis revealed that the C-terminal 42 amino acids of ScRev7 are critical for its binding to the MRX subunits.

(A) Schematic representation of the full-length and truncated ScRev7 variants. The truncated species lacking the indicated number of amino acids (aa) in the N-terminal domain (NTD11) or C-terminal domain (CTD11) is indicated on the right-hand side of the figure. (B) Representative images of spot assays of cells carrying pairwise combination of empty vector, bait and prey plasmids expressing N-terminally truncated species of Rev7 and full-length Rev7, Mre11, Rad50, Xrs2, respectively. (C) Same as panel (B), but with the bait plasmids encoding C-terminally truncated Rev7 variants. Cells were imaged after 48h of growth at 30 °C. Y2H assay was performed as described in the legend to [Figure 1](#). Data are representative of three independent experiments.

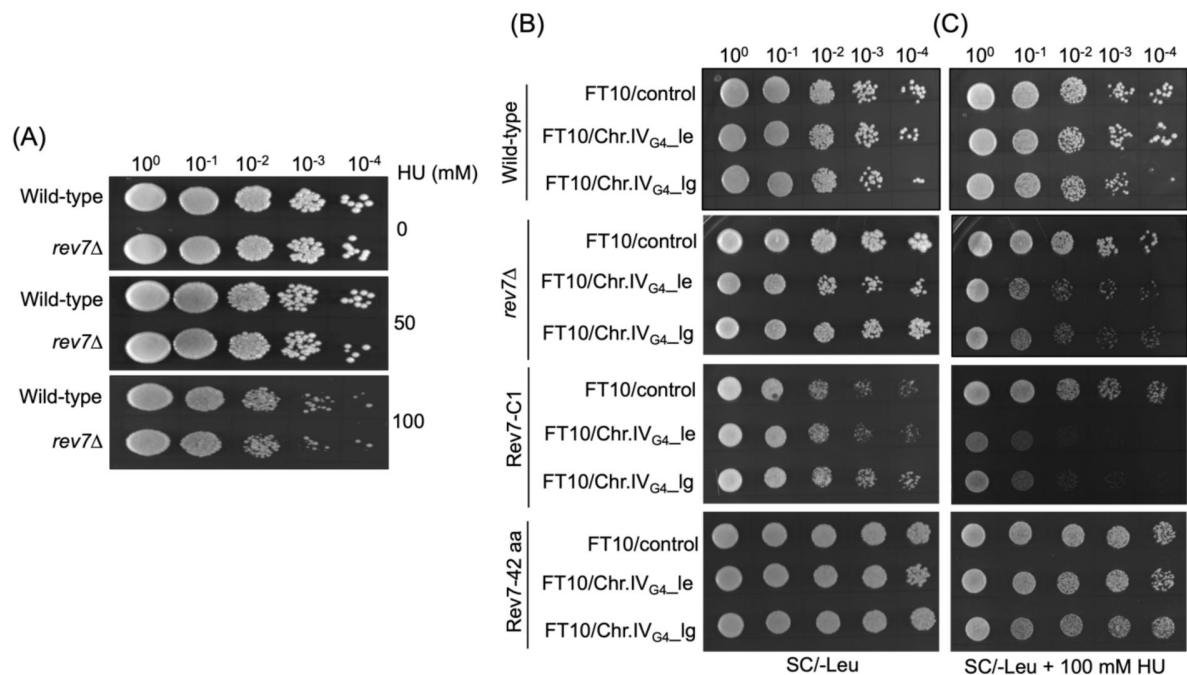


Figure 3.

A 42-amino acid segment at the extreme C-terminus of ScRev7 renders cells resistant to the synergistic adverse effect of G-quadruplex DNA and HU.

(A) *REV7* deletion does not affect HU sensitivity of *rev7Δ* cells. Representative images of YPD plates showing spot assay of wild-type W1588-4C strain and its derivative *rev7Δ* mutant cells, in the absence or presence of 50 or 100 mM HU. (B) Representative images of SC/-Leu agar plates showing spot assay of WT, *rev7Δ*, *rev7-C1* or *rev7-42 aa* cells harboring the indicated plasmids carrying G-quadruplex capable sequences derived from chromosome IV (Chr.IV_{G4}). The cells were grown on SC/-Leu agar plates in the absence of HU. (C) Same as panel (B), but the growth medium contained 100 mM HU. For serial dilutions, each strain was grown in SC/-Leu medium, normalized to OD₆₀₀ = 1.0, and serially diluted using yeast nitrogen base medium. Five μ l aliquots from the serial dilutions were spotted onto the SC/-Leu agar plates with or without HU. FT10/Control plasmid lacks the G-quadruplex DNA insert. The abbreviations "le" and "lg" stand for leading and lagging strands, respectively. Cells were imaged after 4 days of growth at 30 °C. Data are representative of three independent experiments.

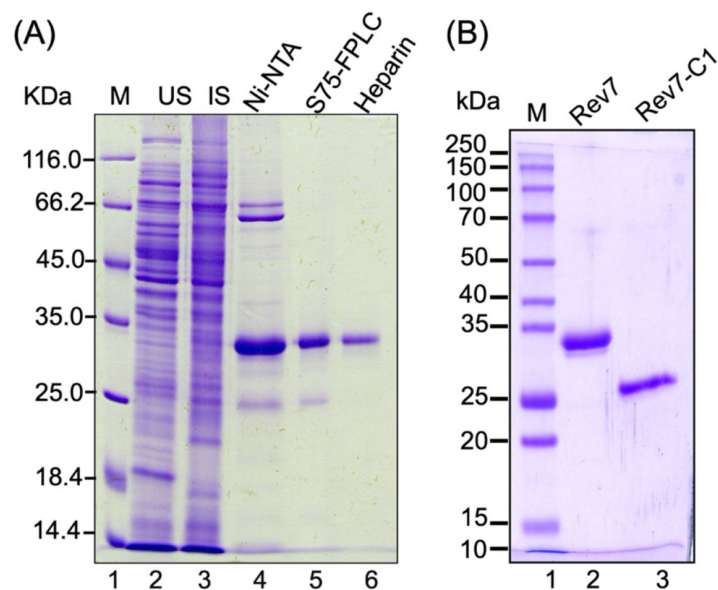


Figure 4.

Purification of ScRev7 and ScRev7-C1 proteins.

(A) SDS-PAGE analysis of protein samples from different stages of ScRev7 purification. Lane 1: standard protein markers; 2, uninduced cell lysate (10 μ g); 3, induced cell lysate (10 μ g); 4, Ni²⁺-NTA column eluate (3 μ g); 5, Superdex S75 column eluate (1 μ g); 6, eluate from heparin column (0.8 μ g). **(B)** SDS-PAGE analysis of purified truncated ScRev7-C1 variant. Lane 1: standard protein markers. Lanes 2 and 3, purified full length ScRev7 and ScRev7-C1 variant.

residues Asp 131 and Arg 181. Interestingly, His 127 in the Rev7 N-terminal HORMA domain also contribute its binding to Asp 131 of Mre11 (**Figure 2—figure supplement 3A** [↗](#), **Supplementary Table S1** [↗](#)). Furthermore, modelling studies of Rev7-Rad50 complex showed that the ScRev7 C-terminal residues Lys 168, Glu 184, Asn 189 and Asp 188 mediate dimerization with ScRad50 via residues Glu 577, Lys 596 and Arg 603 (**Figure 2—figure supplement 3B** [↗](#), **Supplementary Table S2** [↗](#)). Curiously, the AF2-multimer models also revealed that amino acid residues outside of the 42-residue fragment also contribute to pairwise interactions between Rev7 and the Mre11 and Rad50 subunits, although Y2H assays did not identify such interaction. Indeed, similar findings have been previously noted for several other interacting partners (You et al., 2006 [↗](#); Koegl and Uetz, 2007 [↗](#); Hoff et al., 2010 [↗](#)).

The Rev7's C-terminal 42-residue fragment mitigates the G-quadruplex-HU induced toxic effects

Several studies have documented the genome-wide prevalence of G-quadruplex DNA structures in various organisms ranging from viruses to humans, which play regulatory roles in diverse cellular processes (Rhodes and Lipps, 2015 [↗](#); Spiegel et al., 2020 [↗](#); Yadav et al., 2021 [↗](#)). For instance, it has been shown that Rev1-deficient chicken DT40 cells exhibit defects in replicating G quadruplex-forming motifs (Sarkies et al., 2010 [↗](#), 2012 [↗](#)). Similarly, computational and genetic studies in *S. cerevisiae* have revealed that G-quadruplex-forming motifs cause slow growth in replication stressed Pif1-deficient cells and affect genome integrity (Capra et al., 2010 [↗](#); Paeschke et al., 2011 [↗](#), 2013 [↗](#)). Inspired by these findings, we sought to understand whether ScRev7 plays a role in genome maintenance using the assay developed by the V. Zakian's lab (Paeschke et al., 2011 [↗](#)). We first compared the viability of *rev711* mutant cells devoid of exogenous G-quadruplex forming sequences with that of WT under conditions of optimal growth and HU-induced replication stress. This analysis revealed that, like the WT, *rev711* mutant cells grew robustly in the absence or presence of HU (**Figure 3A** [↗](#)).

Next, we next sought to determine whether the G-quadruplex DNA motifs affect cell viability in the absence and presence of HU. To address this question, we tested the effect of exogenous G-quadruplex forming sequences, positioned on either the leading or lagging template strands, on the viability of WT and *rev711* mutant strains, relative to cells expressing the ScRev7 variants (Rev7-C1 or 42-residue C-terminal peptide) on SC/Leu medium lacking HU. We found that all strains used in this experiment showed comparable growth phenotypes on this medium (**Figure 3B** [↗](#)). On the other hand, *rev711* mutant and the strains expressing the variant Rev7-C1 were highly sensitive to exposure to HU when compared to the wild type cells (**Figure 3C** [↗](#)). Remarkably, however, cells expressing the Rev7's 42-residue C-terminal peptide exhibited robust growth in the same medium containing HU (**Figure 3C** [↗](#)). Combining the results presented, we suggest that the 42-residue C-terminal peptide, but not Rev7-C1 variant, confers protection to cells from the toxic effects of HU/G-quadruplex DNA forming sequences, regardless of whether they occur in the leading or lagging template strands.

Rev7 inhibits Mre11 nucleolytic activities

Decades of work has documented that Mre11 is a Mn^{2+} -dependent bifunctional enzyme with both endo- and exonuclease activities, which are critical for DNA end resection (Paull and Gellert, 1998 [↗](#); Tsubouchi and Ogawa, 1998 [↗](#); Ghosal and Muniyappa, 2007 [↗](#); Stracker and Petrini, 2011 [↗](#); Ghodke and Muniyappa, 2013 [↗](#); Paull, 2018 [↗](#); Casari et al., 2019 [↗](#)). Further, it has been demonstrated that Mre11 cleaves non-B DNA structures such as DNA hairpins, as well as intra- and inter-molecular G-quadruplex structures (Trujillo and Sung, 2001 [↗](#); Lobachev et al., 2002 [↗](#); Ghosal and Muniyappa, 2005 [↗](#); 2007 [↗](#)). To further explore the functional significance of complex formation between Rev7 and Mre11, ScRev7 and its truncated derivative ScRev7-C1 were expressed in and purified from *E. coli* whole cell lysates to homogeneity (**Figure 4A** [↗](#) and **4B** [↗](#)). We then performed an experiment in which equimolar amounts of MRX subunits were incubated with 5'-³²P-labelled dsDNA in the presence or absence of ScRev7. The reaction products were

analyzed as previously described (Ghosal and Muniyappa, 2005 [↗](#)). The results informed that (a) ScRev7 has no nuclease activity (**Figure 5A, lane 2** [↗](#)) and (b) Mre11 cleaved ^{32}P -labelled dsDNA, generating a pattern of DNA fragments in a ladder-like pattern (**Figure 5A, lane 3** [↗](#)). Subsequently, we investigated the effect of ScRev7 on MRX nuclease activity by co-incubating various concentrations of ScRev7 and a fixed amount of MRX complex, prior to the addition of 5'- ^{32}P -labelled dsDNA. We detected a notable and substantial inhibition in the MRX complex-mediated DNA cleavage activity, and almost complete inhibition at 2 μM (**Figure 5A, lanes 4-12** [↗](#)). We carried out additional experiments to test the effect Rev7-C1 on Mre11 nuclease activity. By comparison, the efficiency of inhibition was expectedly less than that of full-length ScRev7. Curiously, we note that the ScRev7-C1 does not faithfully recapitulate the Y2H results. One possible reason is that amino acid residues in the ScRev7-C1 fragment interact with Mre11 and cause partial inhibition, in good agreement with the AF2 modeling data (**Figure 2—figure supplement 2** [↗](#)).

Many studies have demonstrated that Mre11 exhibits both endonuclease and exonuclease activities independently of Rad50 and Xrs2 subunits (Cejka and Symington, 2021 [↗](#)). Thus, we next examined the effect of ScRev7 on the Mre11 ssDNA-specific endonuclease activity in reactions lacking Rad50 and Xrs2. In accord with previous studies, Mre11 digested all the input ssDNA substrate into small fragments/nucleotides in a manner dependent on its concentration (**Figure 5C** [↗](#)). Interestingly, we found that the addition of increasing concentrations of ScRev7 coincided with a concomitant decrease in the Mre11 endonuclease activity (**Figure 5D** [↗](#)). To ascertain the specificity, the effect of ScRev7 on Sae2's endonuclease activity was investigated. As expected (Lengsfeld et al., 2007 [↗](#); Ghodke et al., 2016), Sae2 exhibited concentration-dependent nuclease activity on dsDNA (**Figure 5-figure supplement 1A-B** [↗](#)). However, ScRev7 did not inhibit Sae2's nuclease activity, even at the highest concentration tested (**Figure 5-figure supplement 1C and D** [↗](#)). Together, these results support the idea that inhibition of Mre11 endonuclease activity is due to its direct interaction with ScRev7.

ScRev7 impedes Rad50's ATPase activity without affecting its ATP binding ability

Several studies have demonstrated that the ATPase activity of Rad50 plays an important regulatory role in DNA recombination and repair (Cejka and Symington, 2021 [↗](#)). Given that ScRev7 specifically associated with Rad50, we asked whether such association affects the ability of Rad50 to bind ATP and catalyze its hydrolysis. To test this possibility, different concentrations of purified Rad50 were incubated with a fixed amount [ψ - ^{32}P]ATP, and then the reaction mixtures were UV irradiated prior to subjecting the samples to SDS/PAGE. The results showed a single band that migrated as a 153 kDa species corresponding to the position of ScRad50 (**Figure 6A** [↗](#)). Quantitative analyses indicated that Rad50 binds [ψ - ^{32}P]ATP in a manner dependent on its concentration (**Figure 6C** [↗](#)). The results from an accompanying experiment revealed comparable levels of [ψ - ^{32}P]ATP binding by Rad50 in the presence or absence of ScRev7 (**Figure 6B** [↗](#) and **6D** [↗](#)), suggesting that it does not impair the ATP-binding ability of Rad50.

We next explored the potential effect of ScRev7 on ATP hydrolysis catalyzed by Rad50 using thin layer chromatography. Consistent with previous studies (Ghosal and Muniyappa, 2007 [↗](#)), Rad50 catalyzed [ψ - ^{32}P]ATP hydrolysis to ADP and ^{32}Pi in a manner dependent on its concentration in the absence of ScRev7 (**Figure 6E** [↗](#) and **6F** [↗](#)). Interestingly, while ScRev7 itself has no ATPase activity, its addition led to inhibition of ATP hydrolysis catalyzed by Rad50 in a dose-dependent manner (**Figure 6G** [↗](#) and **6J** [↗](#)). The results of a parallel experiment indicated that Rev7 does not affect the ATPase activity of a meiosis-specific *S. cerevisiae* Dmc1, indicating its inhibitory specificity towards Rad50 (**Figures 6H** [↗](#) and **6J** [↗](#)). Similar analysis showed that ScRev7-C1 inhibited the ATP hydrolysis of Rad50 but three-fold less efficiently than its full-length counterpart (**Figures 6I** [↗](#) and **6J** [↗](#)). These results were validated using a colorimetric molybdate/malachite green-based assay (Lanzetta et al., 1979 [↗](#)). While ScRev7 inhibited the ATPase activity of Rad50 to

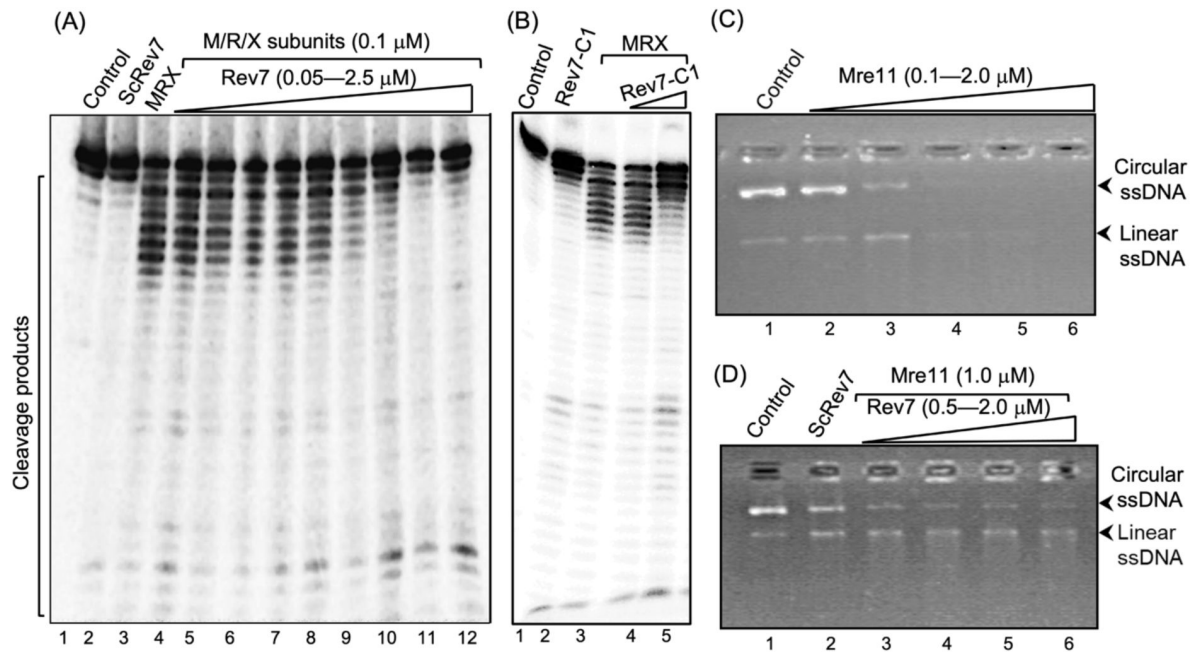


Figure 5.

ScRev7 impedes both exo- and endonucleolytic activities of the MRX subunits.

(A) A representative image showing the effect of ScRev7 on the exonuclease activity of MRX complex. Reaction mixtures containing 5 nM 32 P-labelled 60 bp dsDNA were incubated in the absence or presence of M/R/X subunits and various amounts of ScRev7. Lane 1, 32 P-labelled dsDNA. Lane 2, same as in lane 1, but with 2.5 μ M ScRev7. Lane 3, same as in lane 1, but with M/R/X subunits (0.1 μ M each). Lanes 4-12, same as in lane 3, but with 0.05, 0.1, 0.3, 0.5, 0.7, 1, 1.5, 2 and 2.5 μ M of ScRev7, respectively. (B) A representative image showing the effect of Rev7-C1 variant on the exonuclease activity of the MRX complex. Assay was performed as in panel (A), but with ScRev7-C1 variant. Lane 1, 32 P-labelled dsDNA. Lane 2, same as in lane 1, but with 2.5 μ M ScRev7-C1 variant. Lane 3, same as in lane 1, but with M/R/X subunits (0.1 μ M each). Lanes 4-5, same as in lane 3, but with 1.5 and 3.0 μ M of ScRev7-C1 variant, respectively. (C) A representative image showing Mre11 endonuclease activity on circular ssDNA. Reaction mixtures containing 100 ng M13 circular ssDNA were incubated without (lane 1) or with 0.1, 0.5, 1, 1.5 and 2 μ M Mre11, respectively (lanes 2-6). (D) A representative image showing the effect of ScRev7 on Mre11 endonuclease activity by ScRev7. Reaction mixtures containing 1 μ M Mre11 were pre-incubated with 0.5, 1.0, 1.5 and 2 μ M of ScRev7 (lanes 3-6 respectively), prior to the addition of 100 ng of M13 circular ssDNA. Lane 1, circular ssDNA alone. Lane 2, same as in lane 1, but with 2 μ M of ScRev7. Increasing concentrations of the indicated protein is represented by open triangles at the top of the gel.

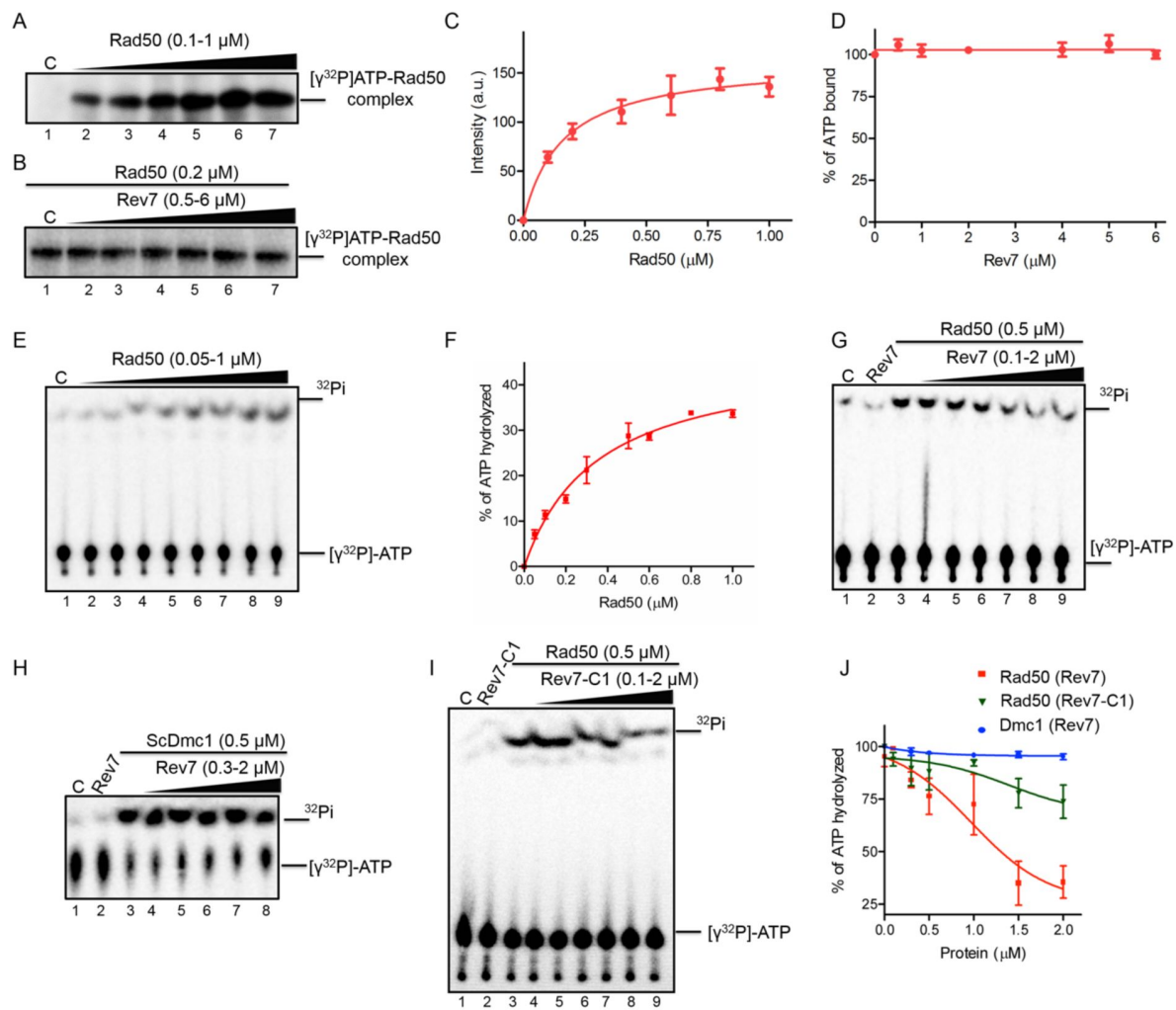


Figure 6.

Rev7 inhibits the ATPase activity of Rad50 without impacting its ability to bind ATP.

(A), Rad50 binds [ψ - 32 P]ATP in a dose-dependent manner. Lane 1, no protein. Lanes 2-7, reactions were performed with 0.1, 0.2, 0.4, 0.6, 0.8 and 1 μ M of Rad50 and 400 pmol [γ - 32 P]ATP. (B) ScRev7 does not affect the ability of Rad50 to bind ATP. Lane 1, Rad50 and 400 pmol [γ - 32 P]ATP. Lanes 2-7, same as in lane 1, but with 0.5, 1, 2, 4, 5, and 6 μ M of ScRev7, respectively. (C) Quantification of ATP binding by Rad50 as a function of its concentration. (D) Quantification of the effect of ScRev7 on ATP binding by Rad50. (E) ATPase activity of Rad50 as a function of its concentration. Reactions were performed in the absence (lane 1) or presence (lanes 2-9) of 0.05, 0.1, 0.2, 0.3, 0.5, 0.6, 0.8 and 1 μ M of Rad50, respectively. (F) Quantification of Rad50 ATPase activity as a function of its concentration. (G) ScRev7 abrogates ATP hydrolysis catalyzed by Rad50. In panels (G) and (I) lane 1 contained 400 pmol [γ - 32 P]ATP; lane 2, same as in lane 1, but with 2 μ M ScRev7/ScRev7-C1 variant; lane 3, as in lane 1, but with 0.5 μ M Rad50; lanes 4-9, as in lane 3, but with 0.1, 0.3, 0.5, 1, 1.5 and 2 μ M ScRev7/ScRev7-C1 variant, respectively. (H) ScRev7 does not impact the ability of ScDmc1 to hydrolyze ATP. Lane 1 contained 400 pmol [γ - 32 P]ATP; lane 2, same as in lane 1, but with 2 μ M of ScRev7; lane 3, as in lane 1, but with 0.5 μ M of ScDmc1; lanes 4-8, as in lane 3, but with 0.3, 0.5, 1, 1.5 and 2 μ M ScRev7, respectively. (I) ScRev7-C1 variant impedes ATP hydrolysis catalyzed by Rad50. In panels (G) and (I) lane 1 contained 400 pmol [γ - 32 P]ATP; lane 2, same as in lane 1, but with 2 μ M ScRev7/ScRev7-C1 variant; lane 3, as in lane 1, but with 0.5 μ M Rad50; lanes 4-9, as in lane 3, but with 0.1, 0.3, 0.5, 1, 1.5 and 2 μ M ScRev7/ScRev7-C1 variant, respectively. (J) Quantification of the inhibitory effect of ScRev7/ScRev7-C1 on ATP hydrolysis catalyzed by Rad50 or ScDmc1. The closed triangles on the top of gel images in panels (A), (B), (E), (G), (H) and (I) represent increasing concentrations of Rad50, ScRev7 or ScRev7-C1. Error bars indicate SEM, and data are representative of three independent experiments.

an extent of 60% at the highest concentration tested, Rev7-C1 was about three-fold less inhibitory than its full-length counterpart at an identical concentration (**Figure 6—figure supplement 1**). Collectively, our results support a model in which ScRev7 negatively regulates the catalytic activities of Mre11 and Rad50 subunits.

REV7 facilitates NHEJ in *S. cerevisiae*

In *S. cerevisiae*, the heterotrimeric MRX complex has been implicated in both Ku-dependent NHEJ and microhomology-mediated end joining repair (Moore and Haber, 1996; Boulton and Jackson, 1998; Ma et al., 2003; Zhang and Paull, 2005). Multiple studies in cancers have shown that Rev7 inhibits DNA end-resection and favors NHEJ over HR (Gupta et al., 2018; Ghezraoui et al., 2018; Dev et al., 2018; Gao et al., 2018; Liang et al., 2020). To our knowledge, it is unknown whether these findings are relevant to other species. However, we note that *Schizosaccharomyces pombe* Rev7 has been shown to inhibit long-range resection at DSBs (Leland et al., 2018). Furthermore, it is unclear whether the *S. cerevisiae* Rev1, Rev3 and Rev7 subunits are required for NHEJ. To investigate this, the efficiency of plasmid-based NHEJ repair was analyzed by transforming BamHI-linearized plasmid pRS416, which has no homology with the genomic DNA, into the WT and isogenic mutant strains as previously described (Boulton and Jackson, 1998; Moreau et al., 1999) (**Figure 7A**). Consistent with a prior study (Boulton and Jackson, 1998), we found that NHEJ was undetectable in the *mre111* mutant (**Figure 7B**, **Supplementary Table S3**). Notably, while the *rev1Δ* and *rev3Δ* strains showed a modest decrease in the efficiency of NHEJ, *rev7Δ* single and *rev1Δ rev3Δ* double mutants exhibited about four- and twofold decrease, respectively, compared with the WT. Further analysis revealed that the NHEJ efficiency in double mutants – *rev1Δ rev7Δ*, *rev3Δ rev7Δ* and *sae2Δ rev7Δ* – was comparable to that of *rev7Δ* mutant (**Figure 7B** and **7C** — **Supplementary Table S3**). Intriguingly, however, we found that ScRev7-42 aa peptide, but not the Rev7-C1 variant, fully restored the NHEJ efficiency in *rev7Δ* cells to the WT levels. It remained possible that the significant reduction in NHEJ efficiency observed in the *rev7Δ* cells could be due to aberrant cell cycle progression. To test this possibility, cell cycle progression in both WT and *rev7Δ* cells was monitored using a fluorescence-activated cell sorter. Compared with the wild-type (WT) cells, we observed a slightly delayed cell cycle progression with *rev7Δ* cells at 30 and 45 min after release from G1 arrest, and there were no differences in their mating type phenotypes. However, after 45 minutes, *rev7Δ* cells exhibited a similar distribution of cells in the G1, S, and G2 phases of the cell cycle as observed in WT cells (**Figure 7—figure supplement 1**). Together, these results indicate that *rev711* cells do not possess aberrant cell cycle or mating type defects as compared with the WT cells.

We next sought to determine whether ScRev7 plays a role in NHEJ pathway at the chromosome level. To address this question, the efficiency of NHEJ was evaluated using a “suicide-deletion” assay in which the I-SceI induced DSB can be repaired via NHEJ (Karathanasis and Wilson, 2002). Briefly, it is based on an approach wherein galactose-induced I-SceI endonuclease inflicts a pair of site-specific DSBs, resulting in the deletion of its own coding region, thereby facilitating the repair of DSB via NHEJ (**Figure 8A**). Using this approach, we determined the frequency of Ade2+ recombinants in the WT and isogenic mutant strains. A critical NHEJ factor Ku70 was used as a control. As expected, while NHEJ was undetectable in strains lacking *MRE11* and *KU70*, deletion of *REV7*, *REV1* and *REV3* led to a 9-fold, 2.3-fold- and 3.2-fold decrease, respectively, in the frequency of NHEJ compared with the WT strain. Of note, a fourteen-fold decrease in the frequency of Ade2+ recombinants was observed in the *rev7-C1* cells, which could be restored to WT levels by expressing the C-terminal 42 amino acid peptide (**Figure 8B**, **Supplementary Table S4**). These results reinforce the notion that Rev7 promotes NHEJ repair at DSBs. The PCR product derived from genomic DNA of Ade2+ recombinants showed a 1.3 kb amplicon, suggesting faithful repair of I-SceI induced DSB (**Figure 8C**).

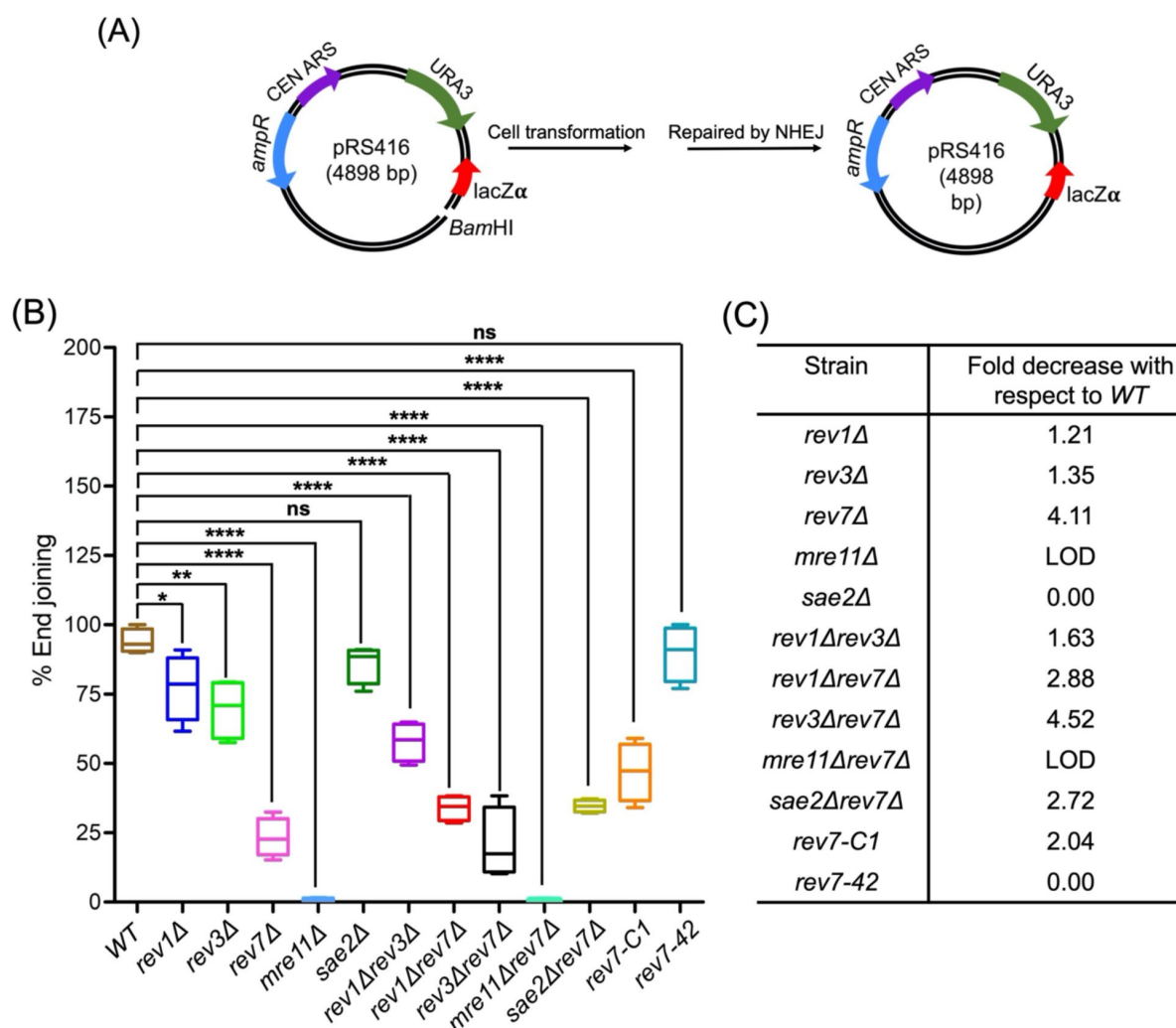


Figure 7.

S. cerevisiae REV7 promotes NHEJ-mediated DSB repair.

(A) Map of the NHEJ reporter plasmid and the experimental workflow. Sixty ng of uncut or linearized pRS416 plasmid DNA was transformed into the WT (W1588-4C) and indicated isogenic mutants carrying single or double deletions. Transformants were selected on SC medium lacking uracil. (B) Quantification of NHEJ efficiency relative to WT cells. (C) Quantification of fold decrease in the efficiency of NHEJ relative to the WT. The boxes represent mean; whiskers, minimum and maximum values. "LOD" denotes below the level of detection compared with the WT. The data are presented as the mean $\pm$ SEM of four independent experiments. n.s.: not significant, * $p < 0.05$, ** $p < 0.005$, **** $p < 0.0001$ vs. control. The exact p-values are presented in [Supplementary Table S3](#).

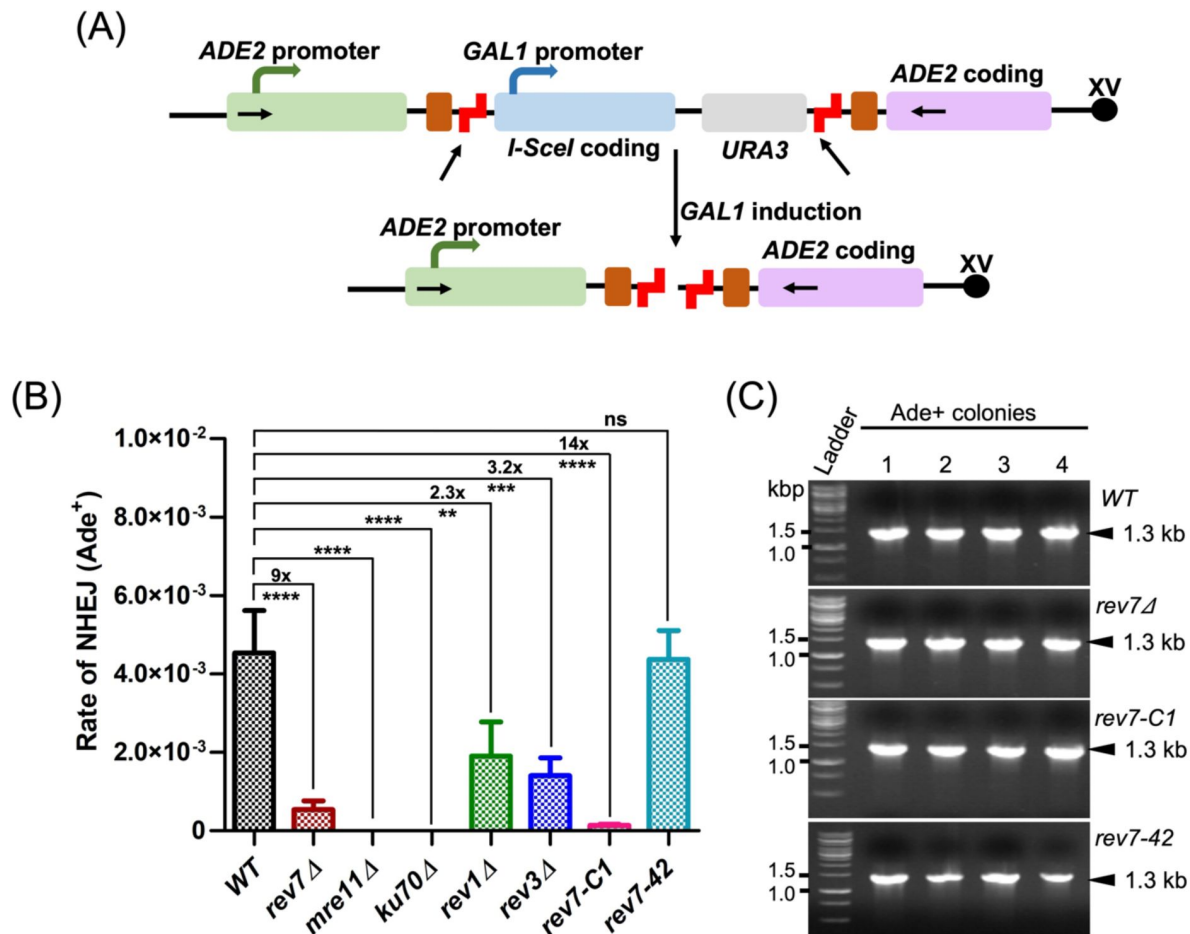


Figure 8.

***S. cerevisiae* REV7 facilitates chromosomal DSB repair via NHEJ pathway.**

(A) A schematic diagram showing the "suicide deletion" cassette at the *ADE2* locus of chromosome XV [redrawn from ref. Karathanasis and Wilson, 2002]. The arrows indicate the locations of PCR primers. Brown and staggered red boxes correspond to direct repeats and *I-SceI* cleavage sites, respectively. (B) Quantification of NHEJ efficiency in the wild-type YW714 and derivative mutant strains. (C) Representative gel images of PCR-amplified DNA products from Ade²⁺ transformant cells. Data are means ± SEM from three independent experiments. n.s.: not significant, **p < 0.005, ***p < 0.001, ****p < 0.0001 vs. control, as assessed by one-way ANOVA Dunnett's multiple comparison test. The exact p-values are presented in [Supplementary Table S4](#).

REV7 plays an anti-recombinogenic role during HR in *S. cerevisiae*

As mentioned above, current evidence suggests that the RIF1/REV7/Shieldin complex blocks DNA end resection and BRCA1-mediated HR, but promotes DSB repair through NHEJ in cancer cells (Gómez-Llorrente et al., 2013 [↗](#); Du Xu et al., 2015 [↗](#); Boersma et al., 2015 [↗](#); Mirman et al., 2018 [↗](#); Findlay et al., 2018 [↗](#); Gupta et al., 2018 [↗](#); Ghezraoui et al., 2018 [↗](#); Dev et al., 2018 [↗](#); Tomida et al., 2018 [↗](#); Gao et al., 2018 [↗](#); Noordermeer et al., 2018 [↗](#); Liang et al., 2020 [↗](#); Malik et al., 2020 [↗](#); Truong et al., 2021 [↗](#)). A flurry of research on Shieldin complex in human cancer cells from other laboratories soon followed (Clairmont and D'Andrea, 2021 [↗](#)). However, the generality of these findings has remained elusive. Therefore, we leveraged a spot assay (Paeschke et al., 2013 [↗](#)) to understand whether *REV7* plays a role in the regulation of HR in *S. cerevisiae*. In this assay, under conditions of optimal growth and replication stress, we measured the frequency of HR between the *ura3-1* allele on chromosome V and *ura3-G4* allele (*ura3* interrupted by G4 motifs) on the pFAT10-G4 plasmid (Figure 9A [↗](#)). In the absence of HU, while no Ura3⁺ papillae were observed in the *ura3-1* strain carrying the empty vector (Figure 9B, top row [↗](#)), all other strains harboring the plasmid pFAT10-G4 formed Ura3⁺ papillae (Figure 9B [↗](#)).

We sought to build on these observations by testing the effect of HU on the formation of Ura3⁺ papillae by the strains (carrying pFAT10-G4 plasmid) used in the experiment. This analysis revealed that the *ura3-1 rev7Δ* double mutant strain formed significantly more Ura3⁺ papillae than *ura3-1* strains, but less than what was seen in the absence of HU. As anticipated, we observed that both *mre11-D56N,H125N* (used as an internal control) and *rev7Δ mre11-D56N,H125N* mutant strains did not form Ura3⁺ papillae (Figure 9B [↗](#)). A similar analysis showed that *ura3-1* strain and the same strain expressing ScRev7-42 amino acid peptide, but not Rev7-C1 variant, displayed a very few, and small Ura3⁺ papillae. As expected, Ura3⁺ papillae formation was not observed in the *ura3-1* strain carrying the empty vector. Quantification indicated that the *ura3-1 rev7Δ* mutant showed 11.5-fold increase in the formation of Ura3⁺ papillae as compared with *ura3-1* strain (Figure 9C [↗](#)). Thus, we envision that the absence of Ura3⁺ papillae in the *ura3-1* strain (with or without empty vector) in the presence of HU might be related to Rev7-mediated suppression; whereas their absence in cells that lack functional Mre11 nuclease is due to HU-induced toxicity. Such an effect has been described previously in the Mre11 nuclease-deficient cells (Tittel-Elmer et al., 2009 [↗](#); Hamilton and Maizels, 2010 [↗](#)). Regardless, these results support a model in which *REV7* gene product plays an anti-recombinogenic role during HR, thus its deletion unleashes the cells to facilitate HR between *ura3G4* and *ura3-1* mutant alleles located on the plasmid and chromosome V, respectively.

To confirm whether the G-quadruplex motifs stimulate HR in the *rev7Δ* strain, the frequency of HR was measured using pFAT10-G4-mut plasmid, which harbors mutations within G-tracts of the *ura3G4* allele. The results revealed that the frequency of HR reduced by about 5.8-fold, as compared to the strain carrying plasmid pFAT10-G4 with unmutated sequences, revealing that reduction in HR may be caused by mutations in the G-quadruplex forming sequences (Figure 9—figure supplement 1 [↗](#)). To assess whether increased frequency of HR is due to the instability of G-quadruplex DNA in *rev7Δ* cells, the length of G4 DNA inserts was assessed in the plasmids isolated from WT and *rev7Δ* cells. The results showed a DNA fragment of 829 bp corresponding to the expected size in both WT and *rev7Δ* cells (Figure 9—figure supplement 2 [↗](#)), raising the possibility that increased frequency of HR in *rev7Δ* cells could be due to the loss of Rev7 function and instability of G-quadruplex forming sequences.

Finally, a qPCR-based assay (Mimitou and Symington, 2010 [↗](#); Ferrari et al., 2018 [↗](#)) was employed to quantify the amounts of ssDNA generated in the *rev711* and *rad51Δ mre11-H125N rev7Δ* strains at a HO endonuclease-induced DSB (Figure 9—figure supplement 3A [↗](#)). The results indicated a significant increase in the percentage of ssDNA 0.7 kb distal to the DSB at 4 h after its induction in the *rev7Δ rad51Δ* cells compared with *rad51Δ* cells (Figure 9—figure supplement 3B [↗](#)). As

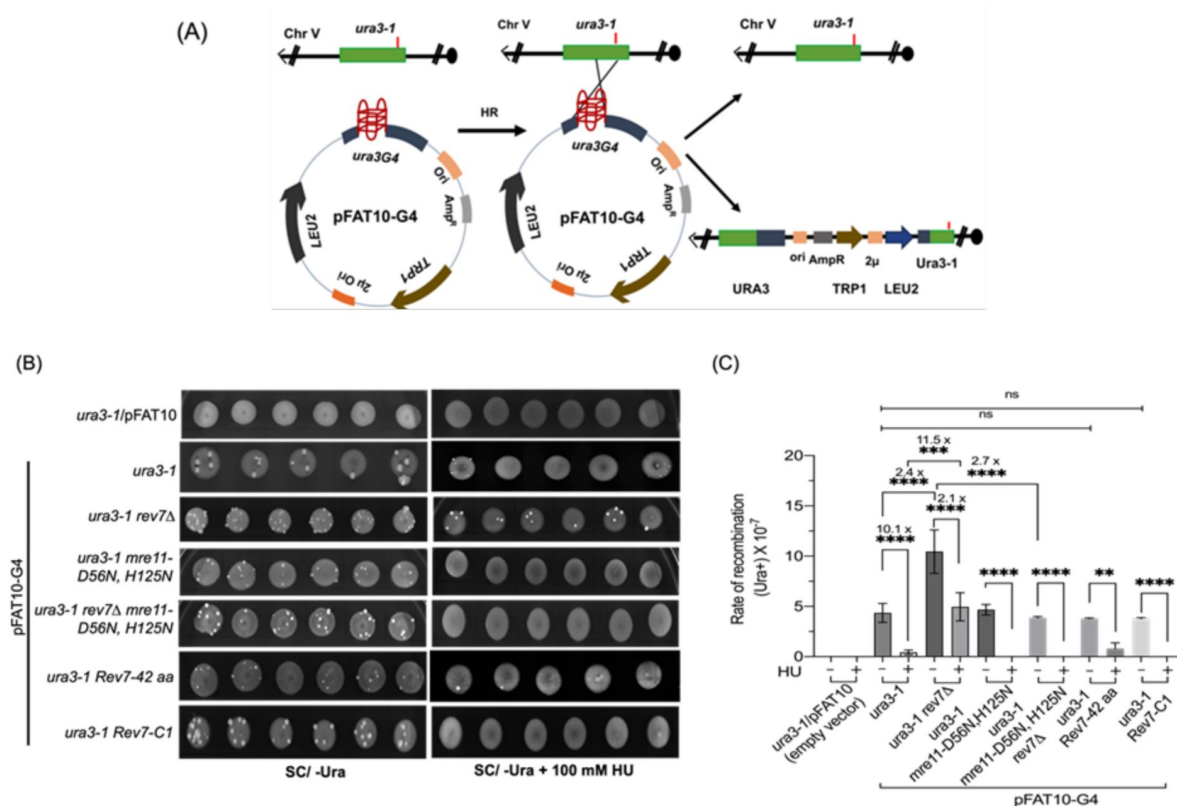


Figure 9.

Deletion of *REV7* increases the frequency of mitotic HR.

(A) Schematic representation of plasmid-chromosome recombination assay. The *ura3-1* and *ura3-G4* alleles are located on chromosome V and plasmid pFAT10-G4, respectively. Recombination between a plasmid borne *ura3-G4* allele and the chromosomal borne *ura3-1* allele would result in Ura⁺ prototrophs. (B) Representative images of Ura3⁺ papillae on SC/-Ura agar plates in the absence or presence of 100 mM HU. (C) Quantification of the rate of HR frequency in different strains. Data are presented as mean $\pm$ SD from three different experiments. ns, not significant, **p < 0.01, ***p < 0.001, ****p < 0.0001 vs. control, as assessed using one-way ANOVA and Tukey's post hoc test.

expected, the DNA end resection rates in the *mre11-H125N* and *mre11-H125N rev7Δ* cells were similar to that in the *rad51Δ* cells. Importantly, at 3 kb distal to the DSB, *rev7Δ* cells showed comparable rates of DNA end resection, in line with both *rad51Δ* and *mre11-H125N* cells (**Figure 9—figure supplement 3C**). Altogether, these results support the notion that Rev7 suppresses HR by inhibiting Mre11-mediated short-range DNA end resection.

Discussion

In this study, we reveal the surprising finding that Rev7 physically associates with the subunits of the MRX complex and also provide unanticipated insights into the mechanism by which it regulates the DSB repair pathway choice between HR and NHEJ in *S. cerevisiae*. Notably, we demonstrate that Rev7 binds to the subunits of the MRX complex, via a 42-residue C-terminal segment (residues 203-245), protects cells from G4 DNA-HU-induced toxicity, facilitates DSB repair via NHEJ while blocking HR. In addition, we present robust evidence that ScRev7 impedes Mre11 nuclease and Rad50's ATPase activities, without affecting the latter's ability to bind ATP. When seen from a teleological perspective, our work is conceptually reminiscent of Shieldin complex-mediated suppression of 5' end resection and repair of DSBs via NHEJ by blocking HR in human cancer cells. It remains plausible that this alternative mechanism of DSB repair pathway choice in *S. cerevisiae* might be conserved across multiple species.

Historically, *REV7* was discovered as playing an important role in DNA damage-induced mutagenesis in *S. cerevisiae* (Lemontt, 1971; Lawrence et al., 1985a). Further investigations showed that Rev7 associates with Rev3 and functions as a regulatory subunit of eukaryotic TLS DNA polymerase Polζ (Prakash et al., 2005; Maiorano et al., 2021; Ling et al., 2022; Paniagua and Jacobs, 2023). While Rev7 has no known enzymatic activity, it acts as a versatile scaffolding protein with pleiotropic functions in diverse cellular processes including, but not limited to, epigenetics, immune signalling, viral mutagenesis and cancer development that were initially considered inconceivable (Decottignies, 2013; de Krijger et al., 2021). However, little is known about the nature of specific effector(s) that associate with Rev7 and regulate DSB repair pathway choice in *S. cerevisiae*. As we discuss below, our work convincingly demonstrates that ScRev7 physically interacts with the individual subunits of MRX complex and regulates their activities. In line with this, MST based protein-protein interaction assays, supported by AF2-multimer modelling, imply tight association between the Mre11 and Rad50 subunits and ScRev7. It is interesting to note that Sae2, which cooperates with the MRX complex to initiate DNA end resection, does not associate with Rev7, suggesting interaction specificity between MRX subunits and Rev7.

However, an intriguing inquiry arises regarding whether the MRX subunits exist as separate entities in the cell. Although, to our knowledge, the existence and relative amounts of monomeric, dimeric and trimeric species of MRX subunits *in vivo* are unknown, current evidence suggests that both Mre11 and Rad50 subunits independently bind DNA (Stracker and Petrini, 2011; Paull, 2018; Casari et al., 2019). Other studies have shown that MRX complex binds and tethers DNA ends, which is thought to be required for DSB repair (Trujillo et al., 2003; Cassani et al., 2015). Furthermore, while data obtained from co-immunoprecipitation experiments implicate that Mre11, Rad50 and Xrs2 subunits exist as a hetero-trimeric complex *in vivo* (Usui et al., 1998), *in vitro* experiments have shown the formation of dimeric Mre11-Rad50 and trimeric MRX complexes (Oh et al., 2016; Arora et al., 2017). Collectively, these results allow us to postulate that the monomeric, dimeric and trimeric species of MRX subunits might exist in a dynamic equilibrium under *in vivo* conditions.

Given that Rev7 and MRX complex play essential roles in DNA repair pathways, our findings provide novel insights into how ScRev7 interacts with the MRX subunits and regulates their functions. Structure-function analysis showed that deletion of 42-amino acid residues (203-245) at

the extreme C-terminus of ScRev7 abolished its ability to interact with the MRX subunits, whereas loss of the N-terminal HORMA domain, an evolutionarily conserved protein-protein interaction module (Muniyappa et al., 2014 [DOI](#); Rosenberg and Corbett, 2015 [DOI](#); de Krijger et al., 2021 [DOI](#);) had no discernible effect on their interactions. Curiously, further analysis indicated that the 42-aminoacid peptide alone was sufficient to bind to the subunits of MRX complex and regulate the DSB repair pathway choice between NHEJ and HR. Reciprocally, future work will be required to determine the regions/domains of MRX subunits that interact with ScRev7 for a comprehensive understanding of the crosstalk between these components. Work is currently in progress to gather insights into these questions.

The data from MST-based protein-protein interaction assays informed that ScRev7 binds to the MRX subunits with sub-micromolar affinity, analogous to the interaction between HORMA-domain protein hMAD2 and hCDC20 (Piano et al., 2021 [DOI](#)). How might ScRev7 attenuate the function of Mre11 and Rad50 subunits? Our finding indicate that nanomolar amounts of ScRev7 besides attenuating the Mre11 nuclease activity also impedes Rad50's ATPase activity without obstructing the ability of the latter to bind ATP. Whilst the specific nuances of the interaction await further research, we surmise that physical interaction between ScRev7 and the Mre11/Rad50 subunits might contribute to the observed effects. Although we provide compelling Y2H data that the C-terminal 42-aminoacid peptide of Rev7 is critical for binding to the MRX subunits and protect cells from G4 DNA-HU-induced toxicity, and regulate the pathway choice between NHEJ and HR; however, we could not demonstrate its function *in vitro* because of technical difficulties associated with its expression and purification. By comparison, we found that the Rev7-C1 variant, which lacks the ability to interact with the MRX subunits in Y2H assays, was capable of blocking, albeit partially, the catalytic activities of Mre11 and Rad50 subunits *in vitro*. A likely possibility is that amino acid residues in the Rev7-C1 fragment might be involved in pairwise interactions between ScRev7 and the Mre11 and Rad50 subunits, as seen in AlphaFold2 models. Conversely, the 42-amino acid fragment as a part of whole protein might act as a lid to block the residues in the Rev7-C1 fragment, thereby enabling it to function effectively as a single site for binding to the MRX subunits in Y2H assays. Future studies are required to test this hypothesis.

Many lines of evidence indicate that G-quadruplex forming sequences are measurably enriched at certain functional regions in the genomes of all organisms, from humans to plants to microbes (Huppert and Balasubramanian, 2005 [DOI](#); Capra et al., 2010 [DOI](#); Castillo Bosch et al., 2014 [DOI](#); Lejault et al., 2021 [DOI](#)). Furthermore, G-quadruplex structures modulate diverse cellular processes, including DNA replication, transcription, translation, and are associated with certain diseases, characterized by high rates of chromosomal instability (Rhodes and Lipps, 2015 [DOI](#); Spiegel et al., 2020 [DOI](#)). Interestingly, we found that *rev711* mutant cells harbouring exogenous G-quadruplex forming sequences exhibit hypersensitivity to HU-induced genotoxic stress and cell death. This is consistent with emerging evidence that DNA replication stress induced by G-quadruplex structures play a prominent role in triggering genomic instability, which is exacerbated in the presence of HU (Sato and Knipscheer, 2023 [DOI](#)). Although the precise mechanism remains unclear, it is tempting to speculate that Rev7 may recruit G4-resolving helicase(s) such as Sgs1 and Pif1 to unwind G-quadruplex structures prior to or during DNA replication (Huber et al., 2002 [DOI](#), Paeschke et al., 2013 [DOI](#)). Alternatively, or in addition, it might generate G4 DNA intermediates which may be processed or cleaved by specific enzymes, including Mus81, Sgs1 or Mre11 (Regairaz et al., 2011 [DOI](#); Sun et al., 1999 [DOI](#); Ghosal and Muniyappa, 2005 [DOI](#); 2007 [DOI](#)). Regardless, our data align with the notion that G-quadruplex structures are endogenous sources of replication stress and their formation and persistence may lead to genomic instability and cell death.

As mentioned above, Shieldin complex facilitates NHEJ-dependent DSB repair, while inhibiting DNA end resection and HR in cancer cells (Clairmont and D'Andrea, 2021 [DOI](#); Paniagua and Jacobs, 2023 [DOI](#)). Since 5' end resection is the primary step in HR-mediated DSB repair, we hypothesized that physical interaction between ScRev7 and MRX subunits might block resection of DNA termini, and then facilitate NHEJ-dependent DSB repair instead of HR. Consistent with this premise, we

found that ScRev7 promotes NHEJ by blocking HR. However, other regulatory components may also play a role in modulating the levels of NHEJ versus HR-mediated DSB repair. For instance, TRIP13 or p31^{comet} inhibits the interaction of Rev7 with the SHLD3 subunit and regulates DNA end resection at DSBs and promotes their repair by HR (Sarangi et al., 2020 [\[link\]](#)). Additionally, interaction between Rev7 and putative binding partners may be regulated by posttranslational modifications and chromatin accessibility under changing physiological conditions.

Although the mechanism underlying Rev7-mediated regulation of DSB repair between *S. cerevisiae* and human cancer cells indicate broad similarities, we also observed some notable differences. A fundamental difference between the results from cancer cells and *S. cerevisiae* is that, while ScRev7 robustly interacts with and suppresses the biochemical activities of both Mre11 and Rad50 subunits to facilitate NHEJ, Rev7-Shieldin complex acts as a downstream effector of 53BP1-RIF1 in restraining DNA end resection to promote NHEJ (Clairmont and D'Andrea, 2021 [\[link\]](#); Paniagua and Jacobs, 2023 [\[link\]](#)). Furthermore, hRev7 interacts with the SHLD3 subunit via the HORMA domain, which is entirely dispensable in the case of ScRev7. Thus, we posit that the Shieldin complex-mediated regulation of DSB repair pathway choice might be a source of evolutionary innovation as an additional layer of regulation. In summary, our data provide novel insights into the alternative mechanism underlying the regulation of DSB repair pathway choice in *S. cerevisiae*.

Materials and methods

S. cerevisiae strains, DNA plasmids and oligonucleotides

All strains and primers used for the construction of DNA substrates in this study are listed in **Supplementary Table S5 and S6** [\[link\]](#), respectively. The plasmids FT10/Chr.IV_{G4}-lg and FT10/Chr.IV_{G4}-le were a kind gift from Dr. Virginia Zakian.

Construction of strains used in the study

The *S. cerevisiae* W1588a haploid strains carrying single/double deletions in the genes *rev111*, *rev311*, *rev711*, *rev111 rev311*, *rev711 mre1111* or *sae211 rev711* were constructed using appropriate pairs of primers as previously described (Sambrook and Russell, 2001 [\[link\]](#); Janke et al., 2004 [\[link\]](#)). The *REV7* gene was deleted using the KanMX4 (pFA6a-KanMX4) cassette utilizing the forward primer OSB11 and reverse primer OSB12, and deletion was confirmed by PCR using gene-specific primers OSB13 and OSB14. Similarly, other strains were generated as follows: *MRE11* was deleted using the hphNT1 cassette (pYM-hphNT1) utilizing the OSB52 and OSB54 primers; deletion was verified by PCR using primer OSB53. The *REV1* was deleted in a similar fashion, using the hphNT1 cassette and the OSB55 and OSB56 primers; deletion was ascertained by PCR using primer OSB57. *REV3* gene was deleted using the KanMX4 cassette (derived from pFA6a-hphNT1, pFA6a-KanMX4 respectively) and the primers OSB58 and OSB59; deletion was confirmed by PCR using primer OSB60. The hphNT1 (pYM-hphNT1) cassette was used in the generation of double mutants (listed in **Supplementary Table S6** [\[link\]](#)). The *rev7-C1* and *rev7-42* mutant strains were constructed by inserting sequences encoding N- and C-terminally truncated forms of Rev7 at the endogenous loci using *rev7-C1-9MYC-hphNT1* and *rev7-42-3MYC-KANMX4* cassettes via overlapping primer-based PCR. The *rev7-C1-9MYC-hphNT1* cassette was generated using primer pairs OSB116, OSB117 and OSB118, OSB70. Likewise, *rev7-42-3MYC-KANMX4* cassette was generated using primer pairs OSB122, OSB120 and OSB121, OSB70. The N- and C-terminal truncation variants were confirmed by PCR using probes OSB13 and OSB14, respectively.

The *mre1111*, *rev111*, *rev311* and *rev711* mutant strains were generated in the strain YW714 using the hphNT1 cassette (derived from pFA6a-hphNT1). The *REV7* truncation variants with the c-myc tag sequence – *rev7-C1-9MYC* and *rev7-42 aa-3MYC* – were generated as described above. The *REV3* gene was deleted using the hphNT1 cassette in the strain PJ694A utilizing primers OSB58 and OSB59 and deletion was confirmed using primer OSB60. Likewise, *mre1111 rad5011 xrs211* triple

mutant was generated in the strain PJ694A as follows: *mre11Δ::KANMX4* was generated by PCR-amplification of KANMX4 cassette (pFA6a-KANMX4) using forward and reverse primers, OSB52 and OSB53, respectively. Gene deletion was confirmed by PCR using primer OSB54. The *rad50Δ::URA3* strain was constructed by PCR amplification of URA3 cassette (pAG60-URA3) using primers OSB133 and OSB75; deletion was confirmed using primer OSB76. Similarly, *xrs2Δ::HphNT1* was generated by PCR amplification of hphNT1 cassette (pFA6a-hphNT1) using primers OSB134 and OSB78, and deletion was confirmed using primer OSB79. The *rev7Δ* mutants were generated in the strain LSY2172-24C and LSY2265-10D, as described above.

Construction of DNA plasmids for expression and recombination assays

The plasmids FT10/Chr.IV_{G4}-lg and FT10/Chr.IV_{G4}-le were a kind gift from Dr. Virginia Zakian. *S. cerevisiae* *REV7* gene was amplified from genomic DNA by PCR using primer pair (forward OSB01 and reverse OSB02) and Phusion DNA polymerase (NEB, Ipswich, MA) as previously described (Sambrook and Russell, 2001 [\[4\]](#)). The reaction yielded an amplicon of expected size, which was digested with BamHI/HindIII, and ligated into a BamHI/HindIII digested pET28a(+) vector (Novagen) using T4 DNA ligase. The resulting expression plasmid was designated pET-28a_*REV7*. Analogously, the truncated plasmid pET28a_*ScREV7*-C1 was generated by PCR amplification using primers OSB01 and OSB125 and pET28a_*ScREV7* plasmid DNA as a template. The PCR product was digested with BamHI/HindIII and ligated into BamHI/HindIII digested pET28a vector. Likewise, *S. cerevisiae* *RAD50* gene was PCR-amplified from genomic DNA using OSB33 (forward) and OSB34 (reverse) primers. The amplicon was digested with BamHI/XhoI and ligated into a BamHI/XhoI digested pE-SUMO Kan vector (Life Sensors, Malvern, PA) using T4 DNA ligase. The resulting expression plasmid was designated pE-SUMO_*RAD50*. The *REV7*-eGFP expression vector was constructed by PCR amplification of *S. cerevisiae* *REV7* gene in the pET-28a_*REV7* plasmid using forward Rev7-eGFP primer and reverse Rev7-eGFP primer (**Supplementary Table S6** [\[4\]](#)). The amplicon of expected size was digested with XbaI/XhoI and ligated into XbaI/XhoI digested pPROEX vector, upstream of eGFP coding sequence. The resulting plasmid encodes ScRev7-eGFP fusion protein. Similarly, the DNA sequence encoding ScRev7-C1 was amplified from pET28a-*REV7* plasmid using primers as shown in Table S6. The amplicon was then cloned into pPROEX::eGFP vector. The pPROEX::eGFP vector was a kind gift from Deepak Saini.

The *ura3-G4* insert was amplified by overlap extension PCR method using OSB80 and OSB82 as forward and OSB81 and OSB83 as reverse primers, respectively. The amplicon was digested with BamHI/SphI and ligated into a BamHI/SphI digested FAT10 vector. The primer OSB82 corresponds to the DNA sequence 362751-362775 of the coding strand of *S. cerevisiae* Chr X. The primer OSB81 is its complementary strand. Similarly, *ura3-G4* mutant was generated using OSB80 and OSB130 as forward, and OSB129 and OSB83 as reverse primers. The PCR product corresponding to full-length *ura3-G4* mutant was digested with BamHI/SphI and cloned into FAT10 vector. The ODN OSB129 sequence 362751-362775 corresponds to mutant version of *S. cerevisiae* Chr X. The OSB130 is its complementary strand.

Construction of DNA plasmids for yeast two-hybrid analysis

To construct the pGBKT7-*REV7*, the *S. cerevisiae* *REV7* orf in the pET28a (+) *REV7* construct was PCR-amplified using Phusion DNA polymerase and forward OSB03 and reverse OSB04 primers, respectively. The amplicon was digested with NdeI/BamHI and ligated into NdeI/BamHI digested pGBKT7 vector. The *S. cerevisiae* *MRE11* gene was PCR-amplified from the genomic DNA using OSB05 forward primer and *MRE11*_RP reverse primer. The amplicon was digested with BamHI/EcoRI and ligated into the pGADT7 prey expression vector at the BamHI/EcoRI site. The same procedure was leveraged for the construction of all other prey expression vectors: *XRS2* was amplified from the genomic DNA using forward OSB36 and reverse OSB37 primers. The amplicon was digested with NdeI/BamHI and ligated into the pGADT7 prey expression vector at the NdeI/BamHI site; *RAD50* was amplified from the pESUMO_*RAD50* construct using forward OSB35

and reverse OSB34 primers. The amplicon was digested with EcoRI/Xho1 and ligated into the pGADT7 prey expression vector digested with EcoRI and Xho1; *REV7* was amplified from the pET-28a_*REV7* construct using the same primers and ligated into pGADT7 prey expression vector as in the case of bait plasmid construction. The *SAE2* gene was amplified from the plasmid pET21a-*SAE2* (Ghodke and Muniyappa, 2013 [DOI](#)) using primers OSD150 and OSD151 and the amplicon was digested with NdeI and EcoRI and cloned into pGADT7 vector.

The N-terminally truncated ScREV7 variants (REV7-N1, REV7-N2, and REV7-N3) were constructed by PCR amplification of relevant portions of the *REV7* gene using OSB61, OSB62, OSB63 as a forward primers and OSB4 as a reverse primer (common for all N-terminal deletions), respectively. The C-terminally truncated ScREV7 variants (REV7-C1, REV7-C2 and REV7-C3) were constructed via PCR amplification of relevant portions of the *REV7* gene in the pET-28a_*REV7* plasmid using OSB3 as a forward primer (common for all C-terminal deletions), and OSB64, OSB65 and OSB66 as reverse primers, respectively. The amplicons, corresponding to the N- and C-terminal variants, were digested with NdeI/BamHI and ligated into NdeI/BamHI digested pGBKT7 vector. *S. cerevisiae* PJ694A strain was used in Y2H analyses.

Cell viability assay

The *S. cerevisiae* WT and isogenic mutant strains were grown in liquid YPD or SC medium to an OD₆₀₀ of 0.5. Ten-fold serial dilutions were spotted on YPD agar plates with or without HU. Similarly, cells were spotted on SC selection medium plates lacking the indicated amino acids, with or without inhibitors, as indicated in the figure legends. The plates were incubated at 30 °C for 3-4 days. The images were captured using epi-illumination at auto-exposure ChemiDoc MP imaging system (Bio-Rad).

Yeast two-hybrid interaction analysis

The Y2H assays were performed as previously described (Fields and Song, 1989 [DOI](#); Thakur et al., 2020 [DOI](#)). Briefly, pairwise combination of plasmids expressing bait proteins, fused to the Gal4 DNA-binding domain (G4BD), and prey proteins, fused to the Gal4 activation domain (G4AD), were co-transformed into the WT strain PJ69-4A, *rev311* single *mre1111 rad5011 xrs211* triple mutant strain. The empty prey vector and a vector expressing Rev7 served as negative and positive controls, respectively. The interactions were analysed by spotting the indicated transformants on SC/-Trp -Leu and SC/-Trp -Leu -His agar plates containing 15 mM 3-aminotriazole (3-AT), which were then incubated for 3-5 days at 30°C. Growth of cells on SC-Leu-Trp-His +3-AT agar plates is indicative of moderate/strong protein-protein interactions between the bait and prey proteins.

Assay for G-quadruplex DNA-HU-induced toxicity

The assay was performed as previously described (Paeschke et al., 2011 [DOI](#)). The plasmid constructs used in this study were identical to the parent plasmid, pFAT10, except that its derivatives contained three tandem arrays of G sequences derived from chromosome IV, inserted in the lagging or leading templates. Briefly, the *S. cerevisiae* WT, *rev711* mutant, and cells expressing ScRev7 truncation variants (ScRev7-C1, ScRev7-42 aa) harboring the pFAT10 plasmid were synchronized in the G1 phase using α -factor at 30 °C and then released from the pheromone block by washing the cells. Subsequently, the cultures were diluted to yield identical A₆₀₀ values. Tenfold serial dilutions of each culture was spotted on SC/-Leu or SC/-Leu/+100 mM HU agar plates, which were incubated at 30 °C for 72 h.

Expression and purification of ScRev7

The *S. cerevisiae* Mre11, Rad50, Xrs2, and Sae2 proteins were expressed and purified as previously described (Ghosal and Muniyappa, 2007 [DOI](#); Ghodke and Muniyappa, 2016 [DOI](#)). The *S. cerevisiae* *REV7* gene was sub-cloned into pET28a(+) expression vector with an N-terminal His₆-tag. The resulting plasmid was designated pET28a(+)_REV7. The *E. coli* BL21(DE3)pLysS cells harboring

pET28a(+)_REV7 plasmid were grown at 37 °C in Luria-Bertani broth (1% tryptone, 0.5% yeast extract, 1% NaCl, pH 7.0) containing 50 µg/ml kanamycin in an orbital shaking incubator at 180 rpm to an OD₆₀₀ of 0.6, and then ScRev7 expression was induced by adding 1-thio-β-D-galactopyranoside (IPTG) to a final concentration of 0.1 mM. Following the addition of IPTG, cultures were grown in an orbital shaking incubator at 25 °C for 12 h. Cells were harvested by centrifugation at 6000 g and the cell paste was resuspended in 50 ml of buffer A (20 mM Tris-HCl (pH 8.0), 150 mM NaCl, 10% glycerol and 5 mM 2-mercaptoethanol containing 1 mM phenylmethylsulfonyl fluoride and 0.05% Triton X-100). The cells were lysed by sonication on ice (7 x 1-min pulses) and subjected to centrifugation at 30000 rpm at 4 °C for 20 min. Solid ammonium sulfate was added (0.472 gm/ml) to the supernatant with continuous stirring for 45 min at 24 °C. The precipitate was collected by centrifugation at 18000 rpm at 4 °C for 1 h. The precipitate was dissolved in buffer A and dialyzed against the same buffer. The dialysate was loaded onto a 5 ml Ni²⁺-NTA column (Qiagen, Valencia, CA). After washing the column with buffer A containing 20 mM imidazole, bound proteins were eluted with a gradient of 20@500 mM imidazole. The fractions containing ScRev7 were pooled and dialyzed against buffer B (20 mM Tris-HCl, pH 8.0, 1.2 M NaCl, 7% glycerol, 5 mM 2-mercaptoethanol). ScRev7 protein was further purified by chromatography using a Superdex S75 gel filtration column, attached to an AKTA Prime FPLC system, which had been equilibrated with and eluted using buffer B. The peak fractions containing Rev7 were pooled and dialyzed against buffer C (20 mM Tris-HCl (pH 8.0), 30 mM NaCl, 30% glycerol, 5 mM 2-mercaptoethanol), and loaded onto a heparin column (5 ml). The bound proteins were eluted with a gradient of 30@350 mM NaCl in buffer C. Aliquots of each fraction were analyzed by SDS-PAGE and the protein bands were visualized by staining the gel with Coomassie brilliant blue. The fractions that contained Rev7 were pooled and dialyzed against buffer D (20 mM Tris-HCl (pH 8.0), 100 mM NaCl, 10% glycerol, 1 mM DTT), and stored at -80 °C.

Expression and purification of ScRev7-eGFP and ScRev7C1-eGFP

The His₆-tagged ScRev7-eGFP and ScRev7C1-eGFP fusion proteins were expressed in and purified from whole cell lysates of *E. coli* BL21(DE3)pLysS host strain harboring the pPROEX/REV7-eGFP plasmid as described above. The His₆-tagged eGFP was purified from the cell lysates of *E. coli* BL21* (DE3) pLysS host strain harboring eGFP pPROEX expression plasmid by Ni²⁺-NTA affinity chromatography. Briefly, the whole cell lysate was loaded onto a 5 ml Ni²⁺-NTA resin column, which had been equilibrated with a buffer A (HEPES, pH 7.5, 50 mM NaCl and 10% glycerol). The column was washed with buffer A containing 70 mM imidazole. The bound proteins were eluted with a gradient of 70 @ 800 mM imidazole in buffer A. Aliquots of each fraction were analyzed by SDS-PAGE and the protein bands were visualized by staining the gel with Coomassie brilliant blue. The protein concentrations were determined by the Bradford assay. The fractions that contained ScRev7-eGFP and eGFP were pooled, dialyzed against buffer A, and stored at -80 °C.

Expression and purification of *S. cerevisiae* Rev1

The GST-tagged Rev1 was purified as described previously with some modifications (Johnson et al., 2006 [\[1\]](#)). Briefly, the *S. cerevisiae* BJ5464 cells carrying plasmid pBJ842-Rev1-GST were selected on SC plates lacking leucine. Single colonies were inoculated into liquid SC/-Leu medium (200 ml) containing 2% raffinose, and the cultures were grown for 12-14 h at 30°C until the A_{600 nm} reached 0.5. Following this, the cultures were centrifuged at 14000 rpm for 10 min and the pellet was washed thrice with MilliQ water. Cells were then resuspended in liquid SC/-Leu medium containing 2% galactose. The cultures were grown at 30 °C in an orbital shaking incubator at 250 rpm for 7 h, harvested by centrifugation at 4000 rpm for 10 min, resuspended in CBB buffer (50 mM Tris-HCl pH 7.5, 10% sucrose, 1 mM EDTA, 10 mM β-mercaptoethanol and 5 µg/mL of protease inhibitor cocktail). Cells were lysed using a FastPrep 24 homogenizer and the cell debris was separated by centrifugation at 10000 rpm for 10 min. The supernatant was centrifuged at 35000 rpm for 30 min. The protein(s) in the supernatant was precipitated using ammonium sulfate (0.208 g/ml), followed by centrifugation at 20000 rpm for 45 min. The pellet was resuspended in the GST-binding buffer (GBB, 50 mM Tris-HCl pH 7.5, 10% glycerol, 1 mM EDTA, 10 mM β-mercaptoethanol

and 5 µg/mL of protease inhibitor cocktail) and dialyzed extensively against the same buffer. The sample was loaded onto a GST-column (GSTrap High performance column, 5 ml, Cytiva Life Sciences) at a rate of 0.5 mL/min. The column was washed with GBB buffer containing 500 mM NaCl. Bound protein was eluted using GBB buffer containing 40 mM glutathione. Fractions (1 mL each) were collected at a flow rate of 1 mL/min, analyzed on 10% SDS-PAGE and staining the gel with Coomassie brilliant blue. The fractions containing pure Rev1-GST protein were pooled and dialyzed against MST buffer (HEPES, pH 7.5, 50 mM NaCl, 10% glycerol) and stored at -80°C. The concentration of ScRev1 was determined by the dye-binding assay.

SDS-PAGE and immunoblot analysis

Western blot analysis was performed as previously described (Mahmood and Yang, 2012). The *S. cerevisiae* PJ69-4A strains were co-transformed with empty vectors pGBKT7 and pGADT7, or a bait vector harboring Rev7 truncations in combination with prey vectors harboring *REV7*, *MRE11*, *XRS2* or *RAD50*. The transformants were selected by plating on SC/-Leu,-Trp agar medium, and single colonies were inoculated into liquid SC/-Trp,-Leu medium (5 ml) containing 2% dextrose. The cultures were incubated at 30°C for 12-14h in a rotary incubator at a speed of 200 rpm to $A_{600nm} = 0.15$, which were then transferred onto 10 ml of liquid SC/-Trp,-Leu medium. Incubation was continued at 30°C with shaking to until the culture reached $A_{600nm} = 0.5$. The cells were harvested by centrifugation at 4000 rpm for 5 min and the pellet was resuspended in lysis buffer (50 mM sodium-HEPES pH 7.5, 200 mM sodium acetate pH 7.5, 1 mM EDTA, 1 mM EGTA, 5 mM magnesium acetate and 5% glycerol). Cells were lysed with 2 min bursts at 4.0 m/s on a FastPrep-24 homogenizer in the presence of 200 ml of acid-washed glass beads. The samples were centrifuged at 7000 rpm for 5 min to remove the beads and cell debris. The supernatant was centrifuged at 13000 rpm for 10 min. Equal amounts of protein (50 µg protein from the supernatants) were separated by 10% SDS-PAGE. The proteins from the gel were transferred onto a polyvinylidene difluoride (PVDF) membrane (0.45 µm). The blot was blocked with 5% nonfat dry milk in TBST (20 mM Tris-HCl buffer, pH 7.5, 150 mM NaCl, 0.1% Tween 20) buffer for 1 h at 25°C, washed, and probed at 25 °C for 1 h with anti-Myc antibodies (dilution 1:3000 dilution). Subsequently, the blots were washed thrice with TBST buffer and incubated with HRP-conjugated anti-rabbit antibodies (dilution 1:20000; Sigma-Aldrich) at 25°C for 1 h. Finally, the blots were washed thrice with TBST buffer and developed using chemiluminescence substrates (Bio-Rad Laboratories, CA, USA) in the Bio-Rad ChemiDoc Imaging systems. Anti-PGK1 antibodies were obtained from Santa-Cruz Biotechnologies, CA, USA.

AlphaFold-Multimer predictions of Mre11-Rev7 and Rad50-Rev7 protein complexes

The sequences of full-length ScRev7, Mre11 and Rad50 were obtained from the *Saccharomyces* Genome Database (<https://www.yeastgenome.org/>). The ScRev7-Mre11 and ScRev7-Rad50 heterodimer models were built using AF2-multimer (Evans et al., 2022; Varadi et al., 2022) via the ColabFold software (<https://github.com/sokrypton/ColabFold>) (Mirdita et al., 2022). The models with the highest confidence were analysed using LigPlot software (Version 2.2) to identify the amino acid residues involved in the interactions between the Mre11/Rad50 subunits and ScRev7. Approximately, 44% and 43% of the residues across the binding interface of ScRev7 and Mre11 displayed confident pLDDT scores greater than 60. Similarly, 67% and 100% of the residues lining the ScRev7 and Rad50 interface exhibited pLDDT scores greater than 60. The PyMOL Molecular Graphics System (Version 2.5.5) was used to visualize and analyze protein structures. Among the five different types of models generated by AlphaFold-Multimer, the top models (based on average pLDDT score) were chosen for display.

Microscale thermophoresis assay

The microscale thermophoresis (MST) assays were carried out on a Monolith NT.115 instrument (NanoTemper Technologies GmbH) according to the manufacturer's instructions. Samples were prepared in 20 μ l MST buffer (HEPES, pH 7.5, 50 mM NaCl, 10% glycerol) containing ScRev7-eGFP or eGFP and different concentrations of ligands in the range: Mre11 (0.00015 to 5 μ M), Rad50 (0.00006 to 2 μ M), Xrs2 (0.00015 to 5 μ M), Mre11-Rad50 (0.00015 to 5 μ M), Rev1 (0.00015 to 5 μ M) or Sae2 (0.00015 to 5 μ M). After incubation at 37 °C for 15 min, samples were transferred into Monolith NT.115 glass capillaries. The measurements were performed using 40% MST power with laser on/off times of 30 s and 5 s, and the MST signals were normalised to fraction bound (X) by $X = [Y(c) - \text{Min}] / (\text{Max} - \text{Min})$, error bars (SD) were normalized by $\text{stdnorm} = \text{std}(c) / (\text{Max} - \text{Min})$. The F_{norm} values or fraction bound were plotted against the ligand concentration to obtain an estimate of binding affinity. The values of the Hill coefficient and equilibrium dissociation constants (K_d) were calculated using the isothermal binding equation model in the MO. Affinity Analysis software provided by NanoTemper. Statistical analysis of data was performed using GraphPad Prism software (v5.0).

Preparation of radiolabeled DNA substrates

The sequences of oligonucleotides (ODNs) used for the preparation of DNA substrates are listed in **supplementary Table S7** [\[4\]](#). The ODNs were labeled at the 5' end using [γ - 32 P]ATP and T4 polynucleotide kinase, as previously described (Sambrook and Russell, 2001 [\[4\]](#)). The unincorporated [γ - 32 P]ATP was removed using a Sephadex G-50 superfine mini-column. The dsDNA substrates were prepared as follows: 41 bp dsDNA by mixing aliquots of 5'-end 32 P-labeled OSB17 (upper strand) with a small excess of unlabeled complementary OSB20; 60 bp dsDNA by mixing OSB41 (upper strand) and OSB42 (lower strand) in 1X SSC buffer (0.3 M sodium citrate buffer, pH 7.0, containing 3 M NaCl), followed by heating at 95 °C for 5 min and then slowly cooling to 24 °C over a period of 90 min. The substrates were resolved by non-denaturing 8% PAGE for 4 h at 4 °C (Thakur et al., 2016). The substrates were eluted from the gel slices in TE buffer (10 mM Tris-HCl, pH 7.5, and 1 mM EDTA) and stored at 4 °C for further use.

Exonuclease assay

The assay was performed as previously described (Arora et al., 2017 [\[4\]](#)) with slight modifications. Two steps are involved in the assay: the first step involves the assembly of the ScRev7-MRX complex, and the second, DNA cleavage. The MRX subunits (100 nM each) were mixed with increasing concentrations of ScRev7 (0.05 to 2.5 μ M) or ScRev7-C1 (1.5 and 3 μ M) and incubated on ice for 15 min, prior to the addition of twenty μ l of reaction mixture that contained 25 mM MOPS (pH 7.0), 20 mM Tris-HCl (pH 7.5), 80 mM NaCl, 8% glycerol, 5 mM MnCl_2 , 5 mM MgCl_2 , 1 mM ATP, 1 mM DTT, 200 μ g/ml BSA and 10 nM of 32 P-labeled 60 bp dsDNA. Subsequently, reaction mixtures were incubated at 37 °C for 1 h, and the reaction stopped by adding a two μ l solution containing two mg/ml proteinase K, 50 mM EDTA and 1% SDS. After incubation at 37 °C for 30 min, five μ l of a solution containing 10 μ g/ml glycogen, two μ l 3 M sodium acetate and 50 μ l of absolute ethanol was added to each sample and were frozen at -80°C. The thawed samples were centrifuged at 15000 rpm at 4 °C for 30 min. The pellets were washed with 70% ethanol and centrifuged at 15000 rpm at 4 °C for 5 min. The dried pellets were resuspended in 10 μ l gel-loading dye (80% formamide, 10 mM EDTA, 0.1% xylene cyanol and 0.1% bromophenol blue) and incubated at 95 °C for 5 min. Aliquots were analyzed on a 8% denaturing polyacrylamide/7 M urea PAGE using TBE buffer (89 mM Tris-borate (pH 8.3) and 1 mM EDTA) for 2 h at 40 W. The gels were dried, exposed to a phosphorimager screen, and scanned using a Fuji FLA 9000 phosphor imager.

Endonuclease assay

The assay was carried out as previously described (Shibata et al., 2014) with slight modifications. Briefly, the reaction mixtures (10 μ l) without ScRev7 contained 100 ng of M13 circular ssDNA, 30 mM Tris-HCl, pH 7.5, 25 mM KCl, 5% glycerol, 1 mM DTT, 200 μ g/ml BSA, 5 mM MnCl_2 and increasing concentrations of ScMre11 (0.1–2 μ M). After incubation at 37 °C for 1 h, the reaction was stopped by adding 2 μ l of stop solution (2 mg/ml proteinase K, 50 mM EDTA and 1% SDS) and incubation was extended for 30 min, followed by the addition of 2 μ l of gel loading solution. The reaction products were separated by electrophoresis on a 0.8% native agarose gel using 44.5 mM Tris-borate buffer (pH 8.3) containing 0.5 mM EDTA at 10 V/cm for 1 h. The gel was stained with ethidium bromide and the image was captured using the UVItect gel documentation system (UVItect, Cambridge, UK). To test the effect of ScRev7 on Mre11 endonuclease activity, the assay was carried out as described above, except that the ScRev7-Mre11 heterodimer was first formed by incubating a fixed amount of ScRev7 with increasing concentration of ScRev7 on ice for 15 min, prior to the addition of 10 μ l of reaction buffer. The reaction mixtures were incubated and analyzed as described above.

ATP crosslinking assay

The assay was carried out as previously described (Thakur et al., 2021b). The reaction mixture (20 μ l) contained 50 mM Tris-HCl (pH 5.0), 10 mM MgCl_2 , 400 pmol [γ - ^{32}P]ATP and increasing concentrations of Rad50 (0.1–1 μ M). Analogously, a fixed concentration of Rad50 (0.2 μ M) was incubated with increasing concentrations of ScRev7 (0.5 – 6 μ M) at 4 °C for 15 min, prior to transferring the sample into 20 μ l buffer containing 50 mM Tris-HCl (pH 5.0), 10 mM MgCl_2 and 400 pmol [γ - ^{32}P]ATP. After incubation at 4 °C for 25 min, samples were exposed to UV irradiation (1.2×10^5 $\mu\text{J}/\text{cm}^2$ in Hoefer UVC 500 ultraviolet crosslinker) at a distance of 2 cm. The reactions were stopped by adding 5 μ l of 5X Laemmli buffer (10 mM Tris-HCl, pH 6.8, 12.5% SDS, 40% glycerol, and 0.1% bromophenol blue). Samples were incubated at 95 °C for 10 min and resolved by SDS/PAGE in 10% polyacrylamide gel at 35 mA for 2 h. The gels were dried, exposed to a phosphorimager screens, and scanned using a Fuji FLA-9000 phosphor imager and the band intensities of radiolabeled species was quantified using UVI-Band Map software (v. 97.04). Data was plotted as mean and SD in GraphPad Prism (v5.0).

ATPase assay

The hydrolysis of [ψ - ^{32}P]ATP by Rad50 was assessed by measuring the release of $^{32}\text{P}_i$ as previously described (Thakur et al., 2021a). A fixed concentration of Rad50 (0.5 μ M) was incubated on ice for 30 min with increasing concentrations of ScRev7 or ScRev7-C1 to allow protein complex formation, prior to the reaction. Analogously, 200 μ M [ψ - ^{32}P] ATP and Rad50 (0.5 μ M) were incubated in 10 μ l reaction mixture containing 20 mM Tris HCl (pH 7.5), 50 mM KCl, 0.2 mg/ml BSA, 0.1 mM DTT, 1.0 mM MgCl_2 and 5 % glycerol, with indicated concentrations of Rad50. The reaction mixtures incubated at 30 °C for 30 min and the reaction was stopped by adding 15 mM EDTA. Aliquots (2 μ l) from each sample were spotted onto a polyethyleneimine-cellulose plate, developed in a solution containing 0.5 M LiCl, 1 M formic acid and 1 mM EDTA. The reaction products were visualized by using a Fuji FLA-9000 phosphor imager, and the band intensities were quantified and plotted using UVI-Band map software. Data was plotted as mean and SD using GraphPad Prism (Version 5.0).

Malachite green phosphate assay

ATP hydrolysis was monitored by measuring the amount of inorganic phosphate released using acidic ammonium molybdate and malachite green assay (Lanzetta et al., 1979). The reaction mixtures (80 μ l) contained 20 mM Tris-HCl (pH 7.5), 50 mM KCl, 5% glycerol, 0.1 mM DTT, 0.2 mg/ml BSA, 1 mM MgCl_2 , 150 μ M ATP, and increasing concentrations of ScRad50 (0.05 – 1 μ M). After incubation at 30°C for 30 min, reaction was stopped by adding 20 μ l of malachite green

reagent (Sigma-Aldrich, MAK307). Incubation was continued for 15 min at 24 °C to allow the formation of phosphomolybdate malachite green chromogenic complex. The absorbance values, as a measure of the extent of ATP hydrolysis, at 620 nm (y-axis) were plotted against increasing concentrations of ScRad50 (x-axis). To test the effect of ScRev7 or ScRev7-C1 on Rad50 ATPase activity, increasing concentrations of ScRev7 or ScRev7-C1 (0.1 – 2 μ M) were incubated with 0.25 μ M of ScRad50 on ice for 20 min, prior to mixing it with the reaction mixture. After incubation at 30 °C for 30 min, ATP hydrolysis was monitored as described above. The data were plotted, and the best-fit line was determined by nonlinear regression incorporating using GraphPad Prism (v. 5.0).

Mass spectrometry analysis

Mass spectrometry analysis of Rev7-GFP was carried out to ascertain its identity using orbitrap mass spectrometry (Zubarev and Makarov, 2013 [DOI](#)). In-gel trypsin digestion was performed as follows: 50 mM ammonium bicarbonate and acetonitrile in 7:3 ratio was used for de-staining; following which, samples were reduced for 30 min by adding 10 mM DTT diluted in 50 mM ammonium bicarbonate. Following reduction, alkylation buffer (55 mM iodoacetamide in 50 mM ammonium bicarbonate) was added and samples were incubated at room temperature for 30 min. Samples were treated with 10 ng trypsin (Mass spectrometry-grade, Sigma Aldrich, India), at 37°C for 12 h. The peptides were eluted with 70% acetonitrile (300 μ L) containing 0.1% trifluoroacetic acid, and the samples were dried in speed vacuum and the pellet was resuspended in 40 μ L resuspension buffer (2% acetonitrile in LC-MS grade Milli-Q). Samples (10 μ L) were injected into the Orbitrap Fusion tribrid mass spectrometer (Thermo Fisher Scientific Inc., USA). Samples were run for 110 min using a nano-spray ionization (NSI) source and static spray voltage. Ions were detected by Orbitrap detector at 60000 resolution using quadrupole isolation. Daughter ions were detected by Ion Trap detector using quadrupole isolation mode. Following data acquisition, peptide spectrum matches (PSMs) and percentage peptide coverage were obtained for the samples. Collectively, these results confirmed that the purified protein is GFP-tagged ScRev7.

Non-homologous end-joining assay

The assay was performed using a linear plasmid as previously described (Zhang and Paull, 2005 [DOI](#); Ghodke and Muniyappa, 2013 [DOI](#)). Briefly, 60 ng of BamHI-digested or undigested plasmid pRS416 was transformed into the WT and isogenic *rev1 Δ* , *rev3 Δ* , *rev7 Δ* , *mre11 Δ* , *sae2 Δ* , *rev1 Δ rev3 Δ* , *rev1 Δ rev7 Δ* , *rev3 Δ rev7 Δ* , *mre11 Δ rev7 Δ* , *sae2 Δ rev7 Δ* *rev7-C1* or *rev7-42* strains. The uncut plasmid pRS416 served as a control. The transformants arising from plasmid re-circularization were selected on SC/-Ura agar plates after incubation at 30 °C for 3-5 days. The efficiency of transformation was calculated as a ratio of the number of transformants with digested plasmid DNA to that with undigested plasmid DNA. The graph was obtained using GraphPad Prism (Version 5.0) and statistical significance was calculated using one-way ANOVA Dunnett's multiple comparisons test.

Cell cycle analysis

Cell cycle analysis was performed using a fluorescence-activated cell sorter (FACS) as previously described (Ghodke and Muniyappa, 2016 [DOI](#)). Cells were cultured to A₆₀₀ of 0.5 and synchronized in the G1 phase by treatment with α -factor (100 ng/ml, Sigma Aldrich) for 2 h at 30°C. The cells were washed three times with 1X PBS and resuspended in fresh liquid YPD medium. Aliquots were collected after release from G1 arrest at 15, 30, 45, 60, 75, 90, 105, and 120 min. The cells were fixed in 70% ethanol and stored at 4°C. Subsequently, samples were treated with RNase A (0.4 mg/mL) at 37°C for 12 h, stained with propidium iodide (15 μ g/mL) and incubated at 37°C for an additional 3 h. FACS analysis was performed on a FACSVerse analyzer (BD Biosciences). Dead and aggregated cells were excluded through gating, and cell cycle distribution was analyzed using FlowJo software (Version 10).

Non-homologous end-joining using a “suicide deletion” reporter assay

The assay was performed as previously described (Karathanasis and Wilson, 2002 [↗](#)). Briefly, *S. cerevisiae* YW714 strains WT, *rev1Δ*, *rev3Δ*, *rev7Δ*, *mre11Δ*, *ku70Δ*, *rev7-C1* and *rev7-42* cells were grown in an orbital shaking incubator at 200 rpm in a liquid SC/-Ura medium overnight at 30 °C. The cells were pelleted by centrifugation at 4000 rpm for 10 min. Cell pellets were washed thrice with MilliQ water and resuspended in liquid SC medium at an OD₆₀₀ of 1. Serial dilutions were plated on SC/-Ura agar plates containing glucose, and SC/-Ade agar plates containing galactose. After 5 days of incubation at 30 °C, the number of colonies was counted. The rate of NHEJ, as a function of Ade⁺ colonies, was calculated using the FALCOR software (Hall et al., 2009 [↗](#)). The differences between experimental results and relevant controls were tested with Dunnett’s multiple comparison test. The graph was generated using GraphPad Prism (Version 5.0). Statistical tests and p-values are mentioned in each figure legend.

Plasmid-chromosome recombination assay

The assay was performed as previously described (Paeschke et al., 2013 [↗](#)). The WT and isogenic *ura3-1*, *ura3-1 rev7Δ*, *ura3-1 mre11-D56N,H125N* *ura3-1 rev7Δ mre11-D56N,H125N*, *ura3-1 rev7-C1* and *ura3-1 rev7-42* strains were transformed with 60 ng of empty vector (pFAT10) or a plasmid bearing G4 DNA forming sequences (pFAT10-G4). The synchronous cells were grown in liquid SC/-Leu medium to an OD₆₀₀ of 0.5. Equal numbers of cells were replica-spotted on SC/-Ura medium and SC/-Ura medium plates containing 0.1 M HU. After incubation for 4-6 days at 30 °C, the number of papillae was counted. The recombination frequency was calculated using FLCOR software (Hall et al., 2009 [↗](#)).

Quantitative PCR analysis

The rate of DNA end resection was determined in *rad5111*, *rad5111 rev711*, *rad5111 mre11-H125N* and *rad5111 mre11-H125N rev711* strains, as previously described (Mimitou and Symington, 2010 [↗](#); Ferrari et al., 2018 [↗](#)). Single colonies grown in liquid YPD medium at 30°C for 12 h were sub-cultured in liquid YP medium containing 2% raffinose at 30°C till A₆₀₀ reached 0.5. At this stage, cells were synchronized at G2/M phase by adding nocodazole (10 µg/mL) and HO expression was induced by adding 2% galactose to the cell culture. For each time point, 20 mL samples were collected for each strain and 0.1% sodium azide was added immediately. Cells were collected by centrifugation at 4000 rpm for 10 min. The bead-beating step was used for isolation of genomic DNA from the resuspended pellets (Amberg et al., 2005 [↗](#)). Samples were incubated with 10 units of StyI-HF and XbaI-HF (New England Biolabs, Ipswich, USA) in a reaction mixture containing 15 µg of genomic DNA, 1X CutSmart buffer at 37 °C for 7 h. In parallel, equal amounts of gDNA was mock-digested as control. DNA was precipitated by adding equal volumes of isopropanol, followed by centrifugation at 13000 rpm for 20 min. Pellets were washed with 1 mL of 70% ethanol, air dried, and resuspended in 50 µL of 1X Tris-EDTA buffer (pH 7.5). In qPCR analysis, equal amounts of DNA (1 ng) was used per reaction in a mixture containing 1X SYBR green master mix (G-Biosciences, St. Louis, MO) and 0.2 µM each of forward and reverse primers, as indicated in the Table S8. The qPCR reaction was performed using a Bio-Rad CFX96 thermocycler and 96-well PCR plates. The data obtained from the qPCR were analysed using the Bio-Rad CFX Maestro 1.1 software (version 2.3). The specificity of the qPCR was ascertained by (a) agarose gel electrophoresis and (b) melting curve analysis of qPCR products. Similarly, primer efficiencies were calculated by performing qPCR analysis using serially diluted DNA samples. The HO cut efficiencies across strains were estimated by qPCR analysis at a single HO site (Coordinates 284324-294420) with a unique upstream sequence on Chromosome III. Upon DSB induction, the percentage of ssDNA generated by end resection at different time points and distance (0.7 kb and 3

kb) from the break was calculated as previously described (Ferrari et al., 2018 [DOI](#)). The *PRE1* gene situated on chromosome V was used as an internal reference. Two-way ANOVA was performed to statistically analyse the datasets and graphs were generated using GraphPad Prism (Version 5.0).

Statistical and data analysis

Differences among groups were analyzed by one-way ANOVA Dunnett's multiple comparison test followed by Tukey's post hoc test (GraphPad Prism 6.07). Statistical significance level was set as follows: *, P-value < 0.05; **, P-value < 0.01; ***, P-value < 0.001; ****, P-value < 0.0001. The statistical methods employed to analyse the data are indicated in the figures legends.

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Author contributions

Sugith Badugu, Kshitiza M. Dhyani, Manoj Thakur: Investigation, Methodology, Software, Validation, Data curation, Writing – original draft and editing; Kshitiza M. Dhyani Manoj Thakur, Kalappa Muniyappa, Conceptualization, Resources, Supervision, Funding acquisition, Formal analysis, Project administration, Writing – review and editing.

Conflict of interest statement

None declared

Supplemental figures

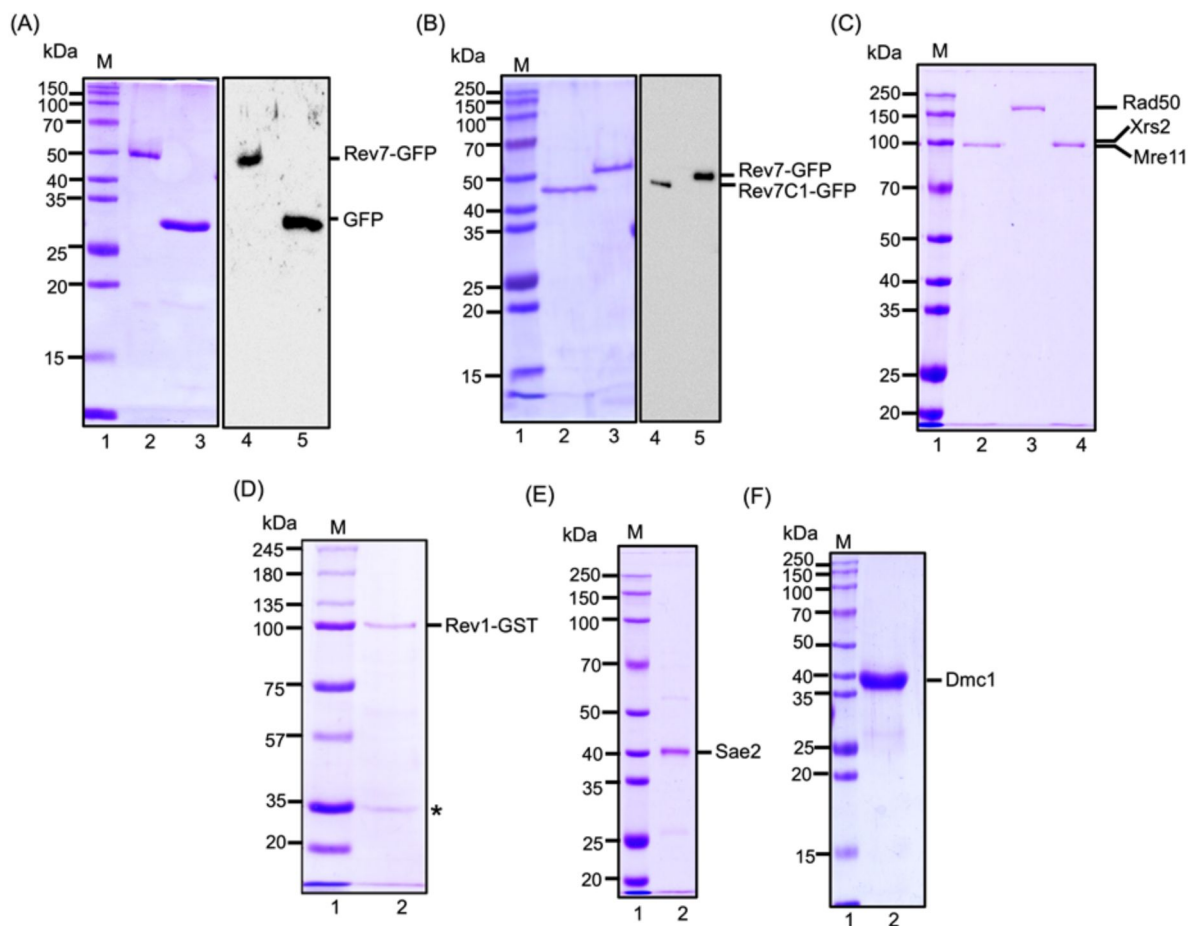


Figure 1—figure supplement 1.

SDS-PAGE analysis of purified proteins used in this study.

(A) SDS-PAGE and Western blot analysis of purified GFP-tagged ScRev7 and eGFP. Lane 1 (in panels A-F) standard molecular weight markers; lane 2, purified GFP-tagged ScRev7; lane 3, purified GFP. Western blot analysis of GFP-tagged Rev7 (lane 4) and eGFP (lane 5) using mouse anti-GFP antibodies. (B) Purified GFP-tagged Rev7-C1 (2) and GFP-tagged Rev7 (lane 3). Western blot analysis of GFP-tagged Rev7-C1 (lane 4) and GFP-tagged Rev7 (lane 5) using mouse anti-GFP antibodies. (C) Purified MRX subunits; lane 2, Mre11; lane 3, Rad50; lane 4, Xrs2; (D) Purified GST-tagged Rev1; The faster migrating band (*) corresponds to ScRev7, which co-purifies with GST-tagged Rev1 (Acharya et al., 2005). (E) Purified Sae2 and (F) Purified Dmc1.

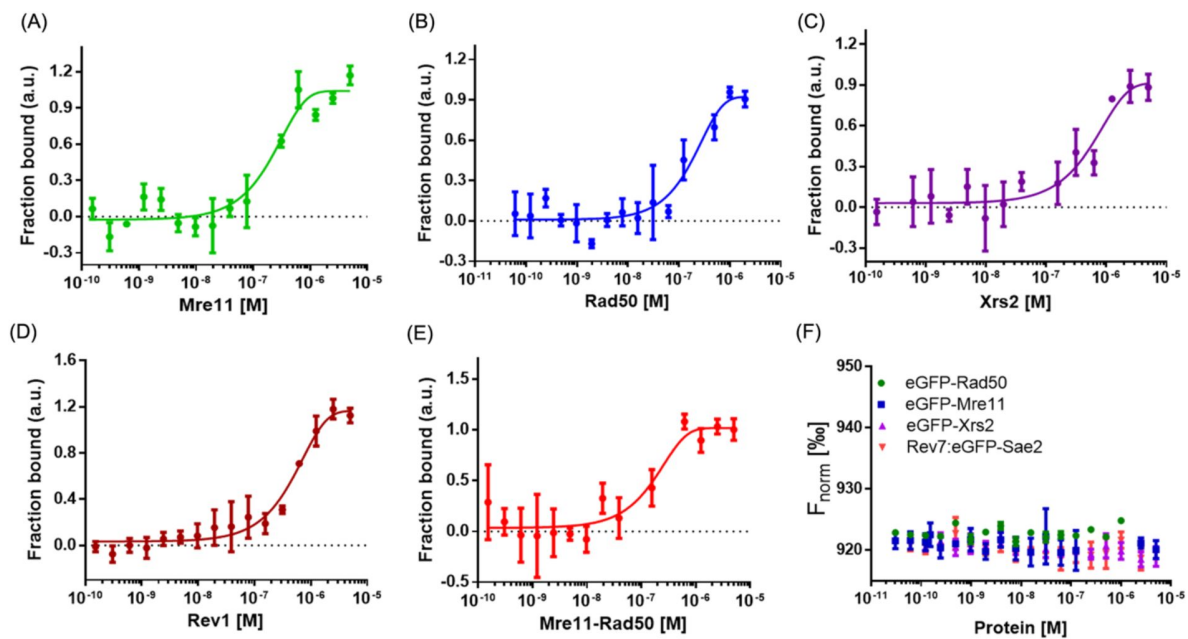


Figure 1—figure supplement 3.

Microscale thermophoresis reveals a direct interaction between Rev7 and MRX subunits.

Normalised MST binding curves were generated for the titration of (A) Mre11, (B) Rad50, (C) Xrs2, (D) Rev1, (E) the Mre11-Rad50 complex, and (F) Sae2 proteins against Rev7-eGFP. Panel F also includes datasets for the binding of Mre11, Rad50, and Xrs2 against eGFP. Error bars represent the standard error of the mean (SEM), and data are representative of three independent experiments.

Figure 1—figure supplement 4.

Rev7-C1 exhibits weak interactions with MRX subunits.

Binding isotherms were obtained by incubating increasing concentrations of (A) Mre11 (0.00015 to 5 μ M), (B) Rad50 (0.000016 to 2 μ M), and (C) MR complex (0.00015 to 5 μ M) with fixed concentration of Rev7C1:GFP protein (0.25 μ M). (D) K_d values obtained upon fitting the curves in K_d model using MO affinity analysis software (Nanotemper). Data is representative of three independent experiments.

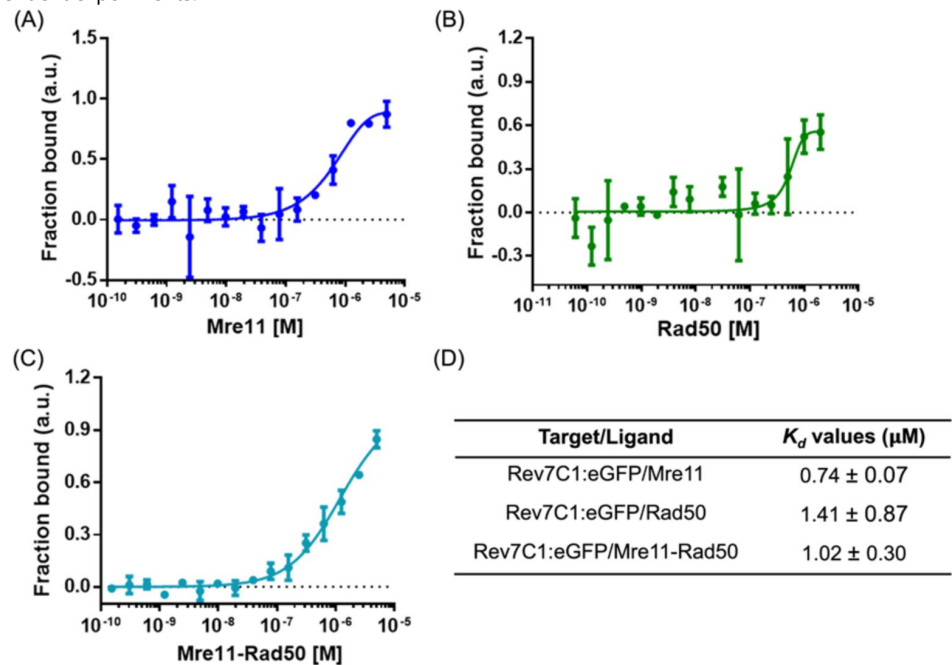
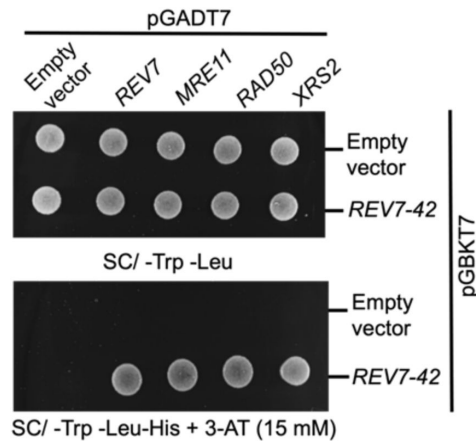


Figure 2 —figure supplement 1.

The C-terminal 42 amino-acid region of Rev7 interacts with M/R/X subunits.

The PJ69-4A cells were co-transformed with empty vectors or vectors expressing Rev7-42 residue peptide or MRX subunits as indicated. Equal number of cells were spotted onto -Trp-Leu and -Trp-Leu-His selection plates containing 15 mM 3-AT and incubated at 30 $^{\circ}$ C for 2 days.



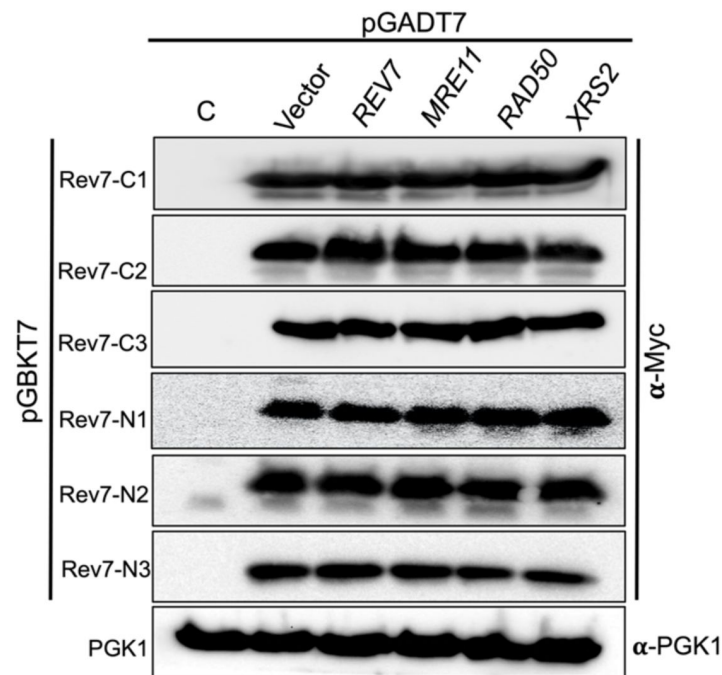


Figure 2—figure supplement 2.

Western blot showing the abundance of N- and C-terminally truncated variants of Rev7.

S. cerevisiae PJ69-4A cells were co-transformed with the indicated bait plasmid expressing c-Myc epitope tagged N- or C-terminally truncated variant of Rev7 (REV7-C1, REV7-C2, REV7-C3, REV7-N1, REV7-N2 or REV7-N3) with and a prey plasmid expressing *REV7*, *MRE11*, *RAD50* or *XRS2*. PGK-1 was used as the loading control and was probed with anti-PGK-1 antibodies. Sample loading was normalized to similar amounts by determining the protein concentration utilizing the dye-binding assay. Lane 'C' represents lysates from cells harboring empty bait and prey vectors.

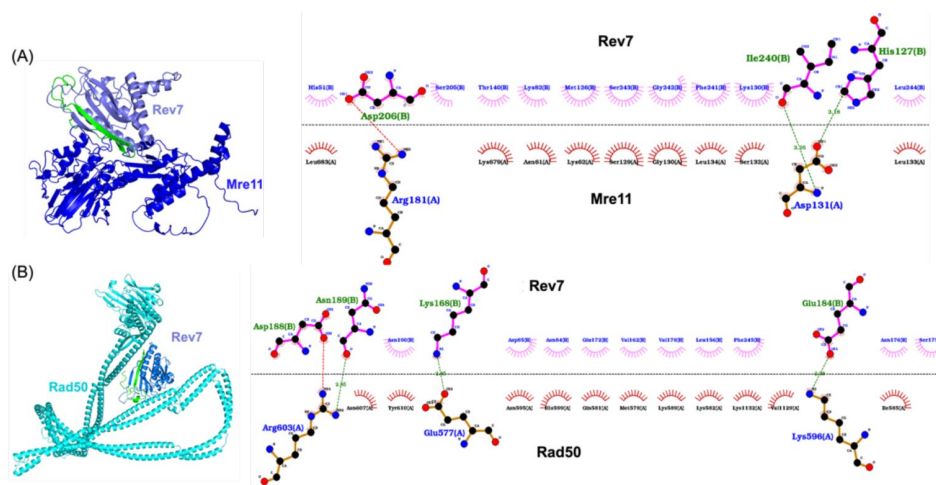


Figure 2—figure supplement 3.

AlphaFold-Multimer generated models of Rev7-Mre11 and Rev7-Rad50 protein complexes.

(A) Model of the Rev7-Mre11 heterodimer, Rev7, light blue, Mre11, deep blue. **(B)** Model of Rev7-Rad50 heterodimer, Rev7, light blue, Rad50, cyan. Alongside is the two-dimensional view of the corresponding protein-protein binding interface as generated by the LigPlot software. Numbers indicated on the dotted lines represent bond-lengths in Angstroms. The C-terminal safety belt region of Rev7 is coloured in green in both complexes. By convention, red, black and blue circles represent oxygen, carbon and nitrogen atoms respectively. The dashed line represents the interaction interface.

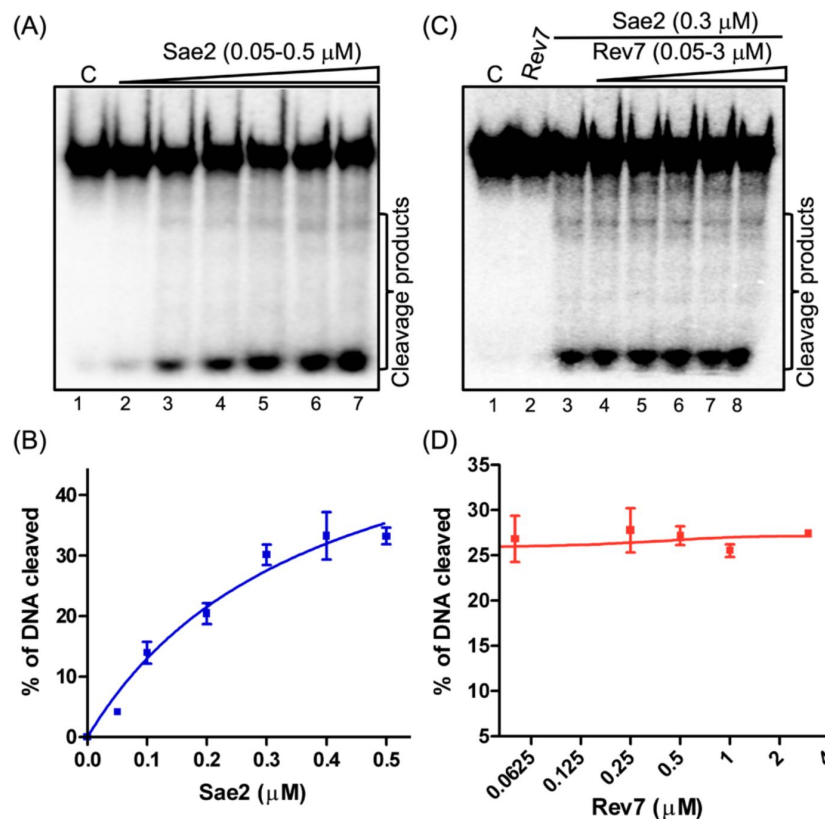


Figure 5—figure supplement 1

Rev7 does not impact the endonuclease activity of Sae2.

(A) Sae2-catalyzed cleavage of 32 P-labeled 60 bp dsDNA. Reaction mixtures containing 5 nM dsDNA were incubated with increasing concentrations of Sae2. Lane 1: control reaction without protein; Lanes 2–7: Sae2 at 0.05, 0.1, 0.2, 0.3, 0.4, and 0.5 μ M, respectively. (B) Quantification of Sae2 endonuclease activity as a function of its concentration. (C) A representative image demonstrating the effect of Rev7 on the endonuclease activity of Sae2. Lane 1: Control reaction of 32 P-labeled dsDNA in the absence of Sae2; Lane 2: same as Lane 1 but with 3 μ M ScRev7; Lane 3: same as Lane 1 but with 0.3 μ M Sae2; Lanes 4–8: same as Lane 3 but with increasing concentrations of ScRev7 at 0.05, 0.25, 0.5, 1.0, and 3.0 μ M, respectively. (D) Quantification of the endonuclease activity catalysed by Sae2 from panel C. The closed triangles on the top of gel images in panels (A), and (C), represent increasing concentrations of Sae2 or ScRev7 respectively. Error bars indicate SEM, and data are representative of three independent experiments.

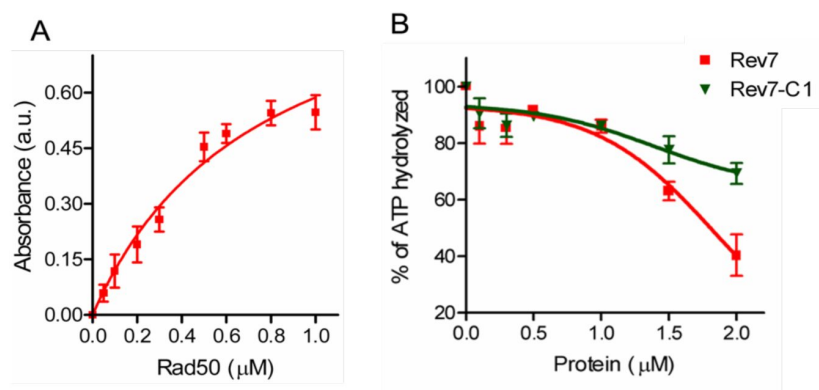


Figure 6—figure supplement 1.

ScRev7 inhibits the ATPase activity of ScRad50.

ATPase activity was measured using malachite green phosphate detection assay that quantifies the amount of Pi released upon ATP hydrolysis. (A) Increasing concentrations of ScRad50 (0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.6, 0.8 and 1 μ M) were incubated with reaction mixtures containing 20 mM Tris-HCl, pH 7.5, 50 mM KCl, 5% glycerol, 0.1 mM DTT, 0.2 mg/ml BSA, 150 μ M ATP and 1 mM $MgCl_2$. Absorbance values were then monitored spectrophotometrically at 620 nm. (B) Same as (A), but fixed concentrations of ScRad50 (0.25 μ M) were pre-incubated with increasing concentrations of ScRev7 or ScRev7-C1 (0, 0.1, 0.3, 0.5, 1, 1.5, 2 μ M). The graphs were generated by non-linear regression analysis using GraphPad Prism (V. 5.0). Error bars represent standard deviation of data from two independent experiments.

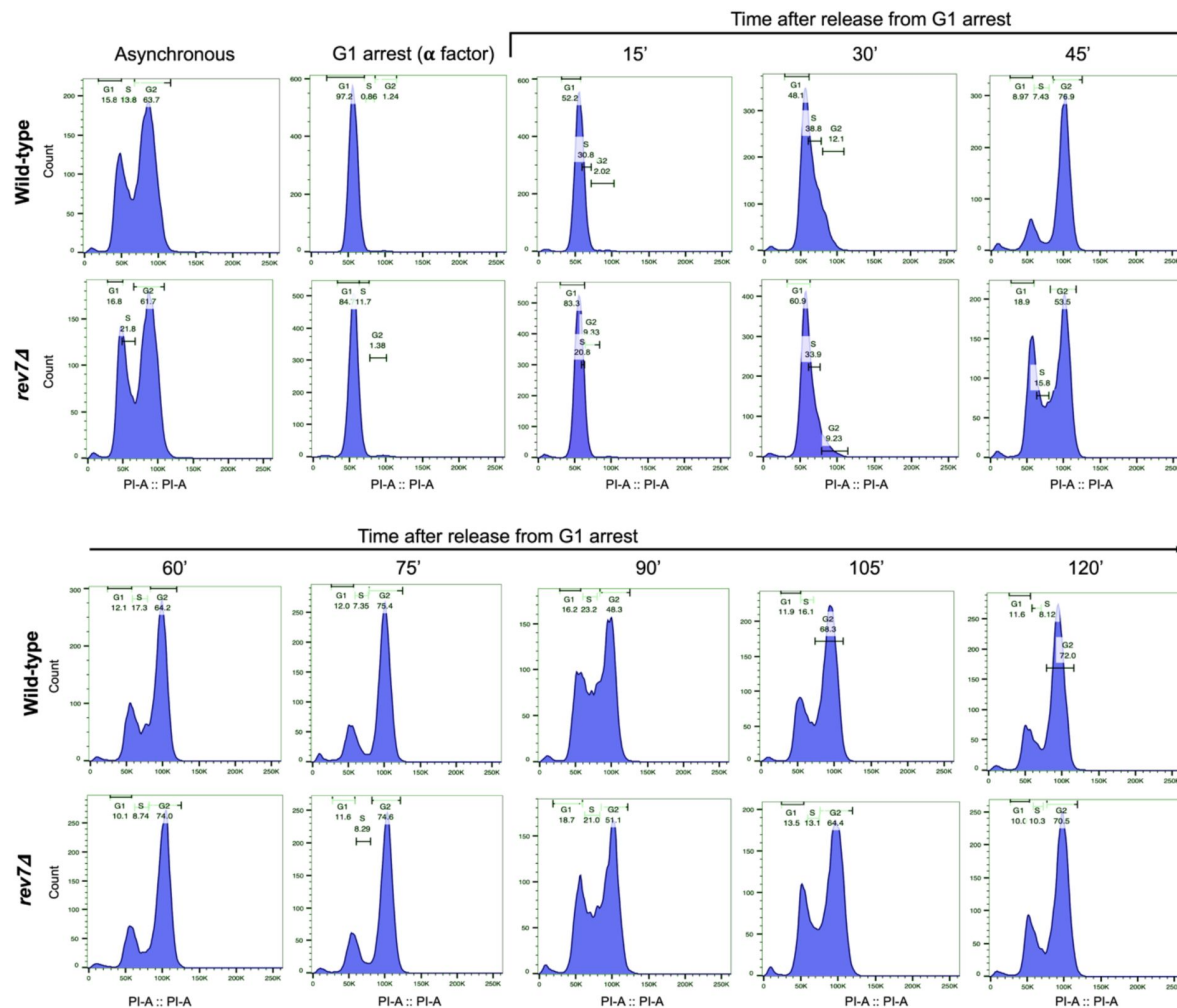


Figure 7—figure supplement 1.

Cell cycle analysis of wild-type and *rev7Δ* cells.

Cell cycle progression in wild-type and *rev7Δ* cells was analyzed by synchronizing an equal number of cells at the G1 phase using α -factor (100 ng/mL) in the growth medium. After releasing cells from G1 arrest, they were collected at specified time points and stained with propidium iodide for fluorescence-activated cell sorting (FACS) analysis. The histograms illustrate the distribution of cells in the G1, S, and G2 phases of the cell cycle for both wild-type and *rev7Δ* strains. Results shown are representative of two independent experiments.

Figure 9—figure supplement 1.

Mutation of G-quadruplex-forming sequences markedly attenuate the rate of HR frequency in *rev711* cells.

(A) Representative images showing Ura³ papillae on SC/-Ura agar plates in the absence or presence of 100 mM HU. Cells were imaged after six days of growth at 30 °C. (B) Quantification of the rate of HR frequency in *ura3-1* and *ura3-1 rev711* strains carrying plasmids with unmutated and mutated *ura3-G4* inserts in the absence of HU. Data are presented as mean $\pm$ SD from three different experiments. n.s., not significant, * $p < 0.05$, and *** $p < 0.001$ vs. control, as assessed using two-way ANOVA and Sidaks multiple comparison test.

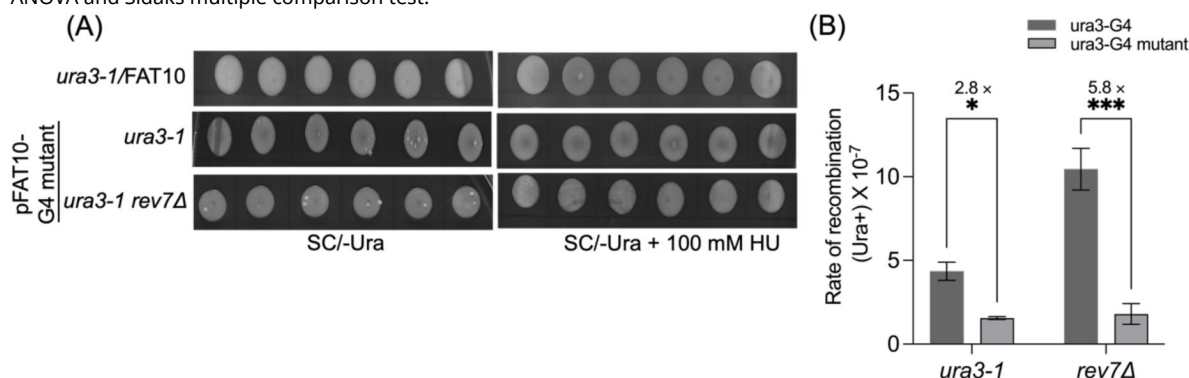
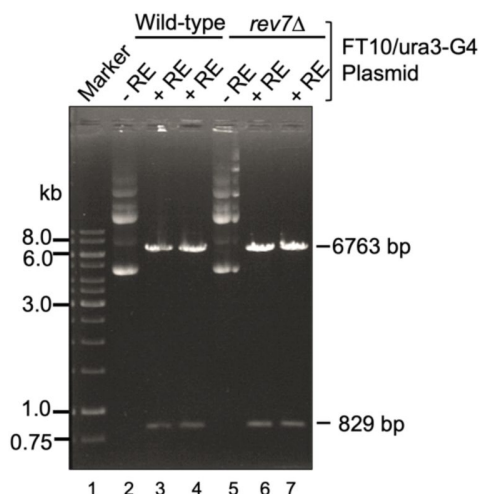


Figure 9—figure supplement 2.

G-quadruplex capable sequences are stable in *rev7Δ* cells during the HR assay.

Total plasmid pFAT10-G4 DNA was isolated from from cultures of WT and *rev7Δ* cells. Samples of undigested and BamHI and Sph1 digested plasmid pFAT10-G4 DNA were analyzed by electrophoresis on a 0.6 % agarose gel and visualized after staining with ethidium bromide. Lane1: DNA marker; 2 and 5, represent undigested plasmid DNA from the WT and *rev7Δ* cells respectively. Lanes 3 and 4, plasmid pFAT10-G4 DNA from the WT cells digested by BamHI and Sph1; 6 and 7, same as lane 3 and 5, but plasmid isolated from *rev7Δ* cells. + RE and -RE indicate DNA digested by BamHI and Sph1 and undigested plasmid DNA respectively.



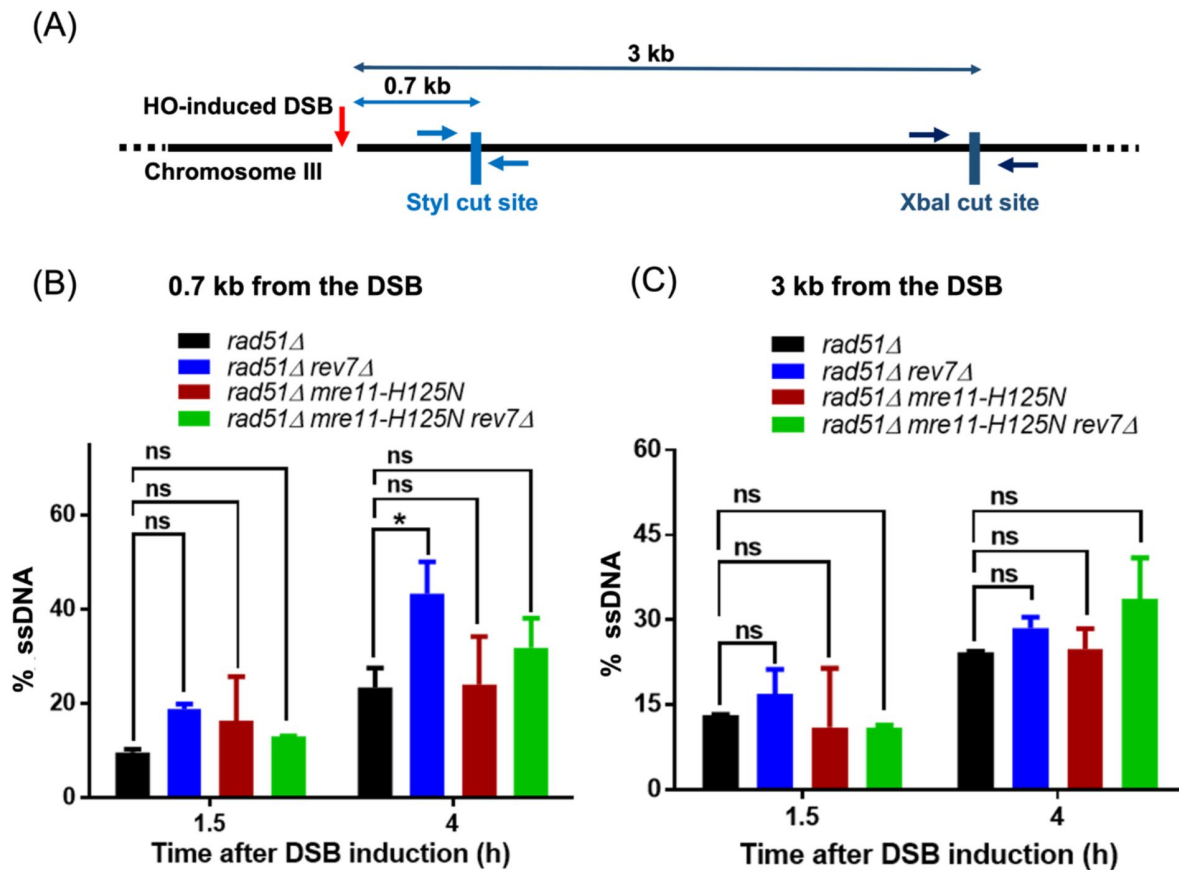


Figure 9—figure supplement 3

Deletion of Rev7 enhances the speed of short-range end-resection in *S. cerevisiae*.

(A) Schematic diagram represents the positions of StyI (0.7 kb) and XbaI (3 kb) restriction sites with respect to HO endonuclease-induced DSB at the Chromosome III *MAT* locus. Arrows flanking the restriction sites mark the binding sites of forward and reverse primers used for qPCR analysis. The 5'—3' end-resection was physically assessed in the indicated strains by quantifying the percentage of ssDNA generated at 1.5 h and 4 h post induction of DSB, at a distance of (B) 0.7 kb and (C) 3 kb from the DSB. Two-way ANOVA was performed to determine the statistical significance of the datasets and graphs were generated using GraphPad Prism (Version 5.0). Data is representative of two independent biological replicates. ns, not significant.

Supplementary Tables

Mre11 (position of amino acid residue)	Rev7 (position of amino acid residue)	Distance (Å)	Mre11-residue	Rev7-residue	Mre11-pLDDT	Rev7-pLDDT
A_0181	B_0206	0.73	ARG	ASP	74.44	65.25
A_0062	B_0126	1.14	LYS	MET	77.81	71.25
A_0130	B_0242	1.27	GLY	GLY	42.81	20.97
A_0134	B_0241	1.36	LEU	PHE	72.12	22.19
A_0131	B_0242	1.38	ASP	GLY	42.41	20.97
A_0132	B_0241	1.86	SER	PHE	51.47	22.19
A_0130	B_0243	2.16	GLY	SER	42.81	20.88
A_0130	B_0241	2.23	GLY	PHE	42.81	22.19
A_0131	B_0241	2.23	ASP	PHE	42.41	22.19
A_0679	B_0082	2.24	LYS	LYS	26.53	81.81
A_0131	B_0240	2.52	ASP	ILE	42.41	26.36
A_0130	B_0130	2.74	GLY	LYS	42.81	72.25
A_0181	B_0207	2.77	ARG	VAL	74.44	63.84

Residues highlighted in green are present in the C-terminal safety-belt region of Rev7 protein.

Table S1.

Inter-atomic distances and confidence parameters of amino acid residues mediating Rev7-Mre11 interactions.

Rad50 (position of amino acid residues)	Rev7 (positions of amino acid residues)	Distance (Å)	Rad50-residue	Rev7-residue	Rad50-pLDDT	Rev7-pLDDT
A_0603	B_0188	1.26	ARG	ASP	73.56	51.81
A_0582	B_0156	1.45	LYS	LEU	73	69
A_0599	B_0084	1.88	HIS	ASN	73.56	68.56
A_0596	B_0184	2.3	LYS	GLU	75.75	62.81
A_1132	B_0245	2.5	LYS	PHE	81	18.19
A_0581	B_0172	2.58	GLN	GLU	72.12	80.44
A_1129	B_0245	2.6	VAL	PHE	79	18.19
A_0599	B_0085	2.78	HIS	ASP	73.56	74.62
A_0577	B_0168	2.85	GLU	LYS	71.31	77.94
A_0603	B_0189	2.85	ARG	ASN	73.56	46.66

Residues highlighted in green are present in the C-terminal safety-belt region of Rev7 protein.

Table S2.

Inter-atomic distances and confidence parameters of amino acid residues mediating Rev7-Rad50 interactions

Table S3.

The exact p-values for Figure 7B. The p-values were obtained by comparing the percentage of non-homologous end-joining observed for the indicated single-or double-gene deletions versus either the wild-type (WT) or *rev7Δ* strain, using non-parametric one-way ANOVA Dunnett test.

Reference strain	Strain	p-values
WT	<i>rev1Δ</i>	0.0461
	<i>rev3Δ</i>	0.0011
	<i>rev7Δ</i>	<0.0001
	<i>mre11Δ</i>	<0.0001
	<i>sae2Δ</i>	0.7111
	<i>rev1Δ rev3Δ</i>	<0.0001
	<i>rev1Δ rev7Δ</i>	<0.0001
	<i>rev3Δ rev7Δ</i>	<0.0001
	<i>mre11Δ rev7Δ</i>	<0.0001
	<i>sae2Δ rev7Δ</i>	<0.0001
	<i>rev7-C1</i>	<0.0001
	<i>rev7-42</i>	0.9910
	<i>rev1Δ rev7Δ</i>	0.2731
<i>rev7Δ</i>	<i>rev3Δ rev7Δ</i>	0.9958
	<i>mre11Δ rev7Δ</i>	0.0029
	<i>sae2Δ rev7Δ</i>	0.2220
	<i>rev7-C1</i>	0.0022
	<i>rev7-42</i>	<0.0001

Table S4.

The exact p-values for Figure 8B. The p-values were obtained by comparing the percentage of non-homologous end-joining observed for the indicated single-or double-gene deletions versus either the wild-type (WT) or *rev7Δ* strain, using non-parametric one-way ANOVA Dunnett test.

Reference strain	Strain	p-values
WT	<i>rev7Δ</i>	<0.0001
	<i>mre11Δ</i>	<0.0001
	<i>ku70Δ</i>	<0.0001
	<i>rev1Δ</i>	0.0046
	<i>rev3Δ</i>	0.0009
	<i>rev7-C1</i>	<0.0001
	<i>rev7-42</i>	0.369
	<i>rev7-C1</i>	0.3664
<i>rev7Δ</i>	<i>rev7-42</i>	0.0002

Strain	Genotype	Source
W1588-4C	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺</i>	Paeschke et al., 2011
KP013	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3</i>	Paeschke et al., 2011
KMY017	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3 rev7Δ::kanMX4</i>	This Study
KMY042	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3 rev1Δ::hphNT1</i>	This Study
KMY061	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3 rev3Δ::KanMX4</i>	This Study
KMY018	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3 mre11Δ::hphNT1</i>	This Study
KMY141	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3 rev7Δ::rev7-C1 (203-245 aaΔ)-9MYC-hphNT1</i>	This Study
KMY143	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3 rev7Δ::rev7-42 (1-203 aaΔ)-3MYC-KANMX4</i>	This Study
LSY1375	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ mre11-D56N, H125N</i>	Krogh et al., 2005
LSY1091	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ sae2Δ::kanMX6</i>	Krogh et al., 2005
KMY058	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3 rev1Δ::hphNT1 rev3Δ::KanMX4</i>	This Study
KMY059	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3 rev7Δ::kanMX4 rev3Δ::hphNT1</i>	This Study
KMY048	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ bar1Δ::HIS3 rev1Δ::hphNT1 rev7Δ::kanMX4</i>	This Study
KMY050	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ rev7Δ::kanMX4 mre11Δ::hphNT1</i>	This Study
KMY053	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ mre11-D56N, H125N rev7Δ::kanMX4</i>	This Study
KMY060	<i>MATa ura3-1 trp1-1 leu2-3 112 his3-11 15 ade2-1 can1-100 ybp1-1 RAD5⁺ sae2Δ::kanMX6 rev7Δ::hphNT1</i>	This Study
KP038	<i>MATa ura3-52 lys2-801 ade2-101 trp1-Δ63 his1-Δ200 leu2-Δ1 pif1-m2</i>	Paeschke et al., 2011
YW714	<i>MATα ade2::SD2::URA3 his31 leu20 ura30</i>	Karathanasis and Wilson,

Table S5.

***S. cerevisiae* strains used in this study.**

		2002
KMY136	<i>MATα ade2::SD2::URA3 his31 leu20 ura30 rev7Δ::hphNT1</i>	This Study
KMY137	<i>MATα ade2::SD2::URA3 his31 leu20 ura30 mre11Δ::hphNT1</i>	This Study
KMY144	<i>MATα ade2::SD2::URA3 his31 leu20 ura30 rev1Δ::hphNT1</i>	This Study
KMY145	<i>MATα ade2::SD2::URA3 his31 leu20 ura30 rev3Δ::hphNT1</i>	This Study
KMY146	<i>MATα ade2::SD2::URA3 his31 leu20 ura30 rev7Δ::REV7-C1 (203-245 aaΔ)-9MYC-hphNT1</i>	This Study
KMY147	<i>MATα ade2::SD2::URA3 his31 leu20. ura30 rev7Δ::REV7-42 (1-203 aaΔ)-3MYC-KANMX4</i>	This Study
PJ69-4A	<i>MATa trp1-901 leu2-3,112 ura3-52 his3-200 ga14Δ ga180Δ LYS2::GALI-HIS3 GAL2-ADE2 met2::GAL7-lacZ</i>	James <i>et al.</i> , 1996
KMY023	<i>MATa trp1-901 leu2-3,112 ura3-52 his3-200 ga14Δ ga180Δ LYS2::GALI-HIS3 GAL2-ADE2 met2::GAL7-lacZ rev3Δ:: hphNT1</i>	This Study
KMY139	<i>MATa trp1-901 leu2-3,112 ura3-52 his3-200 ga14Δ ga180Δ LYS2::GALI-HIS3 GAL2-ADE2 met2::GAL7-lacZ mre11Δ::KANMX4 rad50Δ::URA3 xrs2Δ::HphNT1</i>	This Study
BJ5464	<i>MATα ura3-52 trp1 leu2-Δ1 his3-Δ200 pep4::HIS3 prb1-Δ1.6R can1 GAL</i>	Johnson <i>et al.</i> , 2006
LSY2172-24C	<i>MATa rad51::LEU2 ade3::GAL1-HO</i>	Mimitou and Symington, 2010
LSY2265-10D	<i>MATa rad51::LEU2 mre11-H125N::URA3::mre11-H125N ade3::GAL1-HO</i>	Mimitou and Symington, 2010
KMY170	<i>MATa rad51::LEU2 ade3::GAL1-HO rev7Δ:: kanMX4</i>	This study
KMY171	<i>MATa rad51::LEU2 mre11-H125N::URA3::mre11-H125N ade3::GAL1-HO rev7Δ:: kanMX4</i>	This study

Table S5. (continued)

Primer names	Sequence (5'—3')	Plasmid DNA/primers
OSB01-FP	GCGGATCCATGAATAGATGGGTAGAGAAGTG	pET28a_ScREV7, pET28a_ScREV7-C1
OSB02-RP	GCAAGCTTTTCTTCAAATTTCATTTTTGCAC	pET28a_ScREV7
OSB03-FP	CATATGATGAATAGATGGGTAGAGAAGTG	pGBKT7_ScREV7, pGADT7_ScREV7
OSB04-RP	GGATCCTTCTTCAAATTTCATTTTTGCAC	pGBKT7_ScREV7, pGADT7_ScREV7

Table S6.

Sequences of primers

OSB05-FP	ATGCCT CATATG ATGGACTATCCTGATCCAGACACACAATAAGG	pGADT7_ScMRE11
MRE11_RP	ATGCCT GAATTC CCTATTTTCTTTTCTTAGGAAGGAGGACTTCC	pGADT7_ScMRE11
OSB11-FP	ATCCAAGAAGAAAAAAAAAATAGTAATCGTTGCGTCAGCTTTATGCGTACGCTGCAGGTCGAC	REV7 Knockout
OSB12-RP	ACATTTAATTTTAATTCCATTCTTCAAATTTTCATTTT TGCACCTAATCGATGAATTCGAGCTCG	REV7 Knockout
OSB13-FP	ATATGCATATACGTCTACACTAC	REV7 Knockout
OSB14-RP	CTGCAGCGAGGAGCCGTAAT (KANB)	REV7 Knockout
OSB33-FP	CGC GGATCC ATGAGCGCTATCTATAAATTATC	pESUMO_ScRAD50
OSB34-RP	CCG CTCGAGT CAATAAGTGACTCTGTTAATATC	pESUMO_ScRAD50 & pGADT7_ScRAD50
OSB35-FP	CCG GAATTC ATGAGCGCTATCTATAAATTATC	pGADT7_ScRAD50
OSB36-FP	GGAATTC CATATG ATGTGGGTAGTACGATACCAG	pGADT7_ScXRS2
OSB37-RP	CG GGATCC TTATCCTTTTCTTCTTTTGAA	pGADT7_ScXRS2
OSB52-FP	ATTGACGCAAGTTGTACCTGCTCAGATCCGATAAA ACTCGACTATGCGTACGCTGCAGGTCGAC	MRE11 Knockout
OSB53-RP	TGGTTATAAATAGGATATAATATAATATAGGGATCA AGTACAATAATCGATGAATTCGAGCTCG	MRE11 Knockout
OSB54-FP	AAGGCATCTACAAATCTCATTG	MRE11 Knockout
OSB133-FP	AGGCAAAAATCACAAATTGAGTGGGTCGATATTAA CAGAGTCACTTATCGTACGCTGCAGGTCGAC	RAD50 Knockout
OSB075-RP	TAATTAATCAATCAAAGTCTATCCCTTCGTAGATAT TATGGGGTCTTTATCGATGAATTCGAGCTCG	RAD50 Knockout
OSB076-RP	TCCTTGTTAACAACGGTG	RAD50 Knockout
OSB134-FP	ACGACGACGATGACGACGGTCCGAAGTTTACGTT CAAAAGAAGAAAAGGACGTACGCTGCAGGTCGAC	XRS2 Knockout
OSB078-RP	TGCAAAATATAATTTAATGAAATTGGAAATACTCGG AAAATTTATCATTAAATCGATGAATTCGAGCTCG	XRS2 Knockout
OSB079-RP	TTGGAGTATTCAAAGAGGCTAC	XRS2 Knockout
OSB55-FP	TTCTCAAAATAAATCGATACTGCATTTCTAGGCATA TCCAGCGATGCGTACGCTGCAGGTCGAC	REV1 Knockout
OSB56-FP	AACGCGTGTTTACTGTATGCTGAAATGTTTTTTTT TTTTTAATTCAATCGATGAATTCGAGCTCG	REV1 Knockout
OSB57-FP	TTTACACAGACCAAGACGG	REV1 Knockout
OSB58-FP	TCAATACAAACTACAAGTTGTGGCGAAATAAAAT GTTTGGAATGCGTACGCTGCAGGTCGAC	REV3 Knockout
OSB59-RP	ACAAATAACTACTCATCTTTTGCAGACATATCTG TGTCTAGATTAATCGATGAATTCGAGCTCG	REV3 Knockout
OSB60-FP	ATCCCTGTGGTCTCCTACC	REV3 Knockout
OSB116-FP	ATCCAAGAAGAAAAAAAAAATAGTAATCGTTGCGT CAGCTTTATGAATAGATGGGTAGAGAAGTG	rev7-c1 truncation (rev7-c1-9MYC- hphNT1)
OSB117-RP	GTCGACCTGCAGCGTACGGACTAAAGAAGTGAGT TTTATTTTAGG	rev7-c1 truncation (rev7-c1-9MYC- hphNT1)
OSB118-FP	CCTAAAAATAAACTCACTTCTTTAGTCCGTACGCTG CAGGTCGAC	rev7-c1 truncation (rev7-c1-9MYC-

Table S6. (continued)

		hphNT1)
OSB070-RP	TACTTAGAGACATTTAATTTTAATTCCATTCTTCAAA TTTCATTTTTGCACTTAATCGATGAATTCGAGCTCG	rev7-c1 truncation (rev7-c1-9MYC- hphNT1)
OSB122-FP	ATCCAAGAAGAAAAAAAAAATAGTAATCGTTGCGT CAGCTTTATGGGTTCTGACGTGGGGCC	rev7-42 aa truncation (rev7-42-3MYC- KANMX)
OSB120-RP	GTCGACCTGCAGCGTACGAAACAAAGATCCAAAAA TGCTC	rev7-42 aa truncation (rev7-42-3MYC- KANMX)
OSB121-FP	GAGCATTTTTGGATCTTTGTTTCGTACGCTGCAGG TCGAC	rev7-42 aa truncation (rev7-42-3MYC- KANMX)
OSB61-FP	GGAATTCC CATATG TCATTACCTTGCCGCAGTTC	pGBKT7_REV7-N1
OSB62-FP	GGAATTCC CATATG ACCCACGTTTACAGATTTTCC	pGBKT7_REV7-N2
OSB63-FP	GGAATTCC CATATG TTGGAAGTAGGACATAAGTTGG	pGBKT7_REV7-N3
OSB64-RP	CG GGATCC TTAGACTAAAGAAGTGAGTTTTATTTT AGG	pGBKT7_REV7-C1
OSB65-RP	CG GGATCC TTATTCTGCTTTTTCCTCCAAACTATC	pGBKT7_REV7-C2
OSB66-RP	CG GGATCC TTACAATTCGATCGCATTAAATAACTGC	pGBKT7_REV7-C3
OSB80-FP	CG GGATCC ATGTGCGAAAGCTACATATAAG	pFAT10_ura3-G4
OSB81-RP	CCCGGTCCCCCA ACGACAAC CCCA ATCAACCA ATCGTAACCTTC	pFAT10_ura3-G4
OSB82-FP	GGG TGTTGTCGTT GGGGGG ACC GGG ATGACACC CGGTGTGGG	pFAT10_ura3-G4
OSB83-RP	AC ATGCATG CTTAGTTTTGCTGGCCGCATC	pFAT10_ura3-G4
OSB129-RP	CTCATAGCATCCAACGACAACACGCAATCAACCAA TCGTAACCTTC	pFAT10_ura3-G4 mutant
OSB130-FP	GCGTGTTGTCGTTGGATGCTATGAGATGACACCC GGTGTGGG	pFAT10_ura3-G4 mutant
OSB125-RP	GCA AAGCT TTTAGACTAAAGAAGTGAGTTTTATTTA GG	pET28_ScRev7-C1
REV7GFP-FP	GCT CTAGA ATGAATAGATGGGTAGAGAAG	pPROEX_ScREV7- eGFP
REV7GFP-RP	CCG CTCGAG GTTAATTAACCCGGGGATCCGAAAC AAAGATCCAAAAATGC	pPROEX_ScREV7- eGFP
OSB150-FP	ATGCCT CATATG ATGGTGACTGGTGAAG	pGADT7-SAE2
OSB151-RP	ATGCCT GAATTC TTAACATCTTAGCATATA	pGADT7-SAE2
OSB152-RP	CCG CTCGAG GTTAATTAACCCGGGGATCCGGACT AAAGAAGTGAGTTTTATTTTAGG	pPROEX_ScREV7C1- eGFP

The bold letters correspond to restrictions sites.

Table S6. (continued)

Table S7.

Sequences of oligonucleotides used in the preparation of DNA substrates.

Oligonucleotide name	Sequence (5'—3')
OSB 17 Duplex-41 bp	GCCGTGATCACCAATGCAGATTGACGAACCTTTGCCACGT
OSB 20 Duplex-41 bp	ACGTGGGCAAAGGTTTCGTCAATCTGCATTGGTGATCACGGC
OSB 41 Duplex-60 bp	GGGTGAACCTGCAGGTGGGCAAAGATGTCCTAGCAATGTAATCGT CAAGCTTTATGCCGT
OSB 42 Duplex-60 bp	ACGGCATAAAGCTTGACGATTACATTGCTAGGACATCTTTGCCAC CTG CAGGTTACCC
OSB28 (4G3) G4-DNA	AATTCT <u>GGG</u> TGTGT <u>GGG</u> TGTGT <u>GGG</u> TGTGT <u>GGG</u> TGTGG
OSB 92 (TP) G4-DNA	TGGACCAGACCTAGCAGCTAT <u>GGGG</u> AGCT <u>GGGG</u> AAGGT <u>GGG</u> AA TGTGA
ODN 6G3 G4-DNA	AATTCT <u>GGG</u> TGTGT <u>GGG</u> TGTGT <u>GGG</u> TGTGT <u>GGG</u> TGTGT <u>GGG</u> TGTG T <u>GGG</u> TGTGG
OSB132 (TP-G4mutant)	TGGACCAGACCTAGCACTATCTGCAAGTCAAGTTGACTACGTATAC ATA

Table S8.

Primers used for qPCR analysis.

Primer	Sequence (5'—3')
0.7 kb (Forward)	TTTAGGATACTTTCACGCTTTA
0.7 kb (Reverse)	TTAGCTTGTACCAGAGGAA
3 kb (Forward)	GTCGTTCATATCTAAAGGAGTTAT
3 kb (Reverse)	CCTACCGCACCTTCTAAG
PRE1 (Forward)	AAATCTTTACGGTGGCAAA
PRE1 (Reverse)	CGCTAGAATGACAGAATCC
HO efficiency (Forward)	TCAATGATTAATAATAGCATAGTCGGGT
HO efficiency (Reverse)	CGTCAACCACTCTACAAAACCA

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Author information

Sugith Badugu*

Department of Biochemistry, Indian Institute of Science, Bengaluru, India

*Authors contributed equally to this work and are listed alphabetically

Kshitiza M Dhyani*

Department of Biochemistry, Indian Institute of Science, Bengaluru, India

*Authors contributed equally to this work and are listed alphabetically

Manoj Thakur

Sri Venkateswara College, University of Delhi, New Delhi, India

Kalappa Muniyappa

Department of Biochemistry, Indian Institute of Science, Bengaluru, India

ORCID iD: [0000-0002-9192-9194](https://orcid.org/0000-0002-9192-9194)

For correspondence: kmbc@iisc.ac.in

Editors

Reviewing Editor

Wolf-Dietrich Heyer

University of California, Davis, Davis, United States of America

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Adèle Marston

University of Edinburgh, Edinburgh, United Kingdom

Reviewer #1 (Public review):

Summary:

The mammalian Shieldin complex consisting of REV7 (aka MAD2L2, MAD2B) and SHLD1-3 affects pathway usage in DSB repair favoring non-homologous endjoining (NHEJ) at the expense of homologous recombination (HR) by blocking resection and/or priming fill-in DNA synthesis to maintain or generate near blunt ends suitable for NHEJ. While the budding yeast

Saccharomyces cerevisiae does not have homologs to SHLD1-3, it does have Rev7, which was identified to function in conjunction with Rev3 in the translesion DNA polymerase zeta. Testing the hypothesis that Rev7 also affect DSB resection in budding yeast, the work identified a direct interaction between Rev7 and the Rad50-Mre11-Xrs2 complex by two-hybrid and direct protein interaction experiments. Deletion analysis identified that the 42 amino acid C-terminal region was necessary and sufficient for the 2-hybrid interaction. Direct biochemical analysis of the 42 aa peptide was not possible. Rev7 deficient cells were found to be sensitive to HU only in synergy with G2 tetraplex forming DNA. Importantly, the 42 aa peptide alone suppressed this phenotype. Biochemical analysis with full-length Rev7 and a C-terminal truncation lacking the 42 aa region shows G4-specific DNA binding that is abolished in the C-terminal truncation and with a substrate containing mutations to prevent G4 formation. Rev7 lacks nuclease activity but inhibits the dsDNA exonuclease activity of Mre11. The C-terminal truncation protein lacking the 42 aa region also showed some inhibition suggesting the involvement of additional binding sites besides the 42 aa region. Also, the Mre11 ssDNA endonuclease activity is inhibited by Rev7 but not the degradation of linear ssDNA. Rev7 does not affect ATP binding by Rad50 but inhibits in a concentration-dependent manner the Rad50 ATPase activity. The C-terminal truncation protein lacking the 42 aa region also showed some inhibition but significantly less than the full-length protein. Using an established plasmid-based NHEJ assay, the authors provide strong evidence that Rev7 affects NEHJ, showing a four-fold reduction in this assay. The mutations in the other Pol zeta subunits, Rev3 and Rev1, show a significantly smaller effect (~25% reduction). A strain expressing only the Rev7 C-terminal 42 aa peptide showed no NHEJ defect, while the truncation protein lacking this region exhibited a smaller defect than the deletion of REV7. The conclusion that Rev7 supports NHEJ mainly through the 42 aa region was validated using a chromosomal NHEJ assay. The effect on HR was assessed using a plasmid:chromosome system containing G4 forming DNA. The rev7 deletion strain showed an increase in HR in this system in the presence and absence of HU. Cells expressing the 42 aa peptide were indistinguishable from wild type as were cells expressing the Rev7 truncation lacking the 42 aa region. The authors conclude that Rev7 suppresses HR, but the context appears to be system-specific and the conclusion that Rev7 abolished HR repair of DSBs is unwarranted and overly broad.

Strength:

This is a well-written manuscript with well-executed experiments which suggest that Rev7 inhibits MRX-mediated resection to favor NEHJ during DSB repair. This finding is novel and provides insight into the potential mechanism of how the human Shieldin complex might antagonize resection.

Weaknesses:

The nuclease experiments were conducted using manganese as a divalent cation, and it is unclear whether there is an effect with the more physiological magnesium cation. The data largely support the conclusions, although the effect of Rev7 on HR is less well documented, as only a highly specialized assay is used that does not warrant the broad conclusion drawn. Specifically, the results that the Rev7 c-terminal truncation lacking the 42 aa region still suppresses HR is unexpected and unexplained.

In this revision the authors addressed most of my concerns by text revisions and addition of new data.

The new two hybrid data showing that the 42 amino acid segment interacts with MRN are valuable. However, it may not be clear to which subunit the 42 aa segment binds, as in the yeast 2H system the chromosomally encoded subunits are present or were the 2H experiments conducted in an MRN deletion background?. This could be acknowledged.

The material and methods section was updated to indicate use of 5 mM MnCl₂ and 5 mM MgCl₂ in the exonuclease assay but not the endonuclease assay. Please check if this is correct. Why the difference between both assays? There is a concern that the absence of ATP and Mg affects the endonuclease assay.

The addition of Dmc1 as a specificity control for the ATPase inhibition is nice and shows a specific effect. The use of Sae2 associated nuclease activity as a specificity control for the nuclease inhibition is problematic. There has been considerable debate about the Sae2 associated nuclease activity, which seems to have been solved by the Cejka lab showing that Sae2 is a cofactor of MRN without intrinsic nuclease activity (e.g. <https://pubmed.ncbi.nlm.nih.gov/25231868/>). Or do the authors want to suggest that Sae2 has intrinsic nuclease activity? The control may still be useful mentioning that the nuclease is associated but not intrinsic and citing the relevant papers.

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Reviewer #2 (Public review):

In this study, Badugu et al investigate the Rev7 roles in regulating the Mre11-Rad50-Xrs2 complex and in metabolism of G4 structures. The authors also try to make a conclusion that REV7 can regulate the DSB repair choice between homologous recombination and non-homologous end joining.

The major observations of this study are:

- (1) Rev7 interacts with the individual components of the MRX complex in a two-hybrid assay and in a protein-protein interaction assay (microscale thermophoresis) in vitro.
- (2) Modeling using AlphaFold-Multimer also indicated that Rev7 can interact with Mre11 and Rad50.
- (3) Using a two-hybrid assay, a 42 C terminal domain in Rev7 responsible for the interaction with MRX was identified.
- (4) Rev7 inhibits Mre11 nuclease and Rad50 ATPase activities in vitro.
- (5) Rev 7 promotes NHEJ in plasmid cutting/relegation assay.
- (6) Rev7 inhibits recombination between chromosomal *ura3-1* allele and plasmid *ura3* allele containing G4 structure.
- (7) Using an assay developed in V. Zakian's lab, it was found that *rev7* mutants grow poorly when both G4 is present in the genome and yeast are treated with HU.
- (8) In vitro, purified Rev7 binds to G4-containing substrates.

In general, a lot of experiments have been conducted, but the major conclusion about the role of Rev7 in regulating the choice between HR and NHEJ is not justified.

(1) Two stories that do not overlap (regulation of MRX by Rev7 and Rev7 role in G4 metabolism) are brought under one umbrella in this work. There is no connection unless the authors demonstrate that Rev7 inhibits the cleavage of G4 structures by the MRX complex.

(2) The authors cannot conclude based on the recombination assay between G4-containing 2-micron plasmid and chromosomal *ura3-1* that Rev7 "completely abolishes DSB-induced HR". First of all, there is no evidence that DSBs are formed at G4. Why is there no induction of recombination when cells are treated with HU? Second, as the authors showed, Rev7 binds to G4, therefore it is not clear if the observed effects are the result of Rev7 interaction with G4 or impact on HR. The established HO-based assays where the speed of resection can be monitored (e.g., Mimitou and Symington, 2010) have to be used to justify the conclusion that Rev7 inhibits MRX nuclease activity in vivo.

Comments on the revised version:

I am satisfied with the revision. Specifically, i) the elimination of the G4 part and ii) the implementation of the HO-endonuclease resection assay described in Mimiou and Symington, 2010 significantly improved the clarity of the work and strengthened the conclusion about the Rev7 interference with DNA resection.

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Reviewer #3 (Public review):

Summary:

REV7 facilitates the recruitment of Shieldin complex and thereby inhibits end resection and controls DSB repair choice in metazoan cells. Puzzlingly, Shieldin is absent in many organisms, and it is unknown if and how Rev7 regulates DSB repair in these cells. The authors surmised that yeast Rev7 physically interacts with Mre11/Rad50/Xrs2 (MRX), the short-range resection nuclease complex and tested this premise using yeast two hybrid (Y2H) and microscale thermophoresis (MST). The results convincingly showed that the individual subunits of MRX interacts robustly with Rev7. By AlphaFold Multimer modelling followed by Y2H confirmed that the carboxy terminal 42 amino acid is essential for interaction with MR and G4 DNA binding by REV7. The mutant rev7 lacking the binding interface (Rev7-C1) to MR shows moderate inhibition to the nuclease and the ATPase activity of Mre11/Rad50 in biochemical assays. Deletion of REV7 also causes a mild reduction in NHEJ using both plasmid and chromosome-based assays and increases mitotic recombination between chromosomal *ura3-01* and the plasmid *ura3* allele interrupted by G4. The revision also showed that rev7 deleted cells exhibit mild hyper-resection phenotype at 0.7 and 3 kb from the DSB using qPCR assays. The authors concluded that Rev7 facilitates NHEJ and antagonises HR even in budding yeast, but it achieves this by blocking Mre11 nuclease and Rad50 ATPase.

Weaknesses:

There are several strengths to the studies and the broad types of well-established assays were used to deduce the conclusion. Nevertheless, there are notable discrepancies on the mutant phenotypes that were to test the functionality of Rev7-MRX interaction on the repair outcomes, raising concerns on the validity of the proposed model. The manuscript also needs a few additional functional assays to reach the accurate conclusions as proposed. The revision responded to several comments raised by the reviewers, but they are inadequate to address the key concerns and did not offer sufficient and compelling experimental support to the main premise that Rev7-Mre11/Rad50/Xrs2 interactions regulate MRX activities in cells and thereby modulates DSB repair choice in budding yeast.

(1) AlphaFold model predicts that Mre11-Rev7 and Rad50-Rev7 binding interfaces overlap and Rev7 might bind only to Mre11 or Rad50 at a time. Interestingly, however, Rev7 appears dimerized (Fig.1). Since MR complex also forms with 2M and 2R in the complex, it should still be possible if REV7 can interact both M and R in the MR complex. The author should perform MST using MR complex instead of individual MR components. The authors should also analyze if Rev7-C1 is indeed deficient in interaction with MR individually and with complex using MST assay.

(2) The nuclease and the ATPase assays require additional controls. Does Rev7 inhibit the other nuclease or ATPase non-specifically? Are these outcomes due to the non-specific or promiscuous activity of Rev7? In fig.6, the effect of REV7 on the ATP binding of Rad50 could be hard to assess because the maximum Rad50 level (1 μ M) was used in the experiments. The author should use the suboptimal level of Rad50 to check if REV7 still does not influence ATP binding by Rad50.

(3) The moderate deficiency in NHEJ using plasmid based assay in REV7 deleted cells can be attributed to aberrant cell cycle or mating type in rev7 deleted cells. The authors should demonstrate that rev7 deleted cells retain largely normal cell cycle pattern and the mating type phenotypes. The author should also analyze the breakpoints in plasmid based NHEJ assays in all mutants especially from rev7 and rev7-C1 cells.

(4) It is puzzling why the authors did not analyze end resection defects in rev7 deleted cells after a DSB. The author should employ the widely used resection assay after a HO break in rev3, rev7 and mre11 rev7 cells as described previously.

(5) Is it possible that Rev7 also contributes to NHEJ as the part of TLS polymerase complex? Although NHEJ largely depends on Pol4, the authors should not rule out the possibility if the observed NHEJ defect in rev7 cells are due at least partially to its well-known TLS defect and not all due to their role in MRX activity regulation as the authors proposed. In fact, rev3 or rev1 cells are partially defective in NHEJ (Fig. 7). Rev7-C1 is less deficient in NHEJ than REV7 deletion. These results predict that rev7-C1 rev3 could be more deficient than rev3 or rev7-C1, and such results might indicate that Rev7 contributes to NHEJ by two ways; one by interacting (and modulating) MRX and the other as part of Rev3-Rev7 complex. Additionally, the authors should examine if Rev7-C1 might be deficient in TLS. In this regard, does rev7-C1 reduce TLS and TLS dependent mutagenesis? Is it dominant? The authors should also check if Rev3/Rev1 complexes are stable in Rev7 deleted or rev7-C1 cells by immunoblot assays.

(6) Due to the G4 DNA and G4 binding activity of REV7, it is not clear which class of events the authors are measuring in plasmid-chromosome recombination assay in Fig.9. Do they measure G4 instability or the integrity of recombination or both in rev7 deleted cells. Instead, the effect of rev7 deletion or rev7-C1 on recombination should be measured directly by more standard mitotic recombination assays like mating type switch or his3 repeat recombination. The revision did not address these concerns, which still makes the interpretation of the provided recombination results difficult.

<https://doi.org/10.7554/eLife.96933.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer 1 (Public Review):

Summary

*The mammalian Shieldin complex consisting of REV7 (aka MAD2L2, MAD2B) and SHLD1-3 affects pathway usage in DSB repair favoring non-homologous end-joining (NHEJ) at the expense of homologous recombination (HR) by blocking resection and/or priming fill-in DNA synthesis to maintain or generate near blunt ends suitable for NHEJ. While the budding yeast *Saccharomyces cerevisiae* does not have homologs to SHLD1-3, it does have Rev7, which was identified to function in conjunction with Rev3 in the translesion DNA polymerase zeta. Testing the hypothesis that Rev7 also affects DSB resection in budding yeast, the work identified a direct interaction between Rev7 and the Rad50-Mre11-Xrs2 complex by two-hybrid and direct protein interaction experiments. Deletion analysis identified that the 42 amino acid C-terminal region was necessary and sufficient for the 2-hybrid interaction. Direct biochemical analysis of the 42 aa peptide was not possible. Rev7 deficient cells were found to be sensitive to HU only in synergy with G2 tetraplex forming DNA. Importantly, the 42 aa peptide alone suppressed this phenotype.*

Biochemical analysis with full-length Rev7 and a C-terminal truncation lacking the 42 aa region shows G4-specific DNA binding that is abolished in the C-terminal truncation and with a substrate containing mutations to prevent G4 formation. Rev7 lacks nuclease activity but inhibits the dsDNA exonuclease activity of Mre11. The C-terminal truncation protein lacking the 42 aa region also showed some inhibition suggesting the involvement of additional binding sites besides the 42 aa region. Also, the Mre11 ssDNA endonuclease activity is inhibited by Rev7 but not the degradation of linear ssDNA. Rev7 does not affect ATP binding by Rad50 but inhibits in a concentration-dependent manner the Rad50 ATPase activity. The C-terminal truncation protein lacking the 42 aa region also showed some inhibition but significantly less than the full-length protein.

Using an established plasmid-based NHEJ assay, the authors provide strong evidence that Rev7 affects NEHJ, showing a four-fold reduction in this assay. The mutations in the other Pol zeta subunits, Rev3 and Rev1, show a significantly smaller effect (~25% reduction). A strain expressing only the Rev7 C-terminal 42 aa peptide showed no NHEJ defect, while the truncation protein lacking this region exhibited a smaller defect than the deletion of REV7. The conclusion that Rev7 supports NHEJ mainly through the 42 aa region was validated using a chromosomal NHEJ assay. The effect on HR was assessed using a plasmid:chromosome system containing G4 forming DNA. The rev7 deletion strain showed an increase in HR in this system in the presence and absence of HU. Cells expressing the 42 aa peptide were indistinguishable from the wild type as were cells expressing the Rev7 truncation lacking the 42 aa region. The authors conclude that Rev7 suppresses HR, but the context appears to be system-specific and the conclusion that Rev7 abolished HR repair of DSBs is unwarranted and overly broad.

Strength

This is a well-written manuscript with many well-executed experiments that suggest that Rev7 inhibits MRX-mediated resection to favor NEHJ during DSB repair. This finding is novel and provides insight into the potential mechanism of how the human Shieldin complex might antagonize resection.

We thank Reviewer 1 for their comprehensive summary of our work. The Reviewers' recognition that our manuscript is "well-written" with "many well-executed experiments" and our findings are "novel" is greatly appreciated.

Weaknesses

The nuclease experiments were conducted using manganese as a divalent cation, and it is unclear whether there is an effect with the more physiological magnesium cation. Additional controls for the ATPase and nuclease experiments to eliminate non-specific effects would be helpful. Evidence for an effect on resection in cells is lacking. The major conclusion about the role of Rev7 in regulating the choice between HR and NHEJ is not justified, as only a highly specialized assay is used that does not warrant the broad conclusion drawn. Specifically, the results that the Rev7 C terminal truncation lacking the 42 aa region still suppresses HR is unexpected and unexplained. The effect of Rev7 on G4 metabolism is underdeveloped and distracts from the main results that Rev7 modulated MRX activity. The authors should consider removing this part and develop a more complete story on this later.

We have addressed each point identified as "Weaknesses" by the reviewer, as described below:

The nuclease experiments were conducted using manganese as a divalent cation, and it is unclear whether there is an effect with the more physiological magnesium cation.

We acknowledge the Reviewer's concern and apologize for not having been clear in our first submission. However, several studies have demonstrated that Mre11 exhibits all three DNase activities, namely single-stranded endonuclease, double-stranded exonuclease and DNA hairpin opening only in the presence of Mn^{2+} but not with other divalent cations, such as magnesium or calcium (Paull and Gellert, Mol. Cell 1998; 2000; Usui et al., Cell 1998; Ghosal and Muniyappa, JMB, 2007; Arora et al., Mol Cell Biol. 2017). For this reason, Mn^{2+} was used as a cofactor for the Mre11 nuclease assays. We have clarified this in the revised manuscript. As a side note, Mg^{2+} serves as a cofactor for Rad50's ATPase activity.

Additional controls for the ATPase and nuclease experiments to eliminate non-specific effects would be helpful.

We thank the Reviewer for raising this important point, as it led us to evaluate and confirm the specificity of Rev7 and exclude its potential non-specific effects. To this end, we have performed additional experiments, which showed that (a) the *S. cerevisiae* Dmc1 ATPase activity was not affected by Rev7, contrary to its inhibitory effect on Rad50 and (b) Rev7 had no discernible impact on the endonucleolytic activity of *S. cerevisiae* Sae2, whereas it inhibits DNase activities of Mre11. Thus, the lack of inhibitory effects on the ATPase activity of Dmc1 and nuclease activity of Sae2 confirm the specificity of Rev7 for Mre11 and Rad50 subunits. We have included this new data in Figure 6H and 6J and in Figure 5—figure supplement 1, respectively, in the revised manuscript.

Evidence for an effect on resection in cells is lacking. The major conclusion about the role of Rev7 in regulating the choice between HR and NHEJ is not justified, as only a highly specialized assay is used that does not warrant the broad conclusion drawn.

We agree with the Reviewer that *in vivo* evidence demonstrating the inhibitory effect of REV7 on DNA end resection was lacking in the first submission. Reviewer 2 and 3 have also raised point. We now measured the rate of DNA end resection using a qPCR-based assay (Mimitou and Symington, EMBO J. 2010; Gnugge et al., Mol. Cell 2023). The results revealed that deletion of REV7 led to an enhancement in the rate of DNA end resection at a DSB site inflicted by HO endonuclease (Figure 9—figure supplement 3), providing direct evidence that loss of REV7 contributes to increase in DNA end resection at the DSBs.

Specifically, the results that the Rev7 C-terminal truncation lacking the 42 aa region still suppresses HR is unexpected and unexplained.

This is a fair point, and we thank the reviewer for raising it. Although the interaction of Rev7-C1 in the yeast two-hybrid assays was not apparent, surprisingly, it partially suppressed HR (Figure 9). In line with this, biochemical assays showed that it exerts partial inhibitory effect on the Mre11 nuclease (Figure 5) and Rad50 ATPase (Figure 6) activities compared with the full-length Rev7. Consistent with *vitro* data, the AF2 models revealed that, in addition to the C-terminal 42-aa region, residues in the N-terminal region of Rev7 also interact with the Mre11 and Rad50 subunits (Figure 2—figure supplement 2).

The effect of Rev7 on G4 metabolism is underdeveloped and distracts from the main results that Rev7 modulated MRX activity. The authors should consider removing this part and develop a more complete story on this later.

We agree with the reviewer's comment "that the effect of Rev7 on G4 DNA metabolism is underdeveloped and distracts" from the central theme of the present paper, and suggested that we develop this part as a complete story later. This point has also been raised by

Reviewer 2 and 3 and, therefore, Figures and associated text were removed in the revised version of the manuscript.

Reviewer 2 (Public Review):

In this study, Badugu et al investigate the Rev7 roles in regulating the Mre11-Rad50-Xrs2 complex and in the metabolism of G4 structures. The authors also try to make a conclusion that REV7 can regulate the DSB repair choice between homologous recombination and non-homologous end joining.

The major observations of this study are:

- (1) Rev7 interacts with the individual components of the MRX complex in a two-hybrid assay and in a protein-protein interaction assay (microscale thermophoresis) in vitro.*
- (2) Modeling using AlphaFold-Multimer also indicated that Rev7 can interact with Mre11 and Rad50.*
- (3) Using a two-hybrid assay, a 42 C terminal domain in Rev7 responsible for the interaction with MRX was identified.*
- (4) Rev7 inhibits Mre11 nuclease and Rad50 ATPase activities in vitro.*
- (5) Rev 7 promotes NHEJ in plasmid cutting/relegation assay.*
- (6) Rev7 inhibits recombination between chromosomal ura3-1 allele and plasmid ura3 allele containing G4 structure.*
- (7) Using an assay developed in V. Zakian's lab, it was found that rev7 mutants grow poorly when both G4 is present in the genome and yeast are treated with HU.*
- (8) In vitro, purified Rev7 binds to G4-containing substrates.*

In general, a lot of experiments have been conducted, but the major conclusion about the role of Rev7 in regulating the choice between HR and NHEJ is not justified.

We appreciate Reviewer 2 for comprehensive assessment of our manuscript and their insightful comments. However, we believe that the data (Figure 7-9) in our manuscript, together with new data (Figure 9- figure supplement 2 and 3) in the revised manuscript, clearly demonstrate that Rev7 regulates the choice between HR and NHEJ.

- (1) Two stories that do not overlap (regulation of MRX by Rev7 and Rev7's role in G4 metabolism) are brought under one umbrella in this work. There is no connection unless the authors demonstrate that Rev7 inhibits the cleavage of G4 structures by the MRX complex.*

We agree with the reviewer's point that the themes associated with the regulation of the functions of MRX subunits by Rev7 and its role G4 DNA metabolism do not overlap. This concern has also been expressed by Reviewer 1 and 3. According to their suggestion, we have deleted all figures and text describing the role of Rev7 in G4 DNA metabolism from the revised manuscript.

- (2) The authors cannot conclude based on the recombination assay between G4-containing 2-micron plasmid and chromosomal ura3-1 that Rev7 "completely abolishes DSB-induced HR". First of all, there is no evidence that DSBs are formed at G4. Why is there no induction of recombination when cells are treated with HU? Second, as the authors showed, Rev7 binds to G4, therefore it is not clear if the observed effects are the*

result of Rev7 interaction with G4 or its impact on HR. The established HO-based assays where the speed of resection can be monitored (e.g., Mimitou and Symington, 2010) have to be used to justify the conclusion that Rev7 inhibits MRX nuclease activity in vivo.

We thank the Reviewer for the insightful comments and drawing our attention to the inference “completely abolishes DSB-induced HR”. We have we have rephrased the conclusion, and replaced it with “REV7 gene product plays an anti-recombinogenic role during HR”. Then, the reviewer refers to lack of “evidence that DSBs are formed at G4”. At this point, unfortunately, our attempts to identify DSB at the G4 DNA site in the 2-micron plasmid did not provide a clear answer to this question. This might be related to the existence of myriad DNases in the cell and technical issues associated with the isolation of low-abundant, linearized 2-micron plasmid molecules. Because of these reasons, we cannot provide any data on DSB at the G4 site in the 2-micron plasmid.

The reviewer then correctly points out “Why is there no induction of recombination when cells are treated with HU?” These findings are consistent with previous studies which have shown that Mre11-deficient cells are sensitivity to HU, resulting in cell death (Tittel-Elmer et al., EMBO J. 28, 1142-1156, 2009; Hamilton and Maizels, PLoS One, 5, e15387, 2010). However, a novel finding of our study is that ura3-1 rev7D cells and ura3-1 cells expressing Rev7-42 amino acid peptide (to limited extent) produce Ura3+ papillae. We have included this information in the Results section and adjusted the text to make this point clear to the reader.

In the same paragraph, the Reviewer expresses a concern about the interaction of Rev7 with G4 DNA substrates and its impact on HR. As discussed above, in response to your comment (1) and a similar comment of Reviewer 1 and 3, we have deleted all figures and text describing the role of Rev7 in G4 DNA metabolism in the revised manuscript. The reviewer specifically refers to a study by Mimitou and Symington, 2010 in which the speed DNA end resection at the HO endonuclease-inflicted DSB was quantified. We have carried out the suggested experiment and the results are presented in Figure 9—figure supplement 3.

Reviewer 3 (Public Review):

Summary:

REV7 facilitates the recruitment of Shieldin complex and thereby inhibits end resection and controls DSB repair choice in metazoan cells. Puzzlingly, Shieldin is absent in many organisms and it is unknown if and how Rev7 regulates DSB repair in these cells. The authors surmised that yeast Rev7 physically interacts with Mre11/Rad50/Xrs2 (MRX), the short-range resection nuclease complex, and tested this premise using yeast two-hybrid (Y2H) and microscale thermophoresis (MST). The results convincingly showed that the individual subunits of MRX interact robustly with Rev7. AlphaFold Multimer modelling followed by Y2H confirmed that the carboxy-terminal 42 amino acid is essential for interaction with MR and G4 DNA binding by REV7. The mutant rev7 lacking the binding interface (Rev7-C1) to MR shows moderate inhibition to the nuclease and the ATPase activity of Mre11/Rad50 in biochemical assays. Deletion of REV7 also causes a mild reduction in NHEJ using both plasmid and chromosome-based assays and increases mitotic recombination between chromosomal ura3-01 and the plasmid ura3 allele interrupted by G4. The authors concluded that Rev7 facilitates NHEJ and antagonizes HR even in budding yeast, but it achieves this by blocking Mre11 nuclease and Rad50 ATPase.

Weaknesses

There are many strengths to the studies and the broad types of well-established assays were used to deduce the conclusion. Nevertheless, I have several concerns about the validity of experimental settings due to the lack of several key controls essential to

interpret the experimental results. The manuscript also needs a few additional functional assays to reach the accurate conclusions as proposed.

We are happy that the Reviewer has found “many strengths” in our manuscript and further noted that “*results convincingly showed that the individual subunits of MRX interact robustly with Rev7*”. We greatly appreciate the Reviewer for these encouraging words, and for specific suggestions that helped us to improve the manuscript. As suggested, we have performed additional experiments including key controls and the data is presented in the revised manuscript.

(1) AlphaFold model predicts that Mre11-Rev7 and Rad50-Rev7 binding interfaces overlap and Rev7 might bind only to Mre11 or Rad50 at a time. Interestingly, however, Rev7 appears dimerized (Figure 1). Since the MR complex also forms with 2M and 2R in the complex, it should still be possible if REV7 can interact with both M and R in the MR complex. The author should perform MST using MR complex instead of individual MR components. The authors should also analyze if Rev7-C1 is indeed deficient in interaction with MR individually and with complex using MST assay.

Thank you for the valuable suggestion. As requested, MST titration experiments have been performed to examine the affinity of purified GFP-tagged Rev7-C1 for the Mre11, Rad50 and MR complex. The results revealed that Rev7-C1 binds to the Mre11 and Rad50 subunits with about 3- and 8.8-fold reduced affinity, respectively; whereas it binds to the MR complex with ~5.6-fold reduced affinity compared with full-length Rev7. The data is shown in Figure 1—figure supplement 4A-C.

(2) The nuclease and the ATPase assays require additional controls. Does Rev7 inhibit the other nuclease or ATPase non-specifically? Are these outcomes due to the non-specific or promiscuous activity of Rev7? In Figure 6, the effect of REV7 on the ATP binding of Rad50 could be hard to assess because the maximum Rad50 level (1 mM) was used in the experiments. The author should use the suboptimal level of Rad50 to check if REV7 still does not influence ATP binding by Rad50.

We thank the Reviewer for these valuable comments (Reviewer 1 has raised similar issues). Thus, we performed additional control experiments and the results indicate that (a) the ATPase activity of *S. cerevisiae* Dmc1 was not affected by Rev7 and (b) Rev7 does not inhibit the endonucleolytic activity of *S. cerevisiae* Sae2. The results are depicted in Figure 6H and 6J and Figure 5 –figure supplement 1A-D, respectively.

As suggested by the Reviewer, using suboptimal levels of Rad50 (0.2 mM), we carried out experiments to test the effect of varying concentrations of Rev7 on the ability of Rad50 to bind ATP and catalyse its hydrolysis. The results showed that Rev7 had no discernible effect on its ability to bind ATP, even at concentrations 30 times higher than the concentration of Rad50 (Figure 6B and 6D). However, Rev7 suppresses the ATPase activity of Rad50, but not that of Dmc1, in a concentration-dependent manner (Figure G, 6J).

(3) The moderate deficiency in NHEJ using plasmid-based assay in REV7 deleted cells can be attributed to aberrant cell cycle or mating type in rev7 deleted cells. The authors should demonstrate that rev7 deleted cells retain largely normal cell cycle patterns and the mating type phenotypes. The author should also analyze the breakpoints in plasmid-based NHEJ assays in all mutants, especially from rev7 and rev7-C1 cells.

We appreciate the Reviewer's critical and insightful comment. We monitored cell-cycle progression of both wild-type and *rev7D* cells over time using FACS. The results revealed that the cell cycle profiles and mating type phenotypes *rev7D* cells were similar to the wild type

cells. The data is presented in Figure 7-figure supplement 1. This indicates that *rev7D* cells do not possess aberrant cell cycle or mating type defects as compared with the wild-type cells.

We find the second point raised by the Reviewer although is intriguing, its relevance to the current study is unclear. In our view, identification of breakpoints using plasmid-based NHEJ assays in all the mutants will require a significant amount of time, and the insight that we may gain is unlikely to add to the central theme of this paper. Moreover, we know for sure that Rev7 has no DNA cleavage/nicking activity.

(4) It is puzzling why the authors did not analyze end resection defects in rev7 deleted cells after a DSB. The author should employ the widely used resection assay after a HO break in rev3, rev7, and mre11 rev7 cells as described previously.

Thank you for the suggestion. Reviewer 1 also has raised this point. As suggested, we have analysed end resection in the *rev7D* cells at a HO inflicted DSB site using a qPCR assay (Mimitou and Symington, EMBO J. 2010; Gnugge et al., Mol. Cell 2023). The results revealed that deletion of REV7 led to an enhancement in the rate of DNA end resection at a DSB inflicted by HO endonuclease (Figure 9—figure supplement 3),

(5) Is it possible that Rev7 also contributes to NHEJ as the part of TLS polymerase complex? Although NHEJ largely depends on Pol4, the authors should not rule out that the observed NHEJ defect in rev7 cells is due at least partially to its TLS defect. In fact, both rev3 or rev1 cells are partially defective in NHEJ (Figure 7). Rev7-C1 is less deficient in NHEJ than REV7 deletion. These results predict that rev7-C1, rev3 should be as defective as the rev7 deletion. Additionally, the authors should examine if Rev7-C1 might be deficient in TLS. In this regard, does rev7-C1 reduce TLS and TLS-dependent mutagenesis? Is it dominant? The authors should also check if Rev3 or Rev1 are stable in Rev7 deleted or rev7-C1 cells by immunoblot assays.

We agree with the possibility that Rev7 may play a role in translesion DNA synthesis and TLS-dependent mutagenesis. Accordingly, Rev7-C1 might be deficient in TLS. While we do not rule out such scenarios, we respectfully suggest that this is outside the scope of the current manuscript. This manuscript focuses on the role of Rev7 in NHEJ and HR pathways, not on translesion DNA synthesis. Nevertheless, we recognise the importance of this line of investigation, and we will certainly consider this suggestion in our future work. Thank you.

(6) Due to the G4 DNA and G4 binding activity of REV7, it is not clear which class of events the authors are measuring in plasmid-chromosome recombination assay in Figure 9. Do they measure G4 instability or the integrity of recombination or both in rev7 deleted cells? Instead, the effect of rev7 deletion or rev7-C1 on recombination should be measured directly by more standard mitotic recombination assays like mating type switch or his3 repeat recombination.

We appreciate the Reviewer for highlighting this important point and would like to take the opportunity to clarify the rationale behind plasmid-chromosome recombination assay, as previously described (Paeschke et al., Cell 145, 678, 2011). In this assay, we are measuring the rate of Ura⁺ papillae formation arising from integration of the targeting plasmid into the genome at the *ura3-1* locus of wild-type and *rev7D* cells. Analysis of PCR-generated DNA fragments indicate that pFAT10-G4 plasmid integrates at the *ura3-1* genomic locus of *rev7D* cells, but not in the wild-type cells (Figure 9-figure supplement 2). Further, we also measured the stability of G4 DNA and the results indicate that it is stable in *rev7D* cells.

Recommendations for the authors:

Reviewer 1 (Recommendations for the authors):

(1) Title: The word 'choice' implies a regulator. Is that the model here? Alternatively, is it pathway properties that define the preference of usage?

This is an excellent suggestion. In the revised submission, we rephrased the title “*Saccharomyces cerevisiae* Rev7 promotes non-homologous end-joining by inhibiting Mre11 nuclease and Rad50 ATPase activities and Homologous recombination.”

(2) Line 83, Introduction: Titia De Lange proposed an alternative/complementary model for Shieldin and REV7 to support fill-in by DNA polymerases including Pol alpha. This should be discussed.

We thank the reviewer for pointing out that we have not discussed the work from Titia De Lange’s research group. We have now added new sentences to the Introduction to describe the alternative model involving Pola-primase fill-in synthesis (p3.2.7).

(3) Line 131: The paragraph title needs to change. 2-hybrid assays cannot establish direct interaction especially when analyzing yeast proteins by yeast 2-hybrid. I agree that direct interaction is established by other means later.

Per the Reviewer’s suggestion, we have deleted the word “directly” from the title of the paragraph.

(4) Figure 1 D-F: The purity of the Rev7-GFP fusion is shown in Figure S1, and the purity of the Rad50, Mre11, and Xrs2 subunits as assessed by PAGE should be shown as well.

Following this suggestion, we have included images of Coomassie blue-stained SDS-polyacrylamide gels (Figure 1-figure supplement 1), which show the purity and size of GFP tagged Rev7, Rad50, Mre11, Xrs2, Rev1, Sae2 and Dmc1 proteins.

(5) Please check the Kd values. In the graph in D, the differences between Rad50, Mre11, and Xrs2 look much larger than the values in F suggest.

This is a fair point and we appreciate the reviewer for highlighting. The differences between the binding profiles of the Rad50, Mre11, and Xrs2 with Rev7 as shown in the previous version of the manuscript were not obvious because of cluttering of binding curves. Therefore, the binding profiles of interacting pair of proteins were plotted separately to highlight the differences (Figure 1—figure supplement 3). Further, we rigorously analysed the dataset to ascertain the binding affinities and found that the *Kd* values obtained were in good agreement with the values shown in Figure 1D.

(6) Figure 1S3: Please label the bands.

In the revised manuscript, the protein bands in Figure1-figure (previously Figure 1S3) are identified with their names.

(7) Line 195: Change Figure 1 to Figure 1S4.

We have introduced the correction in the revised manuscript.

(8) Line 202: The minimal interaction domain of 42 aa is only described in the next paragraph. The description anticipates a result about the 42 aa fragment that has not been shown to this point. Please reorder results or descriptions to make this coherent.

We have implemented the change, as per the Reviewer's suggestion.

(9) Figure 2: The two-hybrid analysis in Figures 1 and 2 also identifies Rev7 self-interaction, which is not discussed. This serves as another control against the artifact of the truncation proteins and should be discussed.

We have now discussed the significance of Rev7 self-interaction in the Y2H experiments wherever relevant in the text.

(10) Is the 42 aa fragment sufficient to elicit a two-hybrid signal?

We thank the reviewer for this insightful comment. To test this premise, we expressed the terminal 42 amino acid sequence of Rev7 using bait pGBKT7 vector. The results revealed that the 42 residue fragment of ScRev7 alone is sufficient for a two-hybrid interaction with the MRX subunits (Figure 2-figure supplement 1).

(11) Line 289: Why are the EMSA conditions described as physiological? As per Material and Methods, the reaction mixtures contain 20 mM Tris-HCl (pH 7.5), 0.1 mM DTT, 0.2 mg/ml BSA, and 5% glycerol, which is far from physiological.

As suggested by all three reviewers, the data showing the interaction of Rev7 and its truncation derivative Rev7-C1 with G4 DNA has been deleted in the revised version of the manuscript.

(12) Figure 4C: The figure needs to increase in size. The plotting symbols are not all visible, and it is undefined what the black squares represent.

Following the reviewer's suggestion, Figure 4C has been omitted in the revised version of the manuscript.

(13) Figure 5: The MRX nuclease assays were conducted in the presence of Manganese. Has the more physiological divalent cation magnesium been tested?

This has been addressed in response to the query of Reviewer 1 (Public Review). As noted above, Mre11 exhibits DNase activities only in the presence of Mn^{2+} .

(14) In Figure 5D, lane 2: What is the concentration of Rev7?

We appreciate the reviewer for catching this. The concentration of ScRev7 used for the reaction shown in Figure 5D, lane 2 was 2 μ M, as specified in the Figure legend.

(15) Figure 6 legend: Lane 1620 "same as in lane "Is there a "1" missing?

We thank the reviewer for pointing out the typographical error, which has been corrected in the revised manuscript.

(16) Figure 9: Rev7-C1 lacks the 42 aa peptide that is postulated to mediate anti-resection but shows normal HR here. This seems unexpected based on the premise that the 42 aa fragment supports end-joining. Rev7 seems to suppress HR independent of the function of the 42 aa peptide.

This has been addressed in response to the query posed by Reviewer 1 in the Public Review. We do see that the Rev7-C1 lacking the 42 aa peptide suppresses HR, but the suppression was

only partial as compared with the wild type. This is consistent with biochemical assays suggesting that Rev7-C1 exerts partial inhibition on the Mre11 nuclease (Figure 5) and Rad50 ATPase (Figure 6) activities. Further, the AF2 models indicate that, in addition to the C-terminal 42-aa region, other regions of Rev7 also interact with the Mre11 and Rad50 subunits (Figure 2—figure supplement 2), consistent with biochemical and genetic data.

(17) Line 478: The conclusion that "these findings are consistent with the idea that REV7 completely abolishes DSB-induced HR in *S. cerevisiae*." is overly broad as the assay

We agree with the reviewer's assessment. Accordingly, we have rephrased the sentence to soften the claim.

Line 483ff: Based on the comments on Figure 9, the introductory sentences of the discussion do not seem to be supported by the data, as Rev7 appears to regulate HR independent of the 42 aa peptide.

Please refer to the response of comment #16 above

(18) Line 536: Similarly to above 17, the conclusion about the effect of the 42 aa peptide on HR appears unwarranted.

We have revised the statement to moderate the previously exaggerated claims.

(19) In all figures, please list in the legend, which exact strains have been used referring to Table S5.

We have now included mentions of the strains in the figure legend wherever applicable.

(20) Line 351: linear.

It is corrected in the revised manuscript.

Reviewer 2 (Recommendations For The Authors):

(1) It is very strange and unusual that Rev7 independently binds to all three subunits of the MRX complex, raising a question of how specific these interactions are. At least, it should be a negative control in their Y2H assay and protein-protein interaction assay *in vitro* that Rev7 does not bind to some other proteins. For example, Sae2 and Rev7 interactions can be tested.

The reviewer is right that it is important to validate the specificity of Y2H interactions as well as *in vitro* enzyme assays. These findings are shown in Figure 6 and Figure 5-figure supplement 1. As suggested by the Reviewer, we included SAE2 in Y2H and MST assays, and Dmc1 and Sae2 *in vitro* enzyme assays. Our results clearly showed that Sae2 neither interacts with MRX subunits in Y2H assays (Figure 1A-C) nor inhibits the Sae2's nuclease and Dmc1's ATPase activities *in vitro* (Figure 6 and Figure 5-figure supplement 1)

(2) It is surprising that in the Discussion the authors speculate that Rev7 might recruit Mus81 nuclease for cleavage, completely ignoring their own publication on the cleavage of G4 by MRX.

We agree with the reviewer, and we have added discussion about MRX (mentioned above by the reviewer) in revised version.

(3) How does the AlphaFold-Multimer modeling predict the interaction between Rev7 and MRX as a complex? Are the same regions of MRX accessible for the interaction with Rev7 in this case? Similarly, how are the activities of the MRX complex and phosphorylated Sae2 (see P. Cejka's work) affected by Rev7?

Thank you for pointing this out. In this study, we investigated the interaction between Rev7 and Mre11, and between Rev7 and Rad50 subunits using AF2 algorithm. However, the three-dimensional structure of *S. cerevisiae* MRX-Rev7 complex could not be constructed due to the size limits imposed by AF2 algorithm. Therefore, we are unable to comment on whether the same regions of MRX subunits in the complex are accessible for the interaction with Rev7. That said, AF2 algorithm has recently been used for structural modelling of *S. cerevisiae* Mre11 (1–533)-Rad50 (1–260 + 1,057–1,312) complex (Nicolas *et al.*, *Mol. Cell* **84**, 2223, 2024). As such, there are no AF2 structural models that cover the whole length of Mre11-Rad50 proteins.

Regarding the second point raised by the Reviewer, our results suggest that Rev7 does interact with Sae2 in Y2H assays. However, whether phosphorylated Sae2 could potentially affect the interaction between MRX subunits and Rev7 warrants further studies.

Minor points:

(1) Figure 1. The labeling of the strains in A and B is genes and in C is proteins.

The reviewer is correct. We have now corrected the error in the Figure 1 and 2.

(2) Abstract. Carefully check English grammar.

We thank the Reviewer for spotting this, which has been corrected in the revised manuscript.

(3) Line 322 "Further, it has been demonstrated that Mre11 cleaves non-B DNA structures such as DNA hairpins, cruciforms and intra- and inter-molecular G-quadruplex structures)." It has not been shown that Mre11 cuts cruciform structures.

We thank the referee for spotting this error. Mre11 does not cleave cruciform DNA structures. This error is corrected in the revised manuscript.

(4) Page 14. Lines 452-455. What does "selective and non-selective media" mean? Is it without and with HU treatment?

Thanks very much for the comment. In our manuscript, selective medium is composed of SC/-Leu with HU and non-selective medium is without HU. We have clarified this point in the revised version.

(5) Page 15. Lane 472 "To assess whether increased frequency of HR is due to the instability of G-quadruplex DNA in rev7Δ cells, we examined the length of G4 DNA inserts in the plasmids carrying sequences during HR assay". It is not clear what does mean "during HR assay"? Did you examine the presence of G4 in Ura+ recombinants? If not, this analysis is meaningful.

The reviewer is correct. We measured the presence of G4 DNA insert in Ura+ recombinants. The text has been appropriately edited to reflect these necessary modifications.

(6) What is the nature of the ura3-1 allele? Can it revert to URA3 in rev7 mutants?

The *ura3-1* allele (glycine-to-glutamate substitution) reverts to Ura3⁺ at a low rate of $\sim 2.5 \times 10^{-9}$ in both orientations (Johnson et al., Mol. Cell 59, 163, 2015)

(7) From the way that the recombination process is depicted it seems that the authors believe that plasmid should integrate into the chromosome. In reality, in most cases it should be a gene conversion where the G4 sequence (if it indeed induces DSBs) should be replaced by the wild-type segment from ura3-1, integration is not required since it is 2-micron plasmid.

We apologize for not having made this clearer. The recombination assay with targeting plasmids containing G4 DNA forming sequences was performed as previously described (Paeschke et al., Cell 145, 678, 2011). In this assay, the appearance of Ura⁺ recombinants arise from the integration of the targeting plasmid bearing *ura3G4* allele (with a G4 DNA forming insert) integrates into the genome at the *ura3-1* locus. As shown in Author response image 1B, this is confirmed by PCR amplification of the insert in the genomic DNA of wild type and *rev7D* cells.

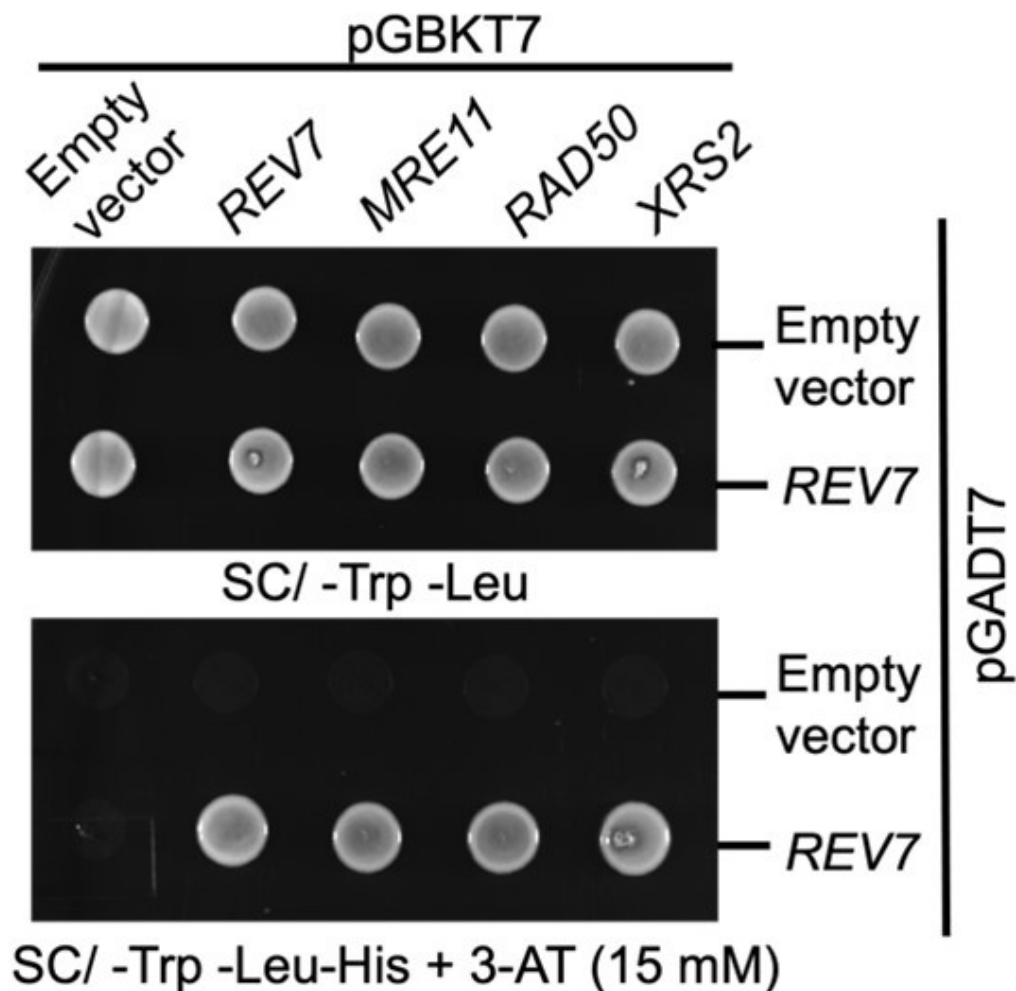
Reviewer 3 (Recommendations For The Authors):

(1) All Y2H experiments were performed with REV7 fusion to pGBKT7 and MRX to pGADT7. It will be helpful to test if pGAD-Rev7 also interacts with pGBK-Mre11 or Rad50 by Y2H.

Following the reviewers' suggestions, we performed Y2H experiments in wild-type PJ69-4a cells co-transformed with the pGBKT7 vector expressing MRX subunits and the pGADT7 vector expressing Rev7. The results indicated that Rev7 interacts with Mre11, Rad50 or Xrs2 subunits, indicating that interactions are vector-independent.

Author response image 1.

Yeast two hybrid analysis suggest interaction between Rev7 and MRX subunits. PJ69-4A cells were co-transformed with bait vector expressing Rev7 or the Mre11, Rad50 or Xrs2 subunits and prey vector expressing Rev7 protein. Equal number of cells were spotted onto –Trp – Leu and –Trp – Leu –His dropout plates containing 3-AT and images were obtained following 48 h of incubation at 30°C. The data is representative of three independent experiments.



(2) G4 studies are under-developed and do not add much or even negatively to the manuscript. The author might consider revising the manuscript to improve their integration with better rationales or logic. Alternatively, the authors should consider removing the G4 part for another paper.

This concern was also raised by Reviewer 1 and 2. Following the suggestions of all reviewers, figures and text related G4 DNA studies have been deleted in the revised manuscript.

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